Analysis note for the paper on the system size and energy dependence of near-side jet-like correlations

By

Christine Nattrass (Yale, nattrass@rhig.physics.yale.edu), Jana Bielcikova (NPI, jana.bielcikova@ujf.cas.cz), Jörn Putschke (Yale, joern.putschke@yale.edu), Helen Caines (Yale, helen.caines@yale.edu)

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Contents

Chapter 1

Particle selection and identification

1.1 Event selection

Data in this paper are from, $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV from RHIC's 2002 run, $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV from RHIC's fourth run (2004) and $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV from RHIC's fifth run (2005.) For all systems and energies, the vertex is limited to within 30 cm of the center of the TPC along the beam axis. The further an event is from the center of the TPC the lower the geometric acceptance; a 30 cm cut allows reasonably uniform geometric acceptance. The production used and the number of good events is listed in Table 1.1 for all data sets.

The centrality bins used in this analysis and their corresponding reference multiplicities, N_{bin}, and N_{part} are given in Appendix C.

Events from $Au + Au$ collisions have a more clearly defined vertex because there are more tracks produced in an $Au+Au$ collision which can be used to reconstruct the vertex position. The 80% most central events were usable in $Au+Au$ collisions. Fewer tracks are produced in $Cu + Cu$ collisions and the luminosity was significantly higher in 2005 than in 2006. The higher luminosity meant that particles from a previous event could still be traversing the TPC when a new event triggered the detector to begin reading out data, a problem called pile-up. For higher multiplicities, it is still possible to determine the primary vertex, but the lower the multiplicity the harder it

Table 1.1: Information about the productions used in the analysis

collision system and energy	production used	number of events
$d + Au$ $\sqrt{s_{NN}}$ = 200 GeV	P03ih	1M
$Cu + Cu$ $\sqrt{s_{NN}}$ = 62 GeV	P05id	24M
$Cu + Cu$ $\sqrt{s_{NN}}$ = 200 GeV	P06ib	38
$Au + Au$ $\sqrt{s_{NN}}$ = 62 GeV	P04id and—P04ie	8M
$Au + Au$ $\sqrt{s_{NN}}$ = 200 GeV	P05ic	24M central, 25M minimum bias

is to determine where the vertex from the event which triggered the detector is. For this reason only the 60% most central data from these events was usable. For more peripheral events, the vertex which triggered the event could not be found in more than 2% of events, in large part because vertices from pile-up events usually had a greater multipicity. Full details of the separation of the vertices from the triggered event and from pile-up events can be found in [1].

1.2 Selection of events and charged tracks

All particles are restricted to $|\eta| < 1.0$. All tracks used in these studies are required to have at least 15 hits in the TPC associated with the track. Tracks are restricted to a DCA < 1.0 cm, limiting contributions from decay particles.

Unidentified hadron efficiency

To determine the efficiency for reconstructing tracks, the response to the particle as it traverses the STAR detector is simulated. For unidentified hadrons, the response of

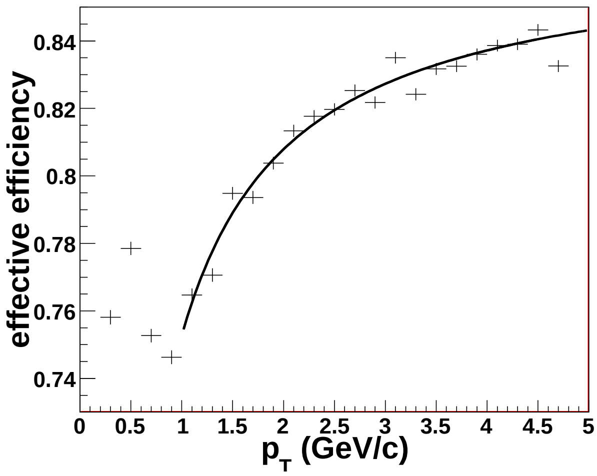

Figure 1.1: The effective efficiency, including acceptance effects, for reconstructing unidentified hadrons in 0-10% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The curve is a fit to the data.

the detector is simulated for pions. There are significant contributions from protons in the data, however, the efficiency for the reconstruction of pions and protons is similar for particles with $p_T > 1$ GeV/c. The reconstructed track must have at least 10 of the simulated hits for the simulated track to be considered reconstructed. The ratio of the number of reconstructed tracks to the number of simulated tracks gives the effective efficiency, including acceptance effects. The effective reconstruction efficiency was determined for tracks meeting the quality requirements applied in the analysis and was determined as a function of p_T and centrality for each data set. This was fit to a curve $a\ exp((\frac{b}{p_T})^c)$. An example of the effective efficiency for the reconstruction of unidentified hadrons in 0-10% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV is shown in Figure 1.1. For $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV, a constant efficiency of 0.89 was used. The parameters for the fits to the efficiency as a function of p_T for each centrality bin for every other data set are in Appendix B.

Chapter 2

Correlation Method

2.1 Correlation technique

In a given event, a high-p_T particle, called the trigger particle, is selected and the distribution of all lower momentum particles, called associated particles, relative to that track in $\Delta\eta$ and $\Delta\phi$ is determined. Tracks are selected as described in Chapter 1, with no additional restrictions on the high momentum track other than the p_T. Multiple tracks in a single event may be counted as trigger particles. The data are symmetric about $\Delta\phi = 0$ and $\Delta\eta = 0$ because the sign of $\Delta\phi$ and $\Delta\eta$ are determined by an arbitrary choice of coordinates. Since many of these measurements are limited by statistics, the data are reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$ to minimize statistical fluctuations. The single track efficiency, described in Chapter 1, is dependent on particle type, collision system and energy, p_T, and centrality and the correction for associated particles is applied on a track-by-track basis. The final distribution is normalized by the number of trigger particles, so no correction for the efficiency of reconstructing the trigger particle is necessary. Two corrections for the acceptance of track pairs in $\Delta\phi$ and $\Delta\eta$, described below, are applied. This is done for the full sample of events to give the average distribution of associated particles relative to a high-p_T trigger particle $\frac{1}{N_{trigger}}\frac{d^2N}{d\Delta\phi d\Delta\eta}$.

Two key features of the STAR TPC affect the geometric acceptance of track pairs. Tracks in the TPC can only be resolved well for -1 < η < 1 because the lower

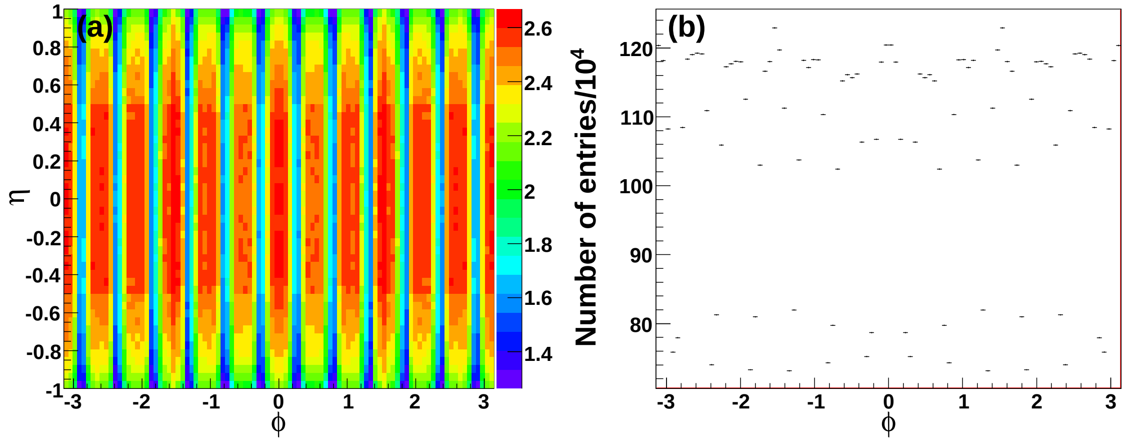

Figure 2.1: Distribution of tracks in (a) ϕ and η and (b) ϕ in $Cu + Cu$ events at $\sqrt{s_{NN}}$ = 200 GeV for 1.5 GeV/c < p_T <6.0 GeV/c. Note the suppressed zero.

geometric acceptance leads to tracks with fewer hits in the TPC, so only tracks within -1 < η < 1 are used for this analysis. Track pairs with $\Delta\eta \approx 0$ are nearly always accepted, however, track pairs with $|\Delta\eta| \approx 2$ are only accepted if the tracks are near opposite edges of the detector. Therefore acceptance is roughly 100% near $\Delta\eta \approx 0$ but roughly 0% near $|\Delta\eta| \approx 2$. This leads to an acceptance in $\Delta\eta$ with a triangular shape.

In addition, there are gaps between each of the 12 TPC sectors. This leads to holes in the acceptance in azimuth. Most tracks will have hits in multiple sectors, however, the loss of hits leads to a decrease in the reconstruction efficiency for tracks at the angles of the sector boundaries. A sample distribution of tracks reconstructed in the TPC is shown in Figure 2.1 showing the lower efficiency at the TPC sector boundaries.

To correct for these effects, the distributions of tracks for both the associated and the trigger particles are recorded. A random trigger (ϕ,η) is chosen from this distribution of trigger particles and a random associated (ϕ,η) is chosen from the distribution of associated particles. In this way a simulated distribution of uncorrelated track pairs that includes detector acceptance effects is attained. This distribution is normalized to 1 at the peak at $\Delta\eta \approx 0$ to give an efficiency for finding a pair of tracks as a function of ($\Delta\phi$,$\Delta\eta$). A sample of the efficiency due to detector effects is shown

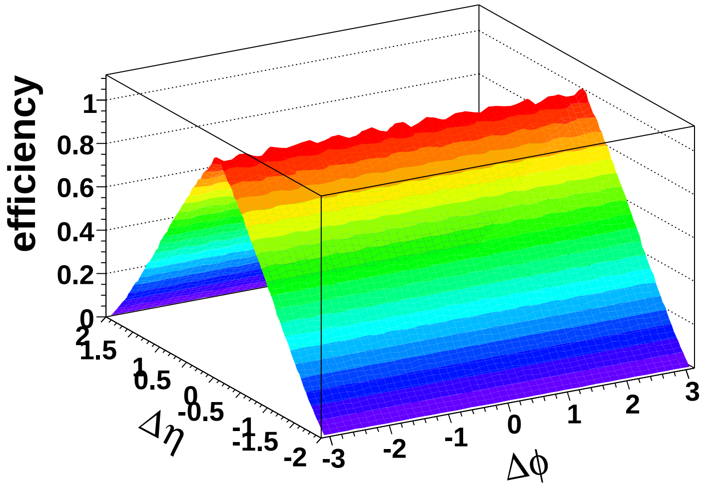

Figure 2.2: Sample efficiency due to detector effects for $Cu + Cu$ events at $\sqrt{s_{NN}}$ = 200 GeV with 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ and $3.0 < p_T^{trigger} < 6.0$ GeV/c

in Figure 2.2 and the effects of the geometric acceptance are evident.

2.1.1 Track merging correction

If two tracks have small separations in pseudorapidity and azimuth, they are more difficult to resolve because hits from each particle may not be resolved by the TPC. Therefore, if the particles have nearly identical momenta, their tracks are not distinguished. This effect is called track merging and causes an artificial dip in the raw correlations centered at ($\Delta\phi$,$\Delta\eta$) = (0,0). Another similar effect is evident in the TPC from the crossing of high-p_T tracks. While the hits are not lost in this case, one track will lose hits near the crossing point and this particle's track gets split into two shorter tracks. Shorter tracks are less likely to meet the track quality cuts in Chapter 1 because they have fewer hits. This effect causes artificial dips near ($\Delta\phi$,$\Delta\eta$) = (0,0) but slightly displaced in $\Delta\phi$. The location of these dips in $\Delta\phi$ is dependent on the relative helicities of the trigger and associated particles. The helicity is given by

$$h = -\frac{qB}{|qB|} \tag{2.1}$$

where q is the charge of the particle and B is the magnetic field [2].

A method for correcting high-p_T triggered di-hadron correlations was developed in [2, 3], building on a method for correcting for track merging in studies of Hanbury-Brown-Twiss correlations [4].

Correction for unidentified hadrons

Figure 2.3 shows a sample of raw di-hadron correlations between unidentified hadrons for all four helicity combinations in $Au + Au$ events at $\sqrt{s_{NN}} = 62$ GeV. When the helicities of the trigger and associated particles are the same, the percentage of overlapping hits is greater. Track pairs are lost whether the pair is part of the combinatorial background or part of the signal so the magnitude of the dip is greater when the background is greater. $Au + Au$ collisions were chosen to demonstrate the effect because the background is largest for these collisions, making the effect easy to see.

The number of hits shared by two tracks is calculated based on the trajectory of the track and if more than 10% of hits are shared between two tracks, the tracks are considered merged and the track pair is discarded. To get an acceptance correction for this effect, the trigger particle is mixed with associated particles from another event. This procedure is called event mixing. The trigger and associated particles from separate events will have no real correlations so any structures in the resulting ($\Delta\phi$,$\Delta\eta$) distribution will result only from detector acceptance effects, including track merging. In the mixed events, the origin of the associated track must be shifted to the new vertex so that the percentage of merged hits can be calculated accurately. In addition, mixed events are required to have vertices within 2 cm of each other along the beam axis in order to ensure similar geometric acceptance in both events. The reference multiplicities of both events are required to be within 10. The calculation of the number of hits which may be shared by the tracks does not describe the performance of the TPC perfectly and some track pairs which were calculated to share more than 10% of their hits are reconstructed. These pairs are discarded even if they were reconstructed so that the percentage of merged track pairs is the same in the data and the mixed events. The mixed events are then used to determine an

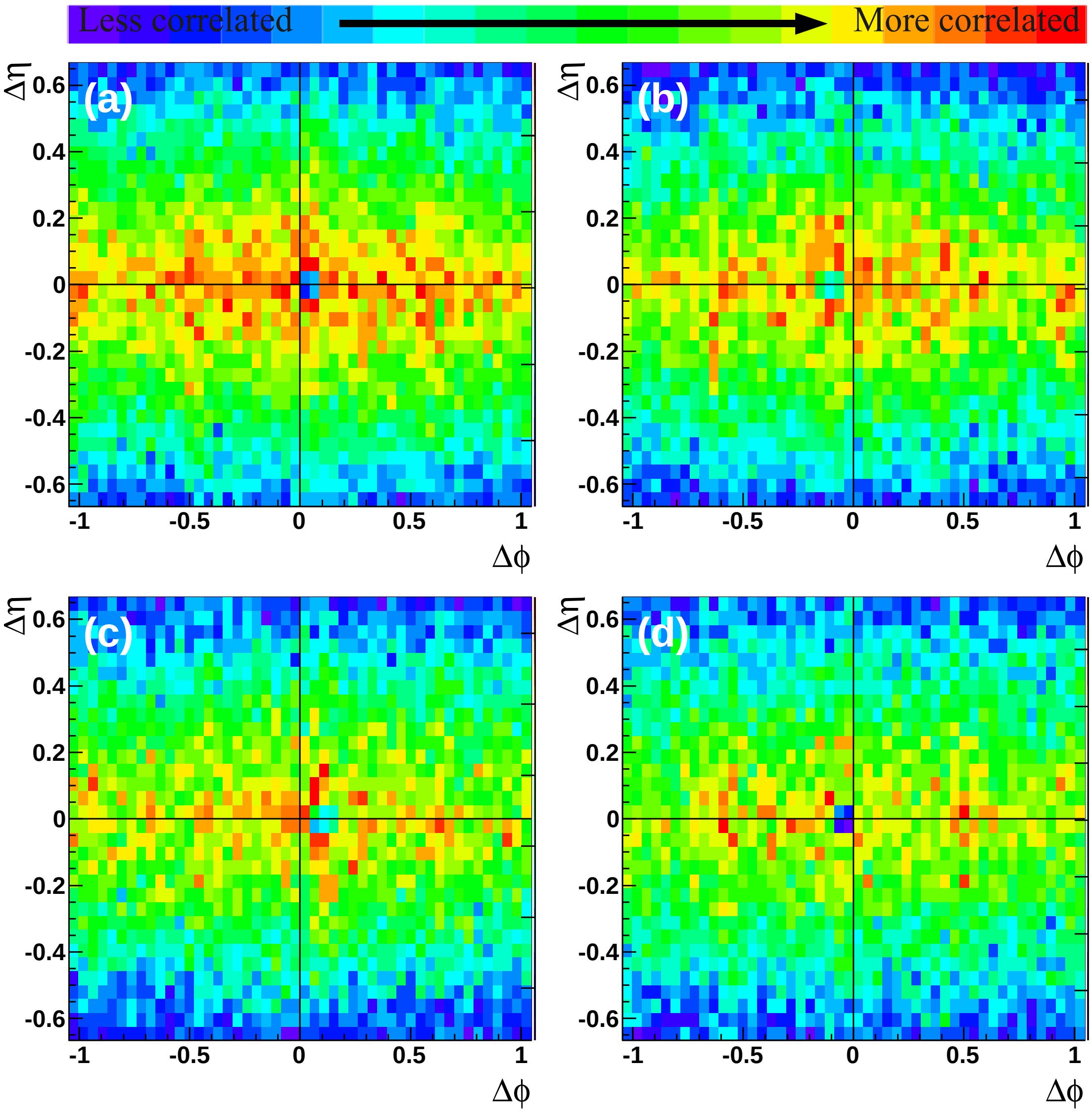

Figure 2.3: Raw correlations for (a) trigger particle helicity 1 and associated particle helicity 1 (b) trigger particle helicity 1 and associated particle helicity -1 (c) trigger particle helicity -1 and associated particle helicity 1 (d) trigger particle helicity -1 and associated particle helicity -1 in h-h correlations in 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV for 1.5 < $p_T^{associated}$ < $p_T^{trigger}$ GeV/c and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c.

acceptance correction as a function of $\Delta\phi$ and $\Delta\eta$ for track pairs. Most of the dip is corrected away when the data are divided by the efficiency from mixed events [2, 3].

While the method discused in Section 2.1 corrects for the geometric acceptance of the detector, this method corrects for both the geometric acceptance and for track merging. This method is much more CPU-intensive and therefore was only applied for small ($\Delta\phi$,$\Delta\eta$).

Figure 2.4(a) shows the data and Figure 2.4(b) shows the mixed events for a sample correlation. The dip has a different shape when the helicities of the tracks are the same and when they are opposite. To do the correction the data and mixed events are rotated so that the dip is in the same location. The same helicity data are divided by the acceptance correction from the same helicity events and the opposite helicity data are divided by the acceptance correction from the opposite helicity mixed events. These data are then added after the acceptance correction. A small residual dip remains because mixed events to do not recreate the dip perfectly. The remaining dip is corrected for using the symmetry of the correlations. Since the data are symmetric about $\Delta\phi = 0$, the data on the same side as the dip are discarded and replaced by the data on the side without the dip. Then the data are reflected about $\Delta\eta = 0$ and added to the unreflected data to minimize statistical fluctuations.

This method is only applied to $|\Delta\phi\ | <1.05$ and $|\Delta\eta\ | <0.67$ because it is CPU-intensive and track merging only affects small $\Delta\phi$ and small $\Delta\eta$. For large $\Delta\phi$ and $\Delta\eta$, the method described in Section 2.1 is applied. Figure 2.4(c) shows the data after the track merging correction and Figure 2.4(d) shows the data using the method described in Section 2.1 - without the track merging correction - for all helicity combinations for 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

2.2 Separation of the jet-like correlation and the ridge

Figure 2.5 shows a sample correlation after detector acceptance and track merging corrections. To study the near-side of di-hadron correlations, the two components are separated and the yield, number of particles in each within limits on $p_T^{associated}$ and $p_T^{trigger}$, is studied. The ridge was previously observed to be roughly independent of

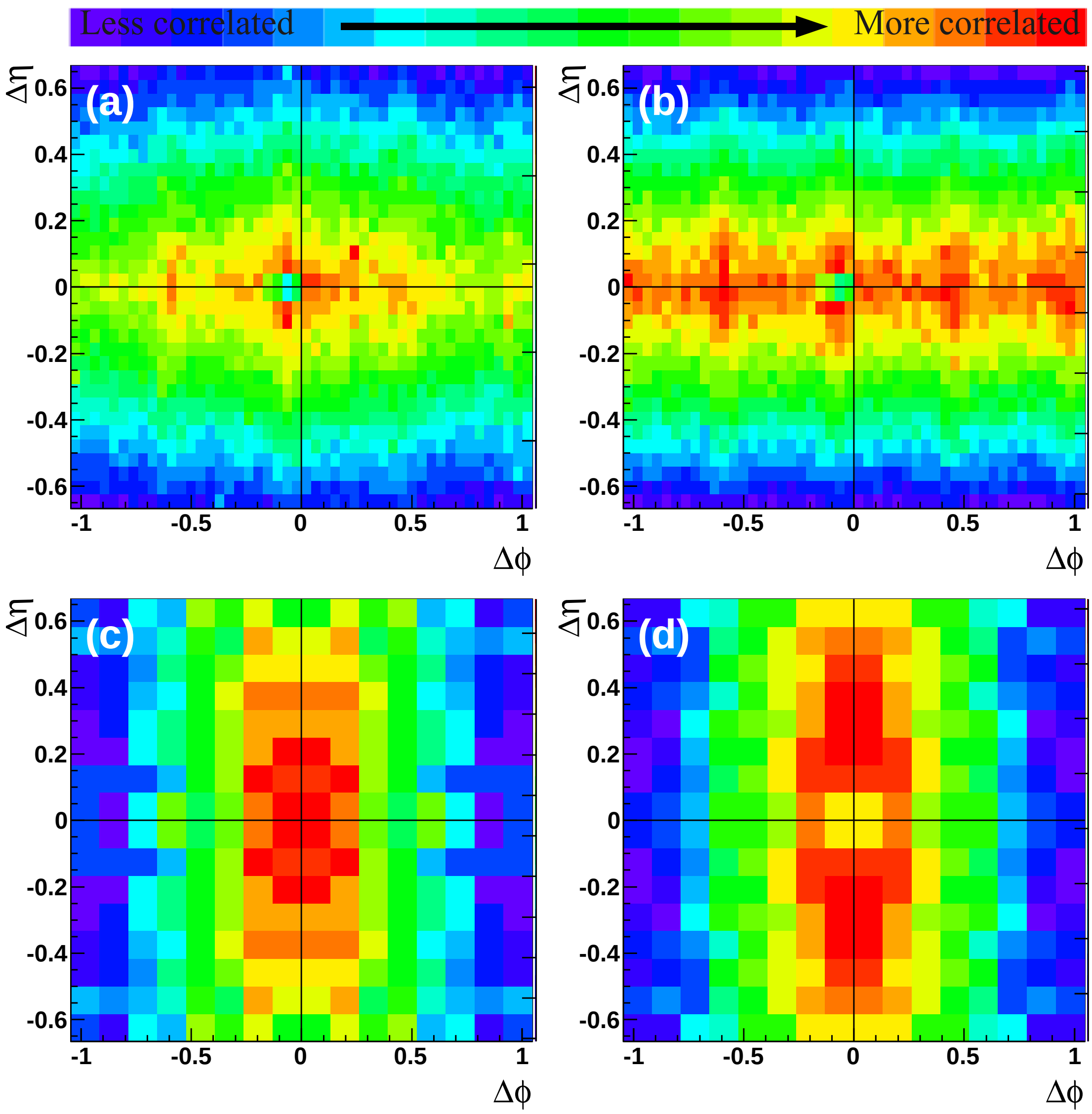

Figure 2.4: The region around the dip for (a) data (b) mixed events (c) data after track merging correction (d) data using the method described in Section 2.1 (no track merging correction) in h-h correlations in 0-40% central $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV for $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c and $3.0 < p_T^{trigger} < 6.0$ GeV/c.

$\Delta\eta$ within the acceptance of the STAR TPC while the jet-like correlation is confined to $|\Delta\eta\,| < 0.75$.

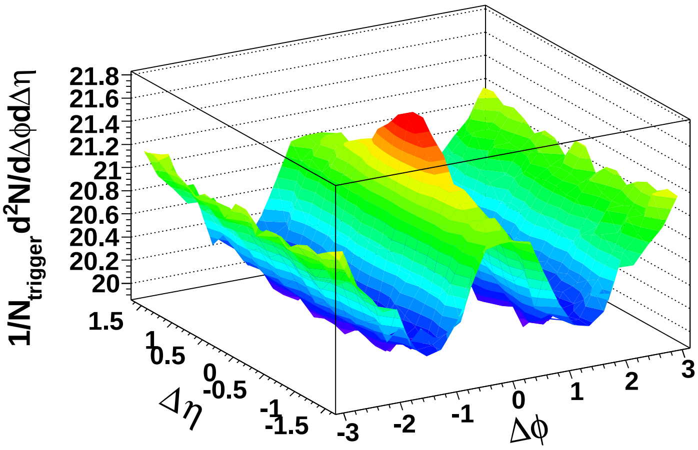

Figure 2.5: Sample correlation after acceptance corrections for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from 0-40% most central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

2.2.1 Extraction of the jet-like yield

The raw correlation consists of a combinatorial background modulated by elliptic flow, v_2, and the signal, which consists of the jet-like correlation and the ridge on the near-side. There are two methods used to extract the jet-like yield from the raw correlation. The first method uses different projections in $\Delta\phi$, the fact that the jet-like correlation is not present at large $\Delta\eta$, and the fact that the combinatorial background and the ridge are both roughly independent of $\Delta\eta$ to isolate the jet-like correlation as a function of $\Delta\phi$. The second method involves taking a projection in $\Delta\eta$ for $-1 < \Delta\phi < 1$ and subtracting a constant background to determine the jet-like correlation as a function of $\Delta\eta$. Elliptic flow, v_2, does not lead to a systematic error on the jet-like yield using either of these methods because v_2 is roughly independent of $\Delta\eta$ within the STAR acceptance [5, 6]. Previous studies in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV have found these methods to be consistent at high-p_T [7], with some indications that there may be greater deviations at lower p_T [8]. Both methods are described in greater detail below.

$\Delta\phi$ method

To determine the jet-like yield, Y_{Jet}, the projection of the distribution of particles $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ is taken in two different ranges in pseudorapidity:

$$\frac{dY_{ridge}}{d\Delta\phi} = \frac{1}{N_{trigger}}\int_{-1.75}^{-0.75}\frac{d^2N}{d\Delta\phi d\Delta\eta}d\Delta\eta + \frac{1}{N_{trigger}}\int_{0.75}^{1.75}\frac{d^2N}{d\Delta\phi d\Delta\eta}d\Delta\eta \quad (2.2)$$

$$\frac{dY_{jet+ridge}}{d\Delta\phi} = \frac{1}{N_{trigger}}\int_{-0.75}^{0.75}\frac{d^2N}{d\Delta\phi d\Delta\eta}d\Delta\eta \quad (2.3)$$

where the former contains only the ridge and the latter contains both the jet-like correlation and the ridge. The jet-like yield, Y_{Jet}, is determined using bin counting in the region $-1 < \Delta\phi < 1$:

$$Y_{jet}^{\Delta\phi} = \int_{-1}^{1}(\frac{dY_{jet+ridge}}{d\Delta\phi} - \frac{0.75}{1}\frac{dY_{ridge}}{d\Delta\phi})d\Delta\phi. \quad (2.4)$$

The factor in front of the second term is the ratio of the $\Delta\eta$ width in the region containing the jet-like correlation and the ridge to the width of the region containing only the ridge. In practice because the data have been reflected about $\Delta\phi = 0$ and $\Delta\eta = 0$, the projections in $\Delta\phi$ and $\Delta\eta$ and the bin counting are done over half of the range and then scaled up to ensure that the errors are correct.

$\Delta\eta$ method

In this method the projection of the distribution of particles $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ on the near-side is taken:

$$\frac{1}{N_{trigger}}\frac{dN}{d\Delta\eta} = \frac{1}{N_{trigger}}\int_{-1}^{1}\frac{d^2N}{d\Delta\phi d\Delta\eta}d\Delta\phi. \quad (2.5)$$

This contains both a background due to combinatorics and the ridge, both of which are independent of $\Delta\eta$. The background is determined first by fitting a Gaussian signal plus a constant background, A, to Equation 2.6 and then using the background from that for background subtraction. If statistics permitted, the background could be determined from $|\Delta\eta\ | > 0.75$, however, many of the correlations presented here have limited statistics and using a fit of the whole region to determine the background prevents statistical fluctuations from skewing the background subtraction. The jet-like yield is:

$$Y_{jet}^{\Delta\eta} = \frac{1}{N_{trigger}} \int\limits_{-1}^{1} (\frac{d^2N}{d\Delta\phi d\Delta\eta} - A) d\Delta\phi. \tag{2.6}$$

The yield is determined by bin counting in the range $|\Delta\eta\ | < 1$. To get the proper errors, the projection and the bin counting are also done over half of the range and scaled because the data have been reflected. The statistical error on A from the fit is used to determine the error on the final yield by adding it in quadrature to the statistical error from bin counting. The magnitudes of these two errors are comparable.

Figure 2.6 shows the jet-like correlation from both the $\Delta\phi$ and the $\Delta\eta$ method after background subtraction for all systems and energies using $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$, the limits used for most studies presented here. For all collision systems and energies the jet-like correlation is within $|\Delta\eta\ | < 0.75$.

2.2.2 Extraction of the ridge yield

There is a large combinatorial background which is dependent on $\Delta\phi$. The raw signal has a background due to particles correlated indirectly with each other in azimuth due to their correlation with the reaction plane. This random background is estimated by

$$\frac{dY_{bkgd}}{d\phi} = B(1 + 2\langle v_2^{trigger}\rangle\langle v_2^{associated}\rangle \cos(2\Delta\phi)) \tag{2.7}$$

where v_2 is the second order harmonic in a Fourier expansion of the momentum anisotropy relative to the reaction plane [9]. The derivation of this parameterization of the background is dependent on the assumption that di-jet production is not correlated with the reaction plane so that the correlation between particles in the jet and random particles in event are not correlated in azimuth through a mutual correlation with the reaction plane. Systematic errors come from the errors on B, $\langle v_2^{trigger}\rangle$ and $\langle v_2^{associated}\rangle$. v_2 is determined from independent measurements. It is assumed that v_2

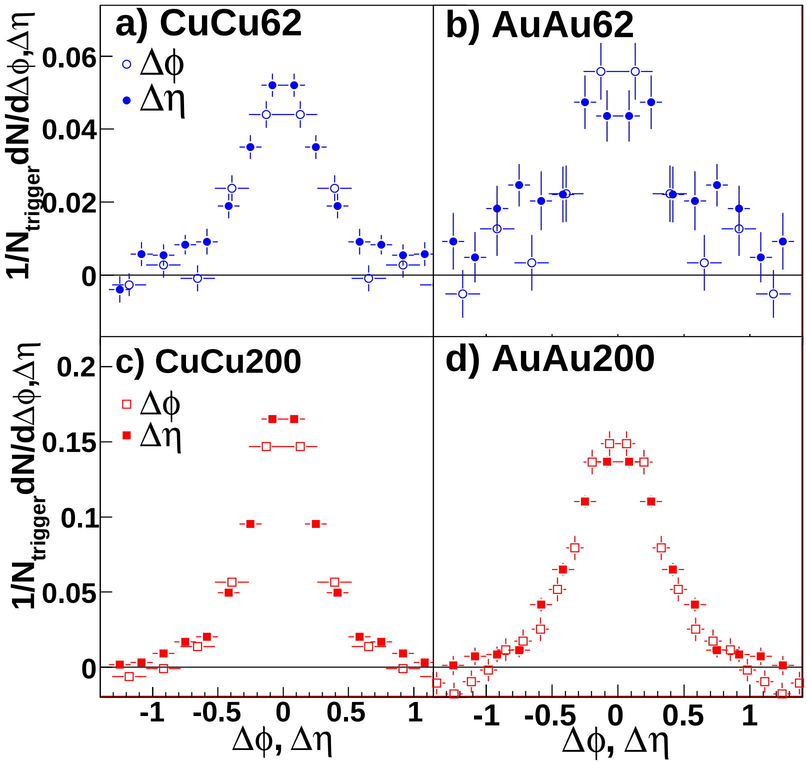

Figure 2.6: Sample jet-like correlations for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from both the $\Delta\phi$ and the $\Delta\eta$ method for (a) 0-60% central $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV (c) 0-60% central $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV and (d) 0-80% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

is the same for events with a trigger particle as for minimum bias events and that v_2 is independent of $\Delta\eta$. For each data set $v_2(p_T)$ was fit in centrality bins to determine $\langle v_2^{trigger}\rangle$ and $\langle v_2^{associated}\rangle$.

Measurements of v_2

The best method for the determination of v_2 depends on the collision system. Since v_2 is a momentum anisotropy in azimuth, the tracks produced in an event can be used to reconstruct the reaction plane. v_2 in Equation 2.7 is the azimuthal momentum anisotropy due to the collective flow of particles, however, there are other sources of azimuthal momentum anisotropy. For particles above 1 GeV/c, jets are the largest known non-flow source of momentum anisotropy. Various methods for measuring v_2 have different non-flow contributions. The systematic error is determined by comparing these different methods. Generally, the higher the multiplicity of an event and the greater the eccentricity of the overlap region, the easier it is to reconstruct the event plane. Therefore that the event plane is more difficult to reconstruct in $Cu + Cu$ collisions.

For $Cu + Cu$ collisions, the nominal for v_2 was taken from a measurement of v_2 using tracks from the FTPCs for reconstruction of the event plane. v_2 measured in $Cu + Cu$ using the event plane from the FTPCs is shown as a function of p_T and centrality in Figure 2.7 [10]. These data were fit to a parameterization which was used for the subtraction of v_2 using Equation 2.7. The fit parameters are given in Table 2.1. The nominal value is given by v_2 from the reaction plane method using the forward TPCs for the determination of the reaction plane and the upper bound is from the upper bound on the statistical error of this measurement. For $Cu + Cu$ collisions measurements using the four particle cumulant method were not possible due to limited statistics. Instead, for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV the lower bound is determined by calculating the v_2 from $p + p$ collisions, scaling it by N_{coll}, and subtracting it from v_2 determined using the reaction plane method. Fits from v_2 {CuCu-pp} are shown in Figure 2.7. For $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV the systematic error is assumed to be the same in $Cu + Cu$ collisions at both energies. The event plane reconstructed from tracks measured in both of the FTPCs is expected to be less biased by jets since tracks in the FTPCs are separated by more than two units of pseudorapidity. Particles produced in di-jets in $p + p$ collisions do not exhibit long range correlations in pseudorapidity.

v_2 and the systematic errors on v_2 in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV were taken from [11]. The dominant systematic errors were determined by comparing the reaction plane method to the four-particle cumulant method. The four-particle cumulant method determines v_2 by looking at correlations between four particles, which significantly reduces its sensitivity to non-flow. v_2 and the magnitude of the negative systematic error on v_2 in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV as a function of p_T and centrality from [11] are shown in Figure 2.9. These were fit to a parameterization

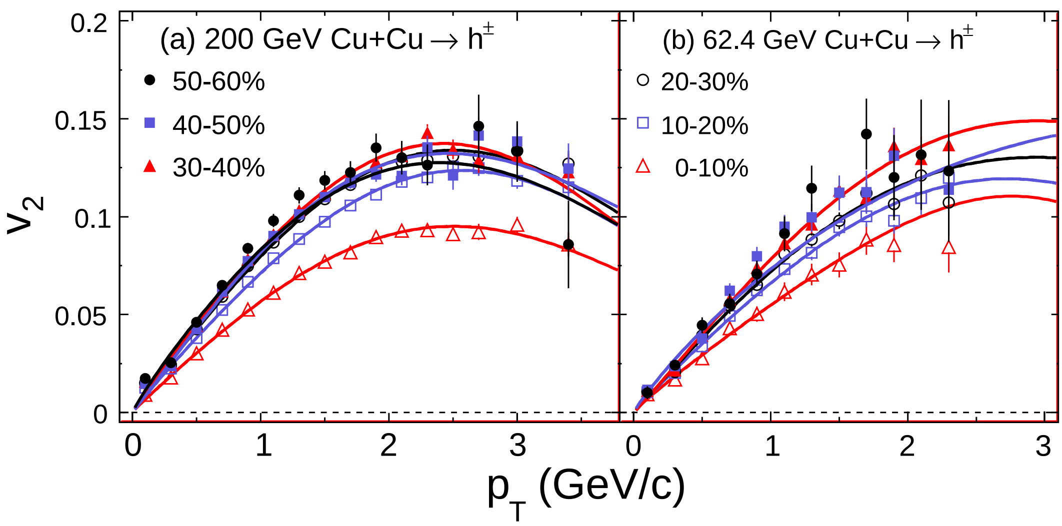

Figure 2.7: v_2 measured using the event plane from the FTPCs in $Cu+Cu$ collisions at (a) $\sqrt{s_{NN}}$ = 200 GeV and (b) $\sqrt{s_{NN}}$ = 62 GeV. Fit parameters are given in Table 2.1

Table 2.1: Fit parameters from fits of v_2 measured using the event plane reconstructed using the FTPC in $Cu+Cu$ shown in Figure 2.7 to v_2 (p_T)=$ap_T^b e^{-(p_T/c)^d}$ $exp(\frac{b}{p_T}^c)$

centrality	a	b	c	d
		200 GeV		
0-10%	0.0594 ± 0.0011	0.9541 ± 0.0192	3.6778 ± 0.0936	2.3449 ± 0.2438
10-20%	0.0735 ± 0.0008	0.9368 ± 0.0127	3.8002 ± 0.0779	2.5779 ± 0.2030
20-30%	0.0836 ± 0.0011	0.9599 ± 0.0141	3.6625 ± 0.0710	2.3440 ± 0.1711
30-40%	0.0863 ± 0.0013	0.9422 ± 0.0163	3.5843 ± 0.0844	2.5421 ± 0.2262
40-50%	0.0910 ± 0.0038	1.0349 ± 0.0364	3.3741 ± 0.1583	1.8515 ± 0.2546
50-60%	0.0888 ± 0.0022	0.8928 ± 0.0247	3.6000 ± 0.0000	2.1461 ± 0.2273
		62 GeV		
0-10%	0.0550 ± 0.0029	0.9076 ± 0.0653	4.0000 ± 0.0000	4.0291 ± 2.7322
10-20%	0.0684 ± 0.0033	0.9539 ± 0.0514	4.0000 ± 0.0000	2.3958 ± 0.6978
20-30%	0.0787 ± 0.0051	1.0126 ± 0.0500	4.0000 ± 0.0000	1.7281 ± 0.4684
30-40%	0.0843 ± 0.0067	1.0519 ± 0.0666	4.0000 ± 0.0000	1.8512 ± 0.6561
40-60%	0.0825 ± 0.0055	0.9039 ± 0.0484	5.6000 ± 0.0000	1.2375 ± 0.2896

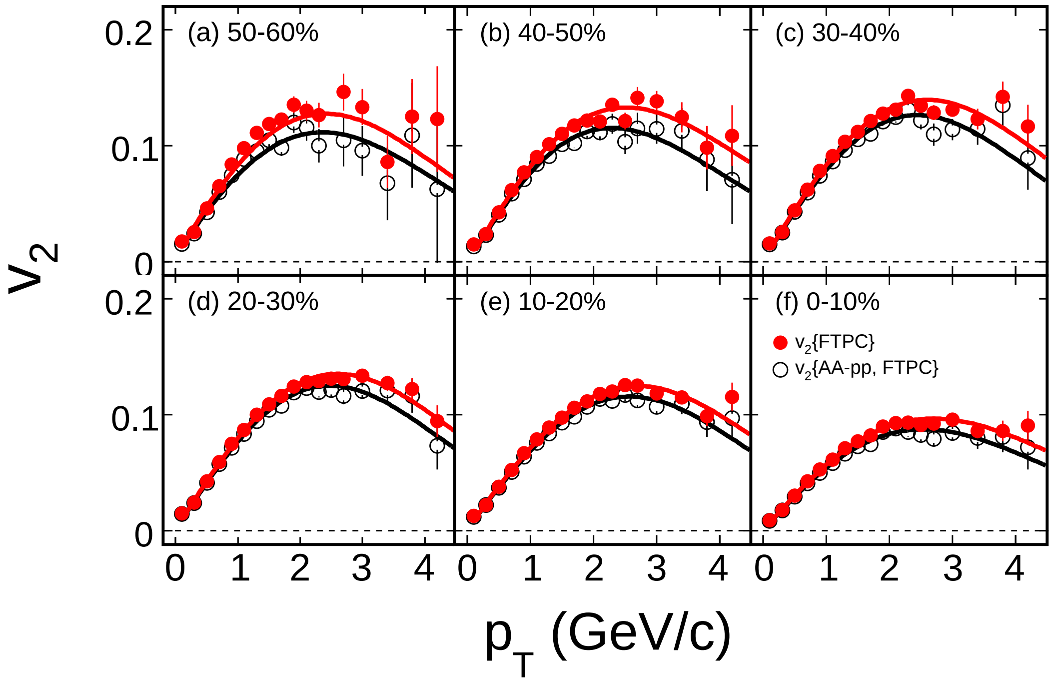

Figure 2.8: v_2 {CuCu-pp} measured in $Cu + Cu$ collisions at $\sqrt{s_{NN}}$ = 62 GeV. Fit parameters are given in Table 2.2

Table 2.2: Fit parameters from fits of v_2 {CuCu-pp} in $Cu + Cu$ $\sqrt{s_{NN}}$ = 200 GeV shown in Figure 2.7 to v_2 (p_T)=$ap_T^b e^{-(p_T/c)^d}$ $exp(\frac{b}{p_T}^c)$

centrality	a	b	c	d
0-10%	0.0588 ± 0.0020	0.9703 ± 0.0307	3.6190 ± 0.1529	1.9036 ± 0.2508
10-20%	0.0722 ± 0.0013	0.9456 ± 0.0188	3.7803 ± 0.1040	2.2305 ± 0.2093
20-30%	0.0822 ± 0.0017	0.9688 ± 0.0205	3.5823 ± 0.1007	2.1063 ± 0.1875
30-40%	0.0848 ± 0.0023	0.9545 ± 0.0266	3.5556 ± 0.1301	2.1025 ± 0.2430
40-50%	0.0878 ± 0.0056	1.0417 ± 0.0523	3.0849 ± 0.2231	1.7623 ± 0.2907
50-60%	0.0801 ± 0.0025	0.8654 ± 0.0319	3.6000 ± 0.0000	2.0979 ± 0.2914

of v_2 and the fit parameters are given in Table 2.3. Statistical errors on the fits are also neglected for $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV since the systematic errors on v_2 are several times larger than the statistical errors. The positive systematic error in $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV is a constant, 0.003, for all centralities and

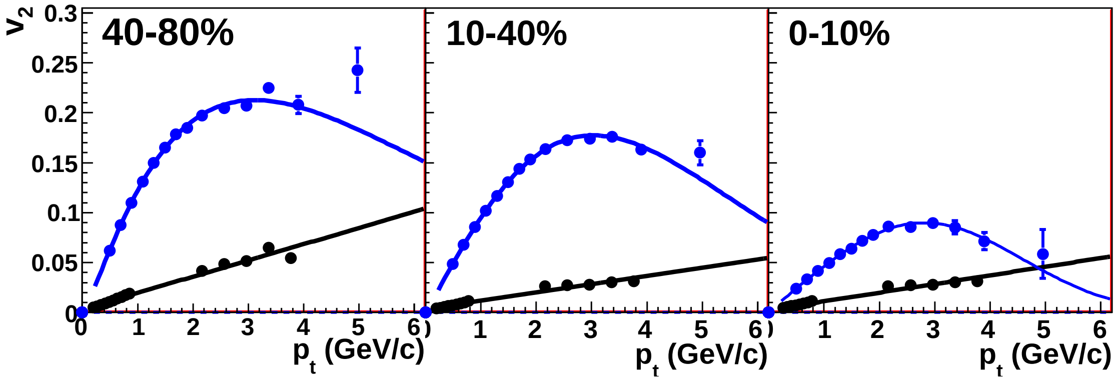

Figure 2.9: v_2 (blue) and the negative systematic error on v_2 (black) in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV. Fit parameters are given in Table 2.3.

Table 2.3: Fit parameters from fits of v_2 from [11] to v_2 (p_T)=$ap_T^b e^{-(p_T/c)^d}$ $exp(\frac{b}{p_T}^c)$ and of the negative systematic error on v_2 to $mp_T + b$

	v_2				systematic error	
centrality	a	b	c	d	m	b
0-10%	0.205	1.30	2.08	0.90	0.0163	0.0035
10-40%	0.105	1.04	4.03	1.67	0.0082	0.0038
40-80%	0.047	0.96	4.11	2.72	0.0087	0.0026

p_T.

v_2 and the systematic errors on v_2 in $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV were taken from *http://www.star.bnl.gov/protected/jetcorr/mvl/v2_compare/auau_compare.html*, where the new parametrization which was agreed upon in the jetcorr PWG in 2009 is described. The v_2 parametrization as a function of p_T and centrality is shown in Figure 2.10. The upper limit is given by the fit to the measurement of $v_2\{FTPC\}$ and the lower limit is the fit to the four-particle cumulant $v_2\{4\}$. For the most peripheral and central bin, the lower limit is determined from an extrapolation. The functional form of the parametrization is $v_2(p_T, cent) = A \cdot p_T^n \cdot e^{B \cdot p_T}$. The values of the parameter B is -0.445 (upper limit) and -0.5 (lower limit). The parameter n depends on centrality as n=1.48-0.5·cent.

B is determined using the Zero Yield At Minimum (ZYAM) method [12, 6]. This

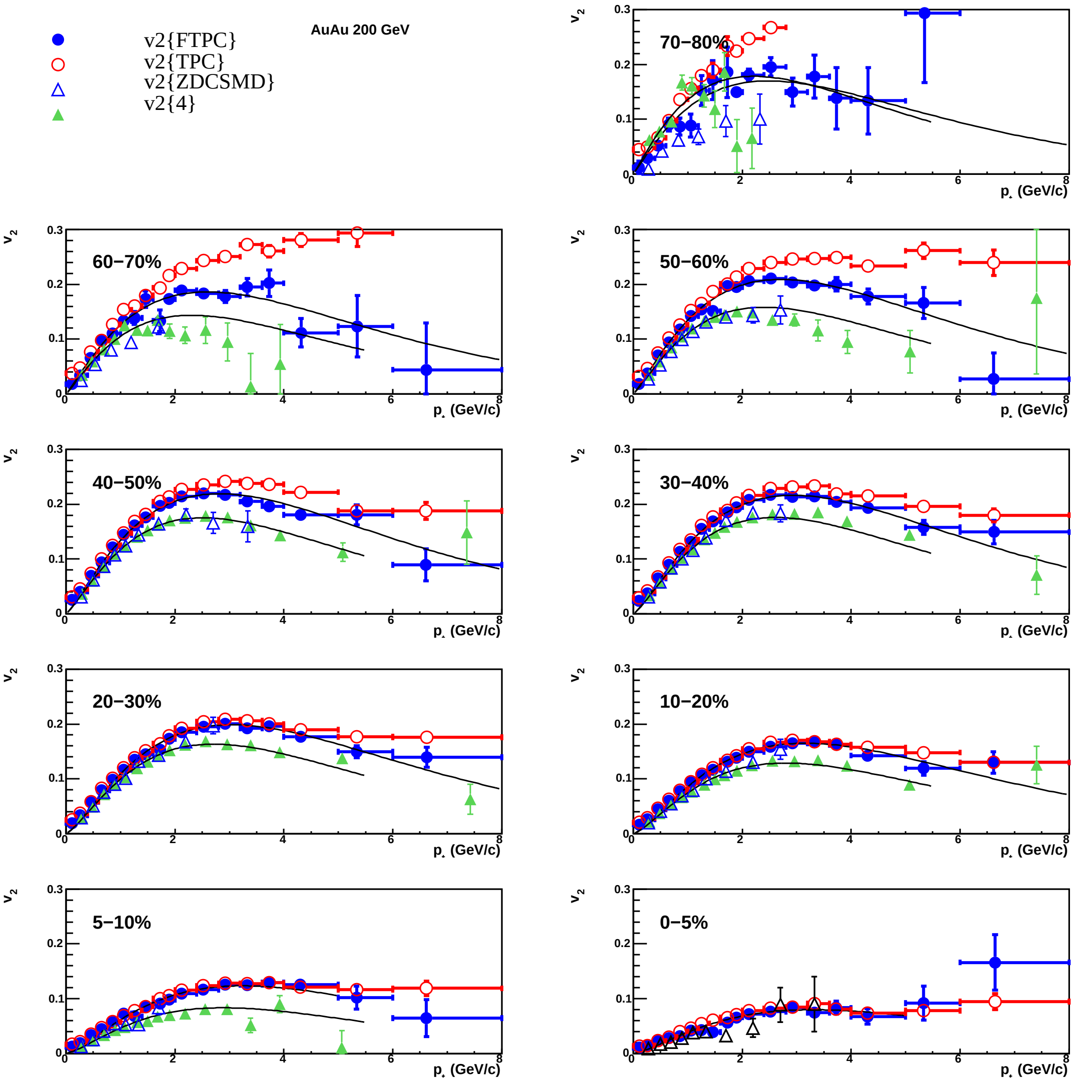

Figure 2.10: Symbols represent v_2 in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV determined by various methods as described in the legend. The black lines are the upper and lower systematic boundaries on v_2 using the fit function $v_2(p_T, cent) = A{\cdot}p_T^n{\cdot}e^{B \cdot p_T}$. The values of the parameters are B=-0.445 (upper curve), B=-0.5 (lower curve), and n=1.48-0.5·cent.

method fixes B by assuming that there is no signal at the minimum of the correlation in azimuth. For measurements with low statistics the statistical error on the minimum data point is a significant contribution to the error on the ridge yield. This error is added in quadrature to the statistical error.

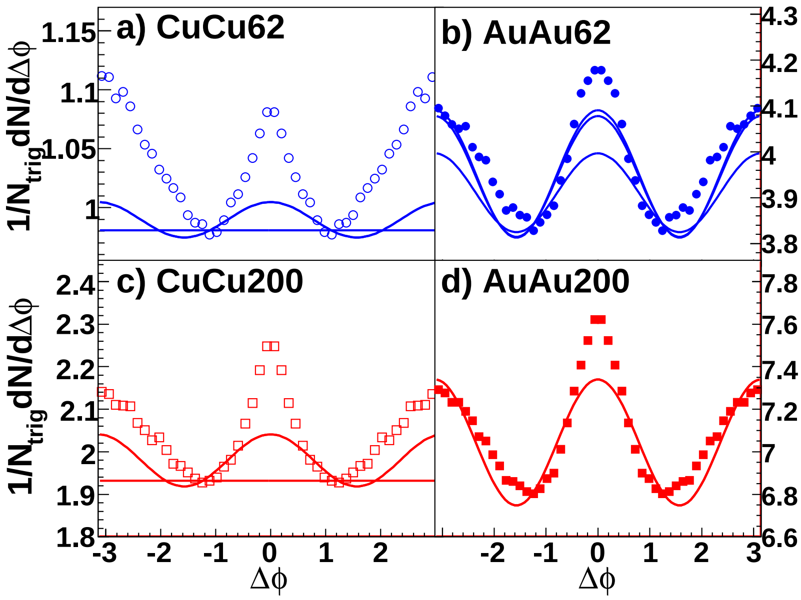

Figure 2.11: Sample correlations in $\Delta\phi$ integrated over $|\Delta\eta\ | < 1.75$ for $3.0 < p_T^{trigger}$ < 6.0 GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from (a) 0-60% central $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV (c) 0-60% central $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV and (d) 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV showing the background level and shape for high and low bounds on v_2 from each data set.

To determine the yield of the ridge, Y_{Ridge}, the di-hadron correlation $\frac{d^2N}{d\Delta\phi d\Delta\eta}$ is projected over the entire $\Delta\eta$ region. To minimize the effects of statistical fluctuations in the determination of the background, $\frac{dY_{bkgd}}{d\phi}$, Y_{Jet} is subtracted after the projection in $\Delta\eta$:

$$Y_{Ridge} = 1/N_{trigger} \int_{-1.75}^{1.75} \int_{-1}^{1} (\frac{d^2N}{d\Delta\phi d\Delta\eta} - \frac{dY_{bkgd}}{d\phi}) d\Delta\phi d\Delta\eta - Y_{Jet}^{\Delta\eta}. \tag{2.8}$$

The integration over $\Delta\phi$ is done by bin counting. $Y_{Jet}^{\Delta\eta}$ is used for the subtraction of Y_{Jet} because it is less sensitive to the assumption that the jet-like correlation is contained within $|\Delta\eta\ | < 0.75$. Sample correlations in azimuth after the projection over $\Delta\eta$ are shown in Figure 2.11 for all data sets. These correlations show the background level and shape for the high and low bounds on v_2.

Chapter 3

Results

3.1 The jet-like correlation

If the jet-like correlation is produced by fragmentation, no dependence on collision system or system size would be expected. The particle ratios of the jet-like correlation should be consistent with what was observed in $p+p$ collisions since $p+p$ collisions are expected to be dominated by jet production at high-p_T. The jet-like yield should increase with $p_T^{trigger}$ because higher p_T trigger particles come on average from higher energy jets and the spectra of particles in the jet-like correlation should be harder than that observed in the inclusive. Studies of the jet-like yield as a function of collision system and energy were done for unidentified hadrons for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV to determine whether the data are consistent with the jet-like correlation being produced by fragmentation.

3.1.1 Energy and system dependence

The dependence of jet-like yield, Y_{Jet}, on $p_T^{trigger}$ is shown in Figure 3.1 for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $d+Au$, $Cu+Cu$, and $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. No ridge is observed in $d+Au$ collisions so it was not necessary to subtract the ridge. The jet-like yields are the same as a function of $p_T^{trigger}$ within errors for all systems at a given collision energy.

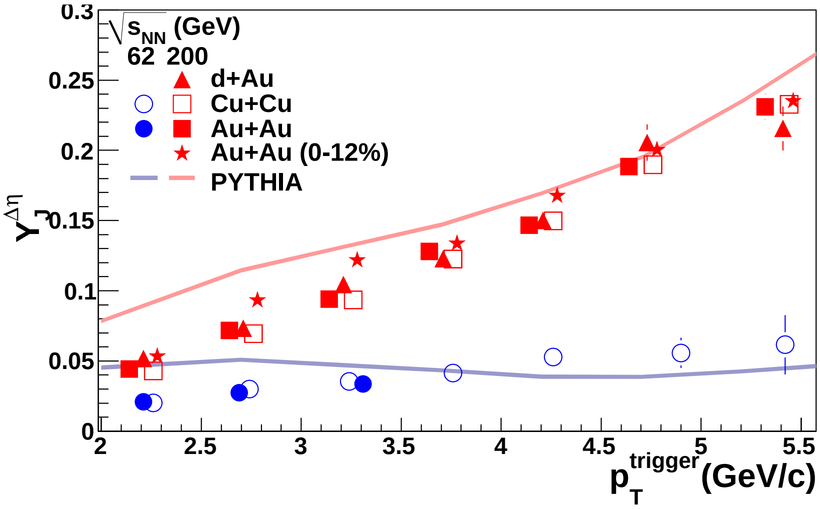

Figure 3.1: $p_T^{trigger}$ dependence of Y_{Jet} for unidentified hadrons for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ for 0-60% central $Cu+Cu$ and 0-80% central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and minimum bias $d+Au$, 0-60% central $Cu+Cu$ and 0-12% central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

Table 3.1: Inverse slope parameter k (MeV/c) of $p_T^{associated}$ for unidentified hadrons for fits of data in Figure 3.2 to A$e^{-p_T/k}$. Statistical errors only.

	$\sqrt{s_{NN}}$ = 62 GeV h-h	$\sqrt{s_{NN}}$ = 200 GeV h-h
$Au+Au$	291 ± 28	478 ± 8
$Cu+Cu$	359 ± 41	424 ± 20
$d+Au$		469 ± 8

The $p_T^{associated}$ dependence of the jet-like yield is shown in Figure 3.2 for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $d+Au$, $Cu+Cu$, and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. The $\Delta\phi$ method is used for Y_{Jet} for $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV because these studies are statistically limited and the jet-like correlation is considerably broader in $\Delta\eta$ in these collisions. This makes the determination of the background difficult in the $\Delta\eta$ method. As for the $p_T^{trigger}$ dependence, the data are within errors for all collision systems at a given energy.

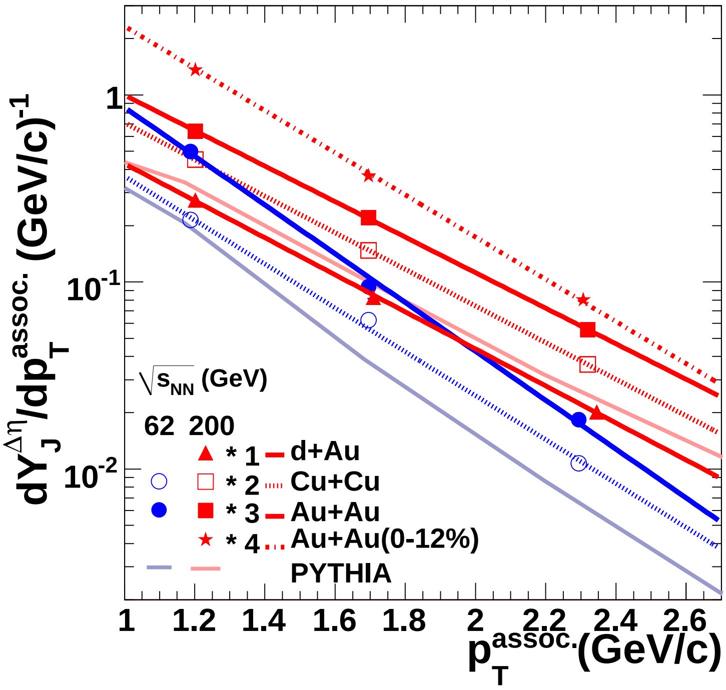

Figure 3.2: $p_T^{associated}$ dependence of Y_{Jet} for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for $3.0 < p_T^{trigger} < 6.0$ GeV/c for 0-60% central $Cu + Cu$ and 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and minimum bias $d + Au$, 0-60% central $Cu + Cu$ and 0-12% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

Figure 3.3 shows the N_{part} dependence of Y_{Jet} for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and for $d+Au$, $Cu+Cu$, and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. There is a slight increase of Y_{Jet} with N_{part} in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. The jet-like yields in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV are slighly higher than those in $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV and Y_{Jet} is about 50% larger in central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV than in $d+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

Figure 3.4 shows the Gaussian widths in $\Delta\eta$ and $\Delta\phi$ as a function of $p_T^{trigger}$, $p_T^{associated}$, and N_{part} for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and $d+Au$, $Cu+Cu$, and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. The widths in $\Delta\phi$ and $\Delta\eta$ are slightly greater in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. Data from $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV are consistent with the same trends observed at $\sqrt{s_{NN}}$ = 200 GeV, however, they are also within error of the data from peripheral $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

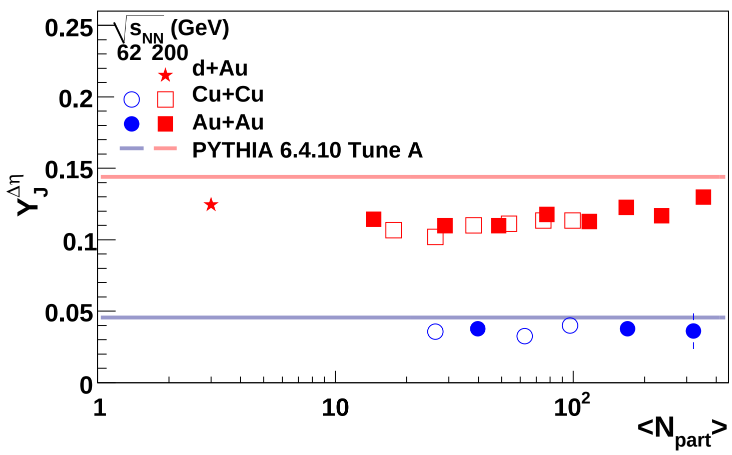

Figure 3.3: N_{part} dependence of Y_{Jet} for unidentified hadrons for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for unidentified hadrons for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and $d+Au$, $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

3.2 The ridge

The ridge is an unexpected phenomenon first observed in $Au+Au$ collisions. In order to understand its origins and to provide a robust test of models for the production of the ridge, the ridge is studied as a function of as many variables as possible. The collision energy, collision system, system size, and trigger particle type dependence of the ridge are presented here.

3.2.1 Energy and system dependence

Figure 3.5 shows the dependence of the ridge yield, Y_{Ridge}, on N_{part} for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}}$ = 200 GeV. No difference is observed between $Cu+Cu$ and $Au+Au$ collisions at either energy. Y_{Ridge} is larger

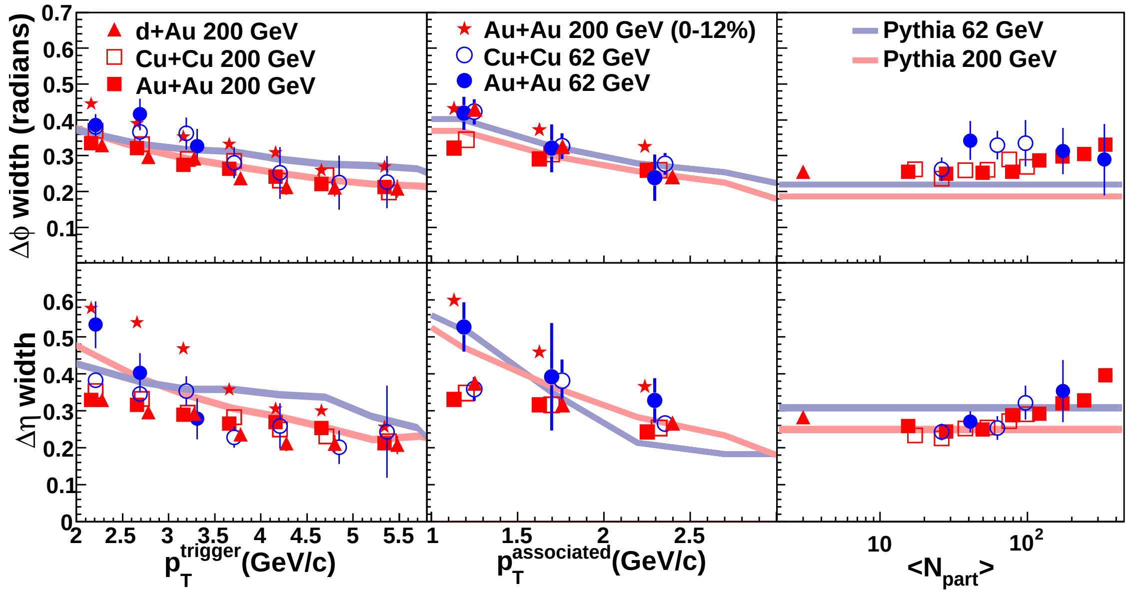

Figure 3.4: Collision energy and system size dependence of Gaussian widths in $\Delta\phi$ and $\Delta\eta$ as a function of $p_T^{trigger}$, $p_T^{associated}$, and N_{part} for unidentified hadrons for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and $d+Au$, $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. Kinematic cuts and centralities are the same as used in Figure 3.1, Figure 3.2, and Figure 3.3 except that the $Au+Au$ data at $\sqrt{s_{NN}}$ = 200 GeV are from 0-80% central collisions for the dependence of Y_{Jet} on $p_T^{trigger}$ and $p_T^{associated}$.

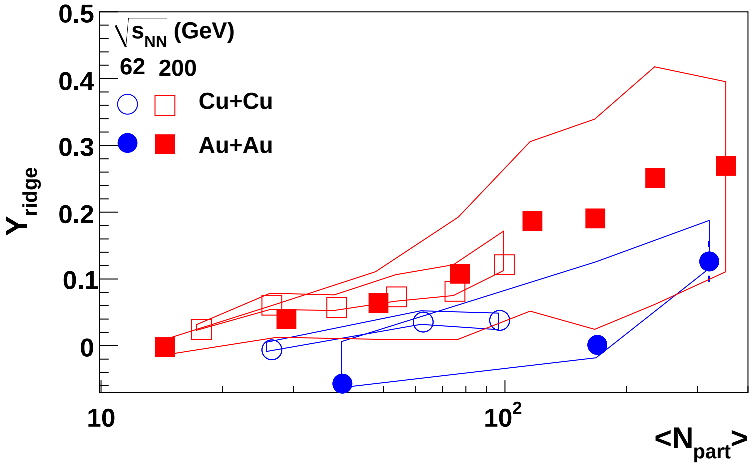

Figure 3.5: N_{part} dependence of Y_{Ridge} for unidentified hadrons for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Data from $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV are from [13, 14, 15, 16, 17, 18]. Lines indicate systematic errors due to v_2.

at $\sqrt{s_{NN}} = 200$ GeV than $\sqrt{s_{NN}} = 62$ GeV, as was observed for Y_{Jet}.

Appendix A

Code documentation

Code for this analysis is stored in $CVSROOT/offline/paper/psn0530/ . The basic code structure is outlined in Figure A.1, Figure A.2, Figure A.3, and Figure A.4. There are six different libraries used. StHighPtTree, StHighPtLoop, StHighPtEvaluator, and StHighPtTools libraries contain classes and functions which can be used by multiple different analyses and StAnalysis and StAnalysisTools libraries contain classes and functions which are used only for the di-hadron correlation analysis.

Numerical codes were used for particle identification. The codes used in this analysis are defined in Table A.1. In addition, sometimes classes of particles were studied separately to improve CPU efficiency. These classes of particles were given numerical codes which are defined in Table A.2.

A.1 StHighPtTree, StHighPtLoop, StHighPtEvaluator, and StHighPtTools libraries

The StHighPtTree, StHighPtLoop, StHighPtEvaluator, and StHighPtTools libraries are diagramed in Figure A.1 and Figure A.2. Each function or class is briefly defined. StHighPtTree comprises container classes used in TTrees for the analysis. StHighPtLoop contains StMakers which can be used to loop over TTrees or MuDsts. The

Table A.1: Codes used for particle identification. Additional numbers were reserved for other particles these particles were later removed from the analysis.

Number	particle
0	h
1	Λ
2	$\bar{\Lambda}$
3	K_S^0
4	Ξ^-
5	Ω^-
7	π^+
8	π^-
9	p
10	$\bar{p}$
11	K^+
12	K^-
13	e^+
14	e^-
18	Ξ^+
18	Ω^+

Table A.2: Codes used for classes of trigger and associated particles.

Number	particles
0	h
1	Λ, $\bar{\Lambda}$, and K_S^0
2	Ξ^-, Ξ^+, Ω^-, and Ω^+
3	dE/dx identified particles ($\pi^\pm$, p, $\bar{p}$, $K^\pm$, $e^\pm$)

primary class in this library is AnalysisMaker, which is a utility class for other analyses. It manages all track and event cuts and passes only events and tracks which pass quality cuts the next maker. TTreeMaker uses AnalysisMaker to create a TTree from a MuDst; this TTree can also be read by AnalysisMaker. StHighPtEvaluator contains classes which evaluate the quality of events and tracks. StHighPtTools contains tools which are used by multiple other functions.

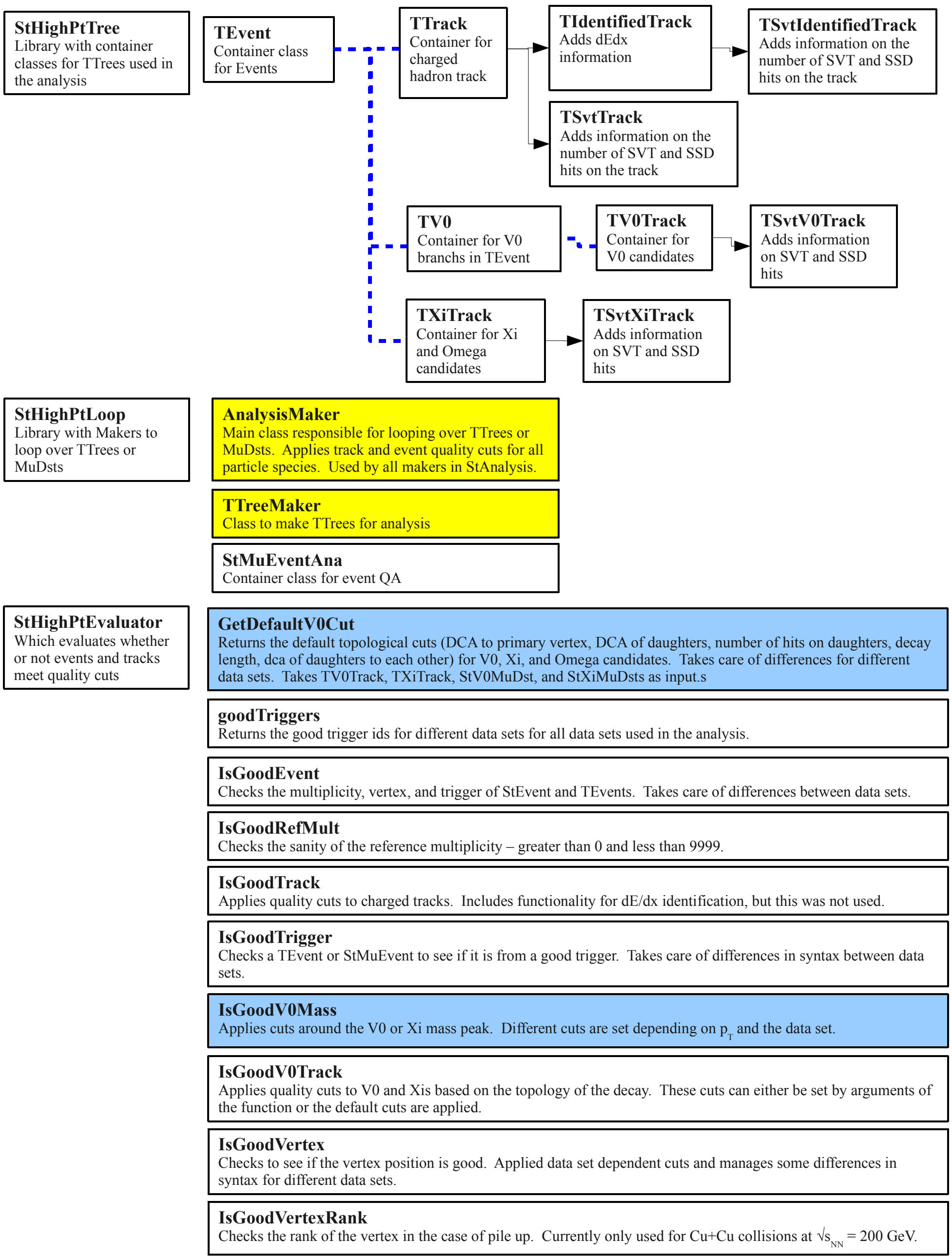

Figure A.1: First part of the code diagram for the StHighPt libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

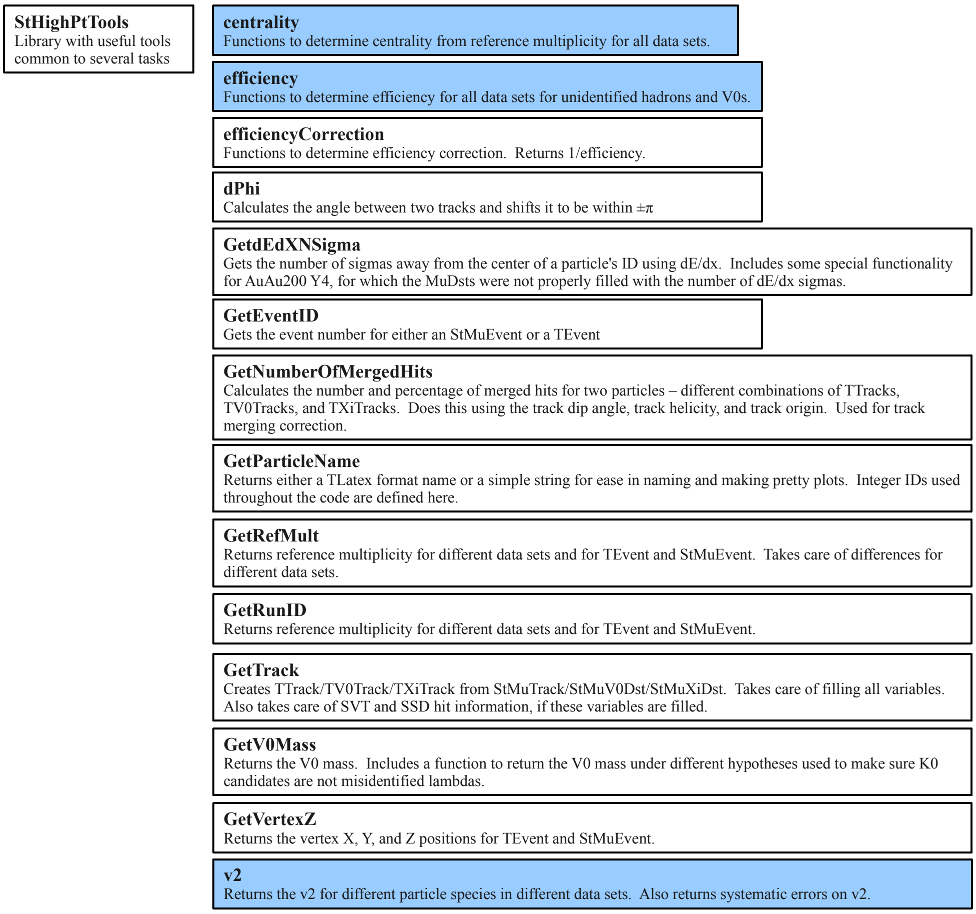

Figure A.2: Second part of the code diagram for the StHighPt libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

A.2 StAnalysis and StAnalysisTools libraries

The StAnalysis and StAnalysisTools libraries are diagramed in Figure A.3 and Figure A.4. StAnalysis contains makers. CorrelationMaker is the maker used for the di-hadron correlation analysis. PhiMaker, RawSpectraMaker, and QAMaker are makers for quality assurance at high-p_T and are provided for convenience. V0CutTuner, XiCutTuner, OmegaCutTuner, and XiNtupleMaker are used for tuning the geometric cuts of V^0s, Ξ^-, and Ω^-. StAnalysisTools contains functions which are used by the makers in StAnalysis. The primary class in this library is CorrelationSummary, which is used to contain all of the relevant information about a di-hadron correlation.

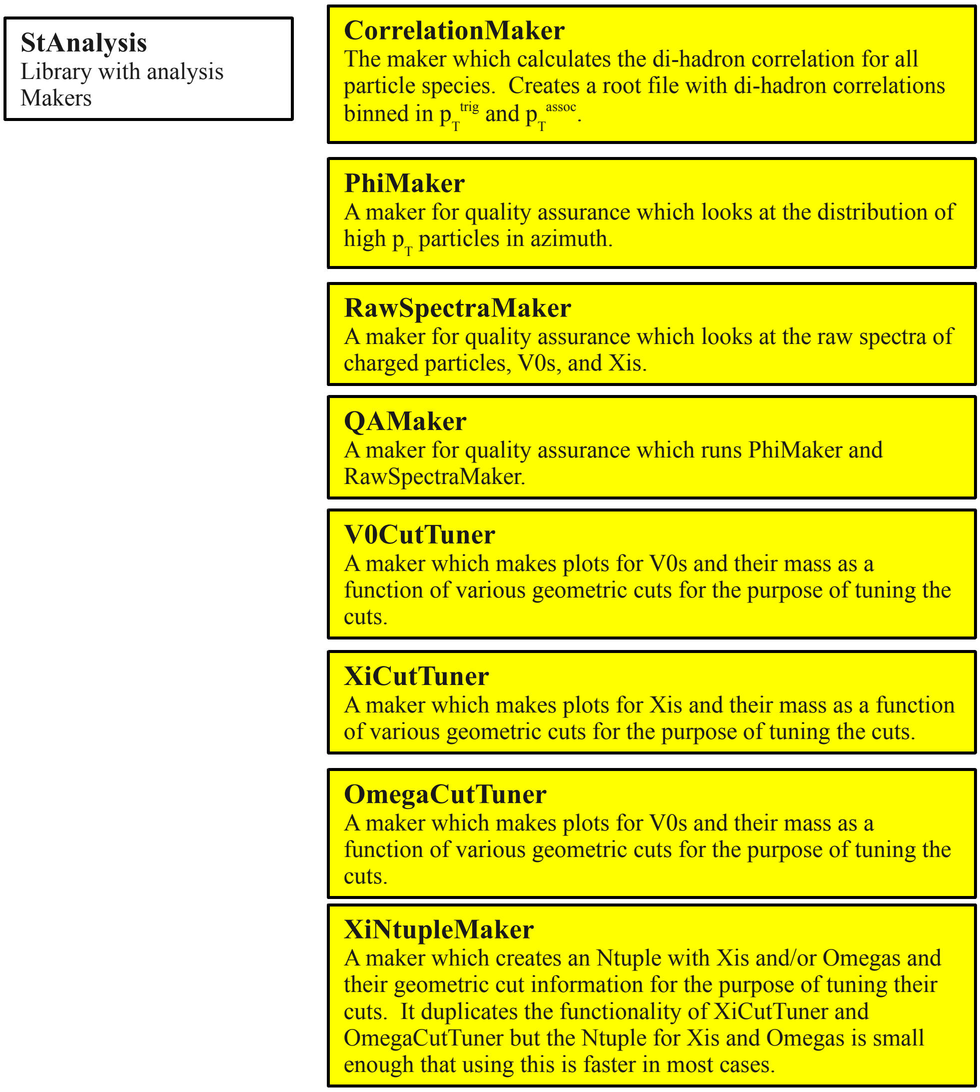

Figure A.3: First part of the code diagram for the StAnalysis libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

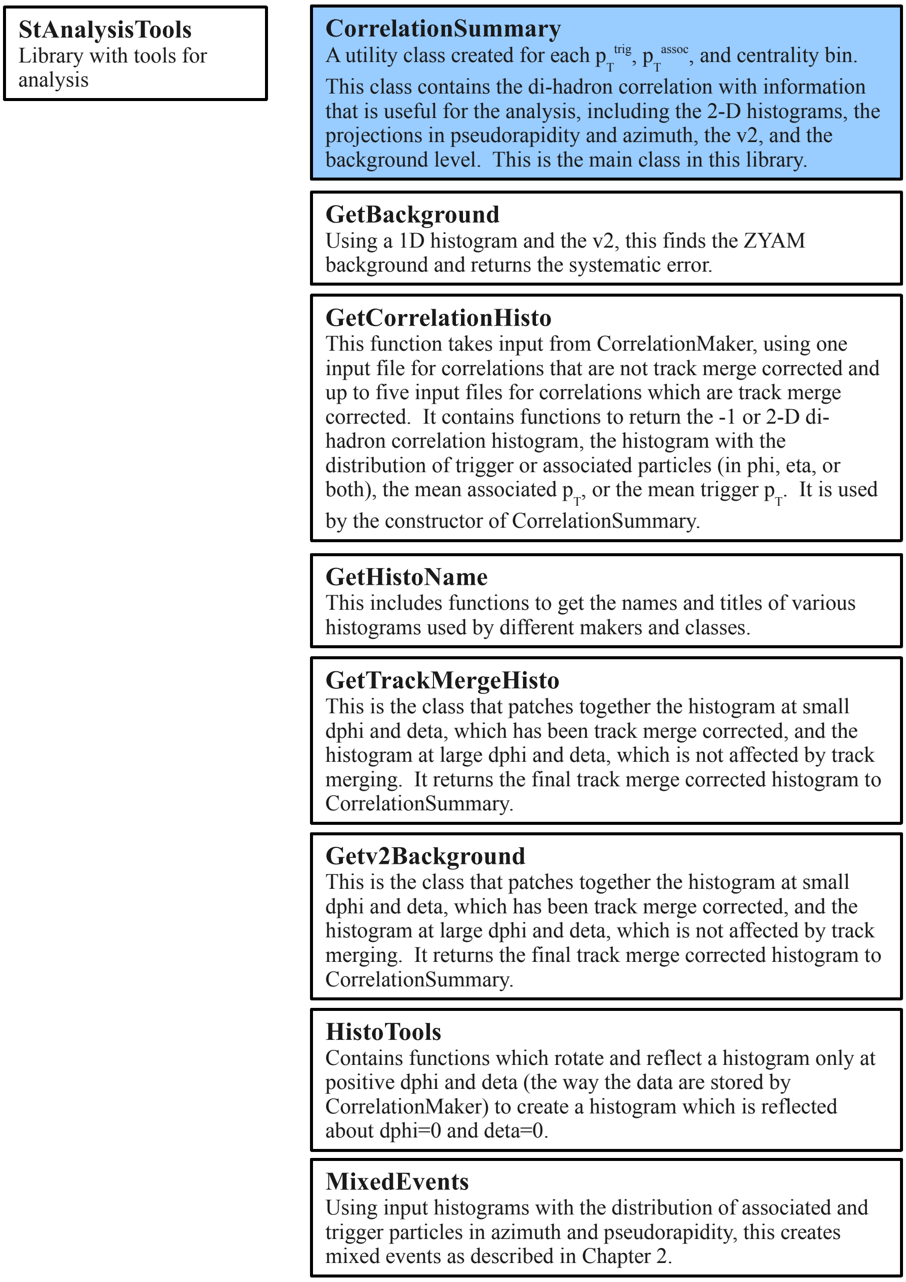

Figure A.4: Second part of the code diagram for the StAnalysis libraries. Arrows indicate inheritence, blue dashed lines indicate objects held in a container class, yellow boxes indicate StMakers, and blue boxes indicate functions which have macros checked in to demostrate these classes.

A.3 Sample macros

Sample macros for each maker and for a selection of functions are given in the macros directory. A brief description of how to use each macro is given. As described in the next section, these macros are not all needed to do the analysis, however, the additional macros are provided for future use.

1. **macros/test/testtools.C** Takes no arguments. Tests efficiency, efficiencyCorrection, centrality, and v2 and prints the result to the screen.

2. **macros/test/TestIsGoodV0Mass.C** Takes no arguments. Tests some of the functionality of IsGoodV0Mass and prints the result to the screen.

3. **macros/test/TestGetDefaultV0Cut.C,**
   macros/test/TestGetDefaultXiCut.C Take no arguments. Print the default cuts out to the screen.

4. **macros/test/TestCorrelationSummary.C**
   Example:
   root4star -b -q macros/test/TestCorrelationSummary.C '("AuAu200","tmp/",1.5,6.0,3.0,6.0)'
   This macro takes output files from CorrelationMaker and makes a CorrelationSummary. The arguments are the data set, the output directory for pictures, the minimum associated p_T, the maximum associated p_T, the minimum trigger p_T, and the maximum trigger p_T.

5. **macros/test/TestCorrelationSummaryTrackMergeCorrected.C**
   Example:
   root4star -b -q macros/test/TestCorrelationSummaryTrackmergeCorrected.C '("AuAu200","tmp/",1.5,6.0,3.0,6.0)'
   This macro takes output files from CorrelationMaker and makes a CorrelationSummary with the track merging correction applied. The arguments are the data set, the output directory for pictures, the minimum associated p_T, the maximum associated p_T, the minimum trigger p_T, and the maximum trigger

p_T. Note that the 1D histograms get filled multiple times when running in the track merge correction mode and therefore have an artificial jump in them. The 1D histogram is not corrected for track merging.

6. **macros/MakeCuCu200P06ibTrees.C**
   Example:
   root4star -b -q macros/MakeCuCu200P06ibTrees.C'("inputMuDst.root","output.root")'
   Creates a TTree from a MuDst using TTreeMaker and AnalysisMaker. This is written to run over 200 GeV Cu+Cu collisions.

7. **macros/QA/RunQAMaker.C**
   Example:
   root4star -b -q macros/QA/RunQAMaker.C '("inputMuDst.root","output.phi.root","output.pt.root")'
   This macro runs QAMaker, which in turn runs PhiMaker and RawSpectraMaker. In the background, it uses AnalysisMaker to filter tracks and events. The output is two root files for quality assurance of high-p_T particles.

8. **macros/geoCuts/RunOmegaCutTuner.C**
   Example:
   root4star -b -q macros/geoCuts/RunOmegaCutTuner.C '("inputTTree.root","output.root")'
   Runs OmegaCutTuner with an input TTree and creates an output file with histograms of omega mass peaks.

9. **macros/geoCuts/RunV0CutTuner.C**
   Example:
   root4star -b -q macros/geoCuts/RunV0CutTuner.C'("inputTTree.root","output.root")'
   Runs V0CutTuner with an input TTree and creates an output file with histograms of V^0 mass peaks.

10. **macros/geoCuts/RunXiCutTuner.C**
    Example:
    root4star -b -q macros/geoCuts/RunXiCutTuner.C'("inputTTree.root","output.root")'

Runs XiCutTuner with an input TTree and creates an output file with histograms of Ξ^- mass peaks.

11. **macros/geoCuts/RunXiNtupleMaker.C**
Example:
root4star -b -q macros/geoCuts/RunXiNtupleMaker.C'("inputTTree.root","output.root")'
Runs XiNtupleMaker with an input TTree and creates an output file with an Ntuple with information about the geometric cuts for Ξ^- and/or Ω^-.

12. **macros/correlations/RunCorrelationMaker.C**
Example:
root4star -b -q macros/correlations/RunCorrelationMaker.C '("inputTTree.root","output.root",0,0)'
This runs CorrelationMaker using a TTree as input and creating a file with di-hadron correlation histograms. The last two arguments are particle classes defined in Table A.2.

13. **macros/correlations/RunCorrelationMakerForTrackMerging.C**
Example:
root4star -b -q macros/correlations/RunCorrelationMakerForTrackMerging.C '("inputTTree.root","output.root",0,0,1,1)'
This runs CorrelationMaker using a TTree as input and creates a file with di-hadron correlation histograms but only for small $\Delta\eta$ and $\Delta\phi$ for one combination of trigger and associated helicities at a time. It also eliminates track pairs where the two tracks share more than 10% of their hits, as described in Chapter 2. The third and fourth arguments are particle classes defined in Table A.2. The fifth and sixth arguments are the helicity of the trigger and associated particles, respectively. Note that it is possible to request an impossible helicity combination and then the histograms will be empty. In this example, unidentified trigger and associated particles are requested, each with a positive helicity.

14. **macros/correlations/RunCorrelationMakerForEventMixingCuCu200.C**

Example:
root4star -b -q macros/correlations/RunCorrelationMakerForEventMixingCuCu200.C '("inputTTree.root","output.root",0,0,1,1)'
This runs CorrelationMaker using a TTree as input and creates a file with di-hadron correlation histograms from mixed events for only for small $\Delta\eta$ and $\Delta\phi$ for only one combination of trigger and associated helicities. It also eliminates track pairs where the two tracks share more than 10% of their hits, as described in Chapter 2. The arguments are the same as for RunCorrelationMakerForTrackMerging.C. Note that the 1D histograms get filled multiple times when running in the mixed event mode and therefore have an artificial jump in them. The 1D histogram is not corrected for track merging.

15. **macros/correlations/RunCorrelationMakerForEventMixingCuCu200InPieces.C**
Example:
root4star -b -q macros/correlations/RunCorrelationMakerForEventMixingCuCu200InPieces.C '(0, "inputTTree.root","output.root",0,0,1,1)'
This does the same thing as RunCorrelationMakerForEventMixingCuCu200.C, however, it will run over only a subset of the input TTree. This is necessary for event mixing to reduce the duration of jobs. This macro is written to run over bunches of 10,000 events. Smaller subsets can be chosen by editing the macro. The arguments are in the same order as RunCorrelationMakerForTrackMerging.C but there is an additional first argument which is an integer defining which part of the TTree to run over. The argument 0 runs over the first 10,000 events.

A.4 Analysis procedure

1. Create TTrees from MuDsts using MakeCuCu200P06ibTrees.C

2. Create output files with di-hadron correlations using RunCorrelationMaker.C. At this point correlations which are not corrected for track merging are available and can be studied with TestCorrelationSummary.C.

3. Create output files with di-hadron correlations at small $\Delta\phi$ and small $\Delta\eta$ using RunCorrelationMakerForTrackMerging.C for each trigger and associated helicity combination. Create corresponding mixed event files using RunCorrelationMakerForEventMixingCuCu200.C or RunCorrelationMakerForEventMixingCuCu200InPieces.C for each trigger and associated helicity combination. For h-h, Ξ^--h, and Ω^--h correlations this corresponds to an additional eight files, since each trigger and each associated particle can have a helicity of either +1 or -1. For V^0-h and h-V^0 correlations there are only four files created at this stage since the V^0 can only have a helicity of 0.

4. Di-hadron correlations can be corrected for track merging are created by using the output file created in the second step and the four or eight input files from the third step to create a track merging corrected di-hadron correlation using TestCorrelationSummaryTrackMergeCorrected.C. CorrelationSummaries can be written to a root file and used later or re-fit.

Appendix B

Efficiencies

Table B.1: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a\ exp(\frac{b}{p_T}^c)$

centrality	a	b	c
0-5%	.767	.1498	1.316
5-10%	.7856	.1419	1.33
10-20%	.7975	.1408	1.464
20-30%	.8113	.1374	1.606
30-40%	.822	.1376	1.743
40-50%	.8321	.1327	1.781
50-60%	.8386	.1357	1.912
60-70%	.8387	.1511	2.381
70-80%	.8415	.1455	2.153

Table B.2: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a\ exp(\frac{b}{p_T}^c)$

centrality	a	b	c
0-10%	0.838435	0.0704	0.9409
10-20%	0.843666	0.0681	1.0000
20-30%	0.843211	0.0610	1.0000
30-40%	0.848090	0.0638	1.0000
40-50%	0.856875	0.0606	0.9496
50-60%	0.851112	0.0584	1.0000

Table B.3: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 62$ GeV to $a\ exp(\frac{b}{p_T}^c)$

centrality	a	b	c
0-10%	0.8655	0.1333	0.9890
10-20%	0.8736	0.1207	0.9330
20-30%	0.8643	0.1503	1.1071
30-40%	0.8693	0.1269	0.9922
40-50%	0.8608	0.1511	1.1616
50-60%	0.8606	0.1971	1.3211

Table B.4: Fit parameters from fits of the efficiencies for unidentified hadrons in $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV to $a\ exp(\frac{b}{p_T}^c)$

centrality	a	b	c
0-5%	.767	.1498	1.316
5-10%	.7856	.1419	1.33
10-20%	.7975	.1408	1.464
20-30%	.8113	.1374	1.606
30-40%	.822	.1376	1.743
40-50%	.8321	.1327	1.781
50-60%	.8386	.1357	1.912
60-70%	.8387	.1511	2.381
70-80%	.8415	.1455	2.153

Appendix C

Centrality

	R	N_{part}	N_{bin}
0-10%	$101 \leq$ R	96	162
10-20%	$71 \leq$ R < 101	72	108
20-30%	$49 \leq$ R < 71	52	68
30-40%	$33 \leq$ R < 49	36	42
40-50%	$22 \leq$ R < 33	25	26
50-60%	$14 \leq$ R < 22	16	15

Table C.1: Centrality definitions for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

	R	N_{part}	N_{bin}
0-10%	$139 \leq$ R	99	189
10-20%	$98 \leq$ R < 139	75	124
20-30%	$67 \leq$ R < 98	54	78
30-40%	$46 \leq$ R < 67	38	48
40-50%	$30 \leq$ R < 46	26	29
50-60%	$19 \leq$ R < 30	17	17

Table C.2: Centrality definitions for $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

	R	N_{part}	N_{bin}
0-5%	$373 \geq$ R	347	904
5-10%	$313 \leq$ R < 373	293	714
10-20%	$222 \leq$ R < 313	229	512
20-30%	$154 \leq$ R < 222	162	321
30-40%	$102 \leq$ R < 154	112	193
40-50%	$65 \leq$ R < 102	74	109
50-60%	$38 \leq$ R < 65	46	57
60-70%	$20 \leq$ R < 38	26	27
70-80%	$9 \leq$ R < 20	13	11

Table C.3: Centrality definitions for $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

	R	N_{part}	N_{bin}
0-5%	$510 \geq$ R	352.4	1051.3
5-10%	$431 \leq$ R < 510	299.3	827.9
10-20%	$312 \leq$ R < 431	234.6	591.3
20-30%	$217 \leq$ R < 312	166.7	368.6
30-40%	$146 \leq$ R < 217	115.5	220.2
40-50%	$94 \leq$ R < 146	76.6	123.4
50-60%	$56 \leq$ R < 94	47.8	63.9
60-70%	$30 \leq$ R < 56	27.4	29.5
70-80%	$14 \leq$ R < 30	14.1	12.3

Table C.4: Centrality definitions for $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

Appendix D

Comparisons to Jörn's figure 4

Cross checks for comparison to http://arxiv.org/pdf/0909.0191 Known differences between these analyses:

- Jörn used the η dependent efficiency for charged hadrons; we used a flat efficiency. (Redoing this analysis with the eta dependent efficiency would take six months.)
- Bin widths in the raw correlations were different. This may effect the final results slightly because it shifts exactly where the background is fixed by ZYAM. It also affects exactly where the end of the jet+ridge section is.
- Jörn had one bin from 3-4 GeV/c in $p_T^{trigger}$; we had two.
- We applied the track merging correction. This should be negligible in this $p_T^{trigger}$ and $p_T^{associated}$ range.
- We used the bin counting method in $\Delta\eta$ for subtracting the jet.
- Jörn used a cut of $\Delta\eta = 1.7$ on the ridge and a cut of $\Delta\phi = 0.7$. We used a cut of $\Delta\eta = 1.708$ and a cut of $\Delta\phi = 0.785398$, due to slight differences in the binning.
- I added the error due to the statistical error on the minimum point when applying ZYAM in quadrature with the statistical error.

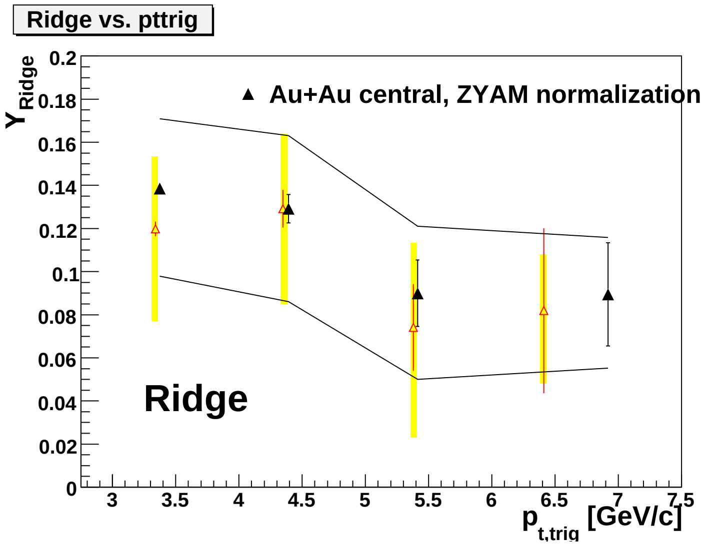

Figure D.1: Comparisons to Jörn's data. Data in this analysis are shown in red with yellow systematic error bands.

For these reasons we should not expect an exact agreement between the two results; however, they should be consistent.

Appendix E

Comparisons to Oana's figure 2a

This is a comparison to Figure 2a in Oana's paper (http://arxiv.org/pdf/0904.1722). Known differences are:

- Oana used the η dependent efficiency for charged hadrons; we used a flat efficiency. (Redoing this analysis with the eta dependent efficiency would take six months.)
- Bin widths in the raw correlations were different. This may effect the final results slightly because it shifts exactly where the background is fixed. It also affects exactly where the end of the jet+ridge section is.

The reason the error bars are larger with the track merging correction is that the statistics are limited for the event mixing for the track merging case. For the track merging mixed events I did not require that the mixed event have any tracks at 3 GeV/c. Comparisons are only made for the $\Delta\eta$ method using bin counting.

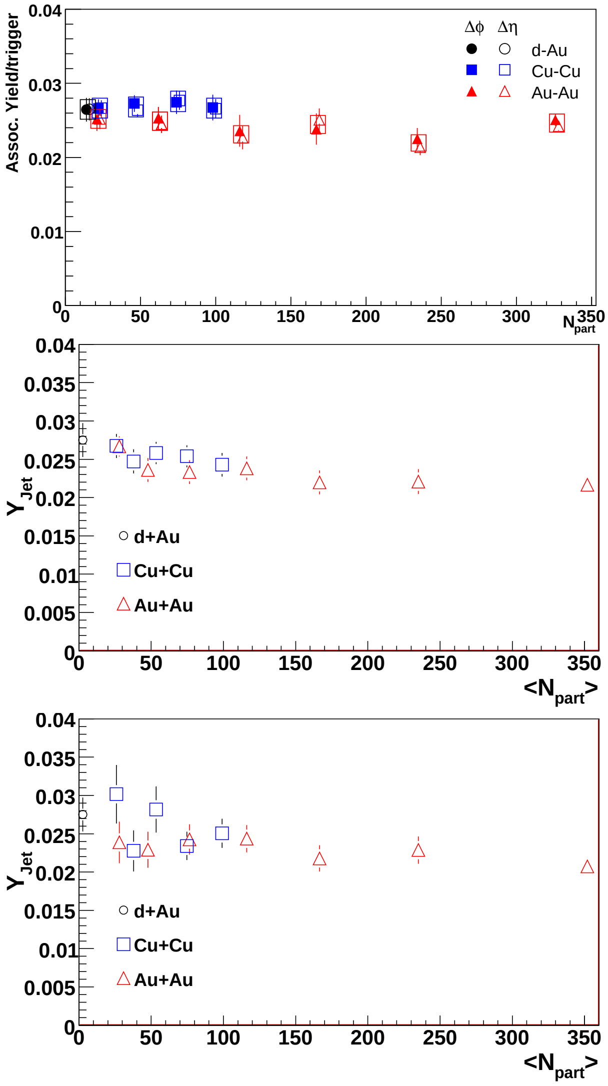

Figure E.1: Top: the figure from Oana's paper. Middle: This analysis, without track merge correction. Bottom: This analysis, with trackmerge correction.

Appendix F

Cross checks using 2D fits

With extensive help from Chanaka DeSilva with the fits and interpretation and from Anthony Timmins with interpretation, cross checks were done using 2D fits to ensure that the data were consistent if v_2 were allowed to be a free parameter. The data were fit to a Gaussian in $\Delta\phi$ on the near-side (the ridge), a Gaussian in $\Delta\phi$ and $\Delta\eta$ on the near-side (the jet-like correlation), a $\Delta\eta$ independent $\cos(\Delta\phi)$ term for the away-side, and a $\Delta\eta$-independent v_2 modulated background.

This was given by:

$$\frac{d^2N}{d\Delta\phi\Delta\eta} = A \quad +2B\langle v_2^{trig}\rangle\langle v_2^{assoc}\rangle \cos 2\Delta\phi + C\cos\Delta\phi \tag{F.1}$$

$$+ \quad \frac{Y_J}{2\sigma^2_{\Delta\phi,J}\sigma^2_{\Delta\eta,J}} e^{-\frac{\Delta\eta}{2\sigma^2_{\Delta\eta,J}}} e^{-\frac{\Delta\phi}{2\sigma^2_{\Delta\phi,J}}} \tag{F.2}$$

$$+ \quad \frac{dY_R}{d\Delta\eta}\frac{1}{\sqrt{2}\sigma^2_{\Delta\eta,R}} e^{-\frac{\Delta\phi}{2\sigma^2_{\Delta\phi,R}}}. \tag{F.3}$$

This is in contrast to the form assumed to describe the near-side in our analysis:

$$\frac{d^2N}{d\Delta\phi\Delta\eta} = B \left(1 + 2\langle v_2^{trig}\rangle\langle v_2^{assoc}\rangle \cos 2\Delta\phi\right) \tag{F.4}$$

$$+ \frac{Y_J}{2\sigma^2_{\Delta\phi,J}\sigma^2_{\Delta\eta,J}} e^{-\frac{\Delta\phi}{2\sigma^2_{\Delta\phi,J}}} e^{-\frac{\Delta\phi}{2\sigma^2_{\Delta\phi,J}}} \tag{F.5}$$

$$+ \frac{dY_R}{d\Delta\eta}\frac{1}{\sqrt{2}\sigma^2_{\Delta\eta,R}} e^{-\frac{\Delta\phi}{2\sigma^2_{\Delta\phi,J}}}. \tag{F.6}$$

We do not consider the away-side in our analysis and make no assumptions about the shape of the away-side.

The fits were done in four different centrality bins in $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. The jet-like yield is consistent with the method used in this paper. The widths in $\Delta\phi$ and $\Delta\eta$ are somewhat smaller in the fit method. v_2 is smaller with the fit method. For the 0-12% bin, v_2 from the fit is negative.

Qualitatively the results are the same - the ridge yield is larger in central collisions than peripheral collisions. This is the behavior we are interested in demonstrating in our paper. Quantitatively the ridge yields are quite different. For comparison with the other method, the $\cos(\Delta\phi)$ term is added to the ridge in order to get comparable terms. The $\cos(\Delta\phi)$ term appears to lead to a higher ridge yield. In the fit method, the $\cos(\Delta\phi)$ term is considered to be an off-set from the v_2-modulated background, leading to a smaller background from the $\cos(2\Delta\phi)$ term. This leads to a higher ridge yield. In addition, the ridge was fit to have a width of roughly 0.5 radians using the fit; this is likely because the dip created by the $\cos(\Delta\phi)$ term must be compensated for by the Gaussian for the ridge.

The projections of the different terms in the fits are shown in Figure F.2. The $\cos(\Delta\phi)$ term leads to a structure which dramatically changes shape underneath the near-side peak. It isn't clear how to interpret these shapes.

0-12%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0045 (low), 0.0070 (avg.), 0.0100 (high)	-0.0021
cos($\Delta\phi$) term coefficient	N/A	-0.097±0.001
near-side $\Delta\phi$ width	0.330 ± 0.006	0.318±0.002
near-side $\Delta\eta$ width	0.395 ± 0.008	0.363±0.003
jet-like yield	0.1298 ± 0.0015	0.134
ridge yield (width)	0.270 ± 0.006 $^{+0.126}_{-0.160}$	0.664 (0.52731)
v_3*multiplicity (v_3/B) of ridge term		0.106 (0.0324)
10-20%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0148 (low), 0.0189 (avg.), 0.0236 (high)	0.0078
cos($\Delta\phi$) term coefficient	N/A	-0.109±0.003
near-side $\Delta\phi$ width	0.304 ± 0.013	0.271±0.004
near-side $\Delta\eta$ width	0.328 ± 0.014	0.310±0.005
jet-like yield	0.1168 ± 0.0029	0.114
ridge yield (width)	0.251 ± 0.012 $^{+0.167}_{-0.189}$	0.719 (0.56238)
v_3*multiplicity (v_3/B) of ridge term		0.101 (0.0438)
20-30%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0242 (low), 0.0294 (avg.), 0.0352 (high)	0.019
cos($\Delta\phi$) term coefficient	N/A	-0.088±0.003
near-side $\Delta\phi$ width	0.298 ± 0.011	0.273±0.003
near-side $\Delta\eta$ width	0.321 ± 0.012	0.296±0.004
jet-like yield	0.1226 ± 0.0027	0.117
ridge yield (width)	0.191 ± 0.011 $^{+0.149}_{-0.166}$	0.507 (0.56978)
v_3*multiplicity (v_3/B) of ridge term		0.0731 (0.0466)
30-40%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0278 (low), 0.0343 (avg.), 0.0414 (high)	0.029
cos($\Delta\phi$) term coefficient	N/A	-0.056±0.003
near-side $\Delta\phi$ width	0.287 ± 0.012	0.260±0.003
near-side $\Delta\eta$ width	0.293 ± 0.012	0.261±0.003
jet-like yield	0.1128 ± 0.0026	0.105
ridge yield (width)	0.187 ± 0.011 $^{+0.119}_{-0.135}$	0.248 (0.54699)
v_3*multiplicity (v_3/B) of ridge term		0.0456 (0.0441)

Table F.1: Fit parameters for 0-12% central $Au+Au$ at $\sqrt{s_{NN}} = 200$ GeV using both ZYAM and a fit

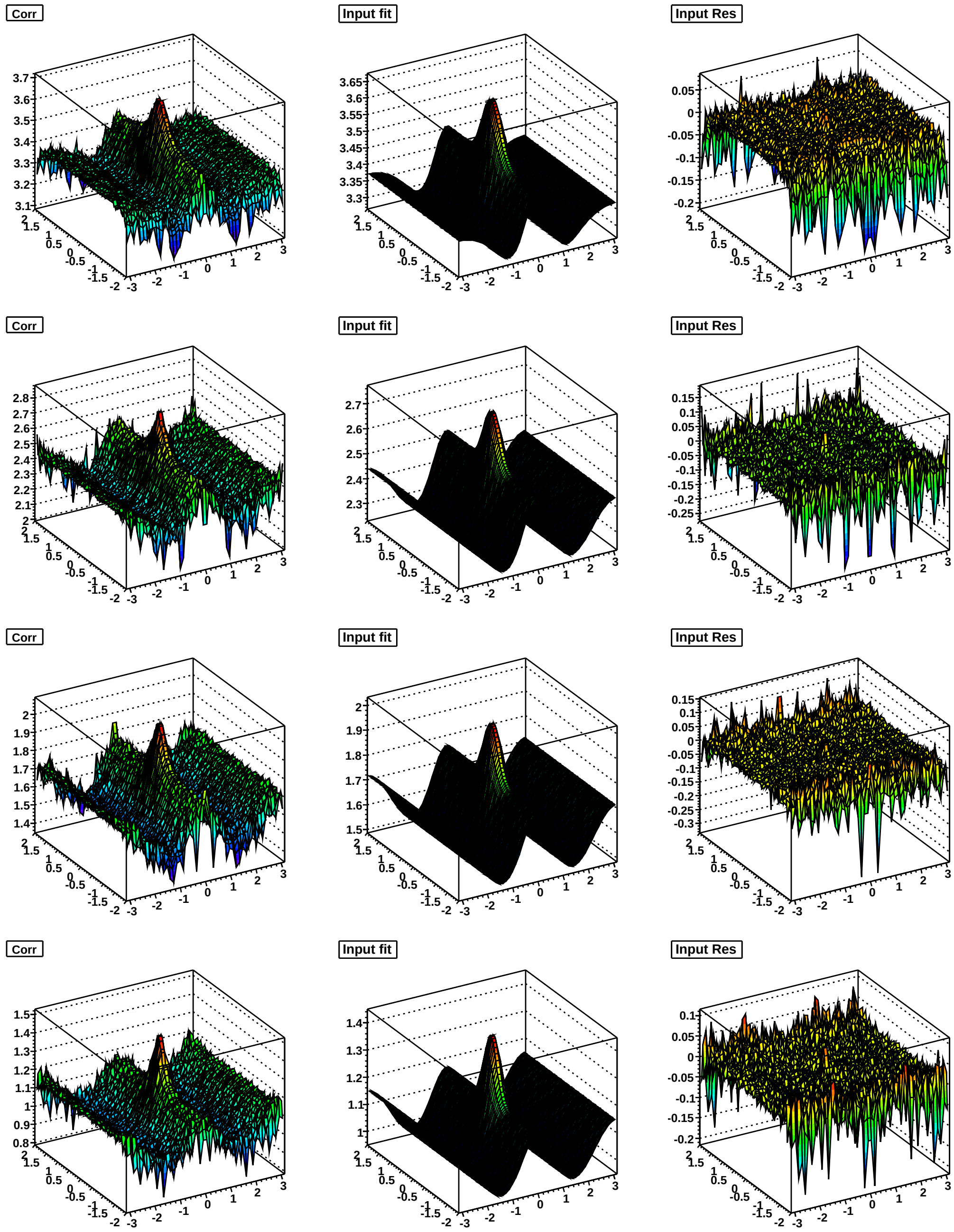

Figure F.1: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ and $3.0 < p_T^{trigger} < 6.0$ GeV/c in 0-12% (top), 10-20% (second down), 20-30% (third down), and 30-40% (bottom) $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV

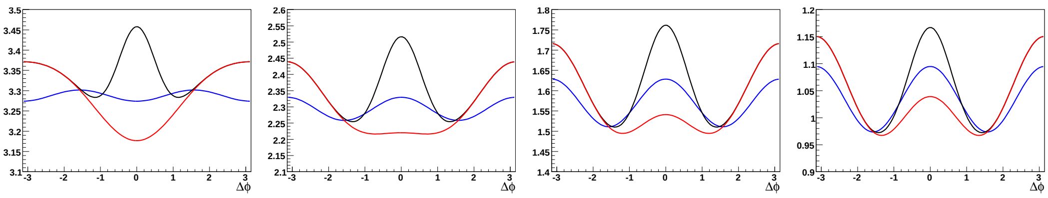

Figure F.2: Projections of ridge, $\cos(\Delta\phi)$, and v_2 modulated background terms from fits from left to right for for 0-12%, 10-20%, 20-30%, and 30-40% $Au + Au$ $\sqrt{s_{NN}}$ = 200 GeV. The black line shows the total, the blue line shows the v_2 modulated background, and the red line shows the v_2 modulated background plus the $\cos(\Delta\phi)$ term. Note that the jet-like correlation has been omitted from these projections.

F.0.1 Extraction of $(\frac{v_3}{v_2})^2$

These fits are also used to extract $(\frac{v_3}{v_2})^2$. The data are fit to

$$\begin{aligned}\frac{d^2N}{d\Delta\phi\Delta\eta} = & A \quad (1 + 2v_1^2 \cos\Delta\phi + 2v_2^2 \cos 2\Delta\phi + 2\frac{v_3^2}{v_2^2} v_2^2 \cos 3\Delta\phi + 2\frac{v_4^2}{v_2^2} v_2^2 \cos 4\Delta\phi) \\ + & A_J e^{-\frac{\Delta\eta}{2\sigma^2_{\Delta\eta,J}}} e^{-\frac{\Delta\phi}{2\sigma^2_{\Delta\phi,J}}}. \end{aligned} \tag{F.7}$$

The output of these fits is given in Table F.2 and Table F.3 and the fits are shown in Figure F.6, Figure F.7, Figure F.8, Figure F.9, Figure F.10, and Figure F.12 for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$. The results from these fits were cross checked by using fits with the ridge described as a Gaussian and the v_3 calculated from the Fourier decomposition of the fit function, omitting the jet-like correlation term. Proper error propagation was not done for this form as this was done mainly as a cross check. The output of these fits is given in Table F.4 and Table F.5 and the fits are shown in Figure F.13, Figure F.14, Figure F.15, Figure F.16, Figure F.17, and Figure F.18 for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated}$ $< p_T^{trigger}$. Figure F.3 and Figure F.4 show the $(\tilde{v}_1/\tilde{v}_2)^2$ and $(\tilde{v}_4/\tilde{v}_2{}^2)^2$, respectively, from the fit.

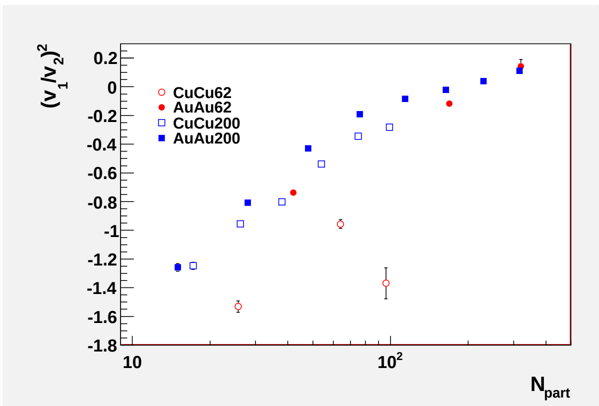

Figure F.3: The ratio of $(\tilde{v}_1/\tilde{v}_2)^2$ from the fit.

Figure F.5 shows the $(\frac{v_3}{v_2})^2$ determined from both functional forms. These two fit forms are numerically consistent for every data point. Comparisons are only done for the case when $v_3^2 > 0$. Figure F.19 shows the difference between the $\tilde{v}_2$ determined

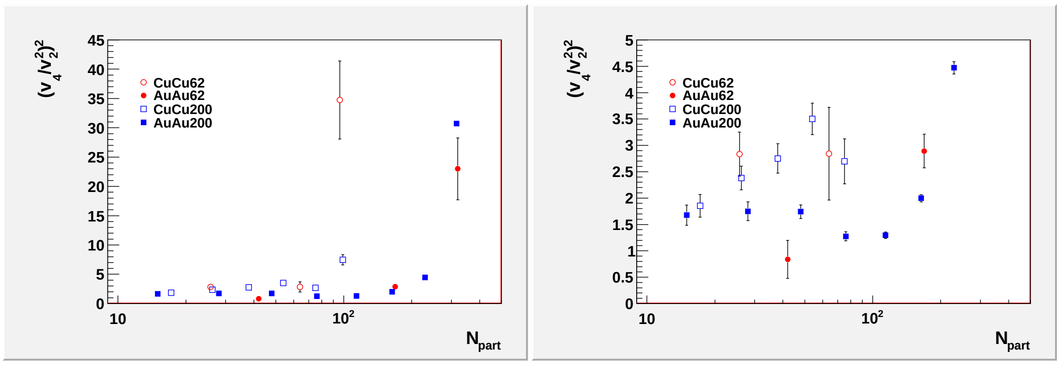

Figure F.4: The ratio of $(\tilde{v_4}/\tilde{v_2}^2)^2$ from the fit showing both the full range (left) and zoomed (right) in to smaller values.

from the fit and the standard $v_2^{trigger}v_2^{assoc}$ used in the analysis as a function of N_{part}. Statistical errors on the fit have been added in quadrature to the systematic errors on $v_2^{trigger}v_2^{assoc}$. The values used in this plot are given in Table F.6. As expected, these values are in agreement for large N_{part} and the values diverge for smaller npart.

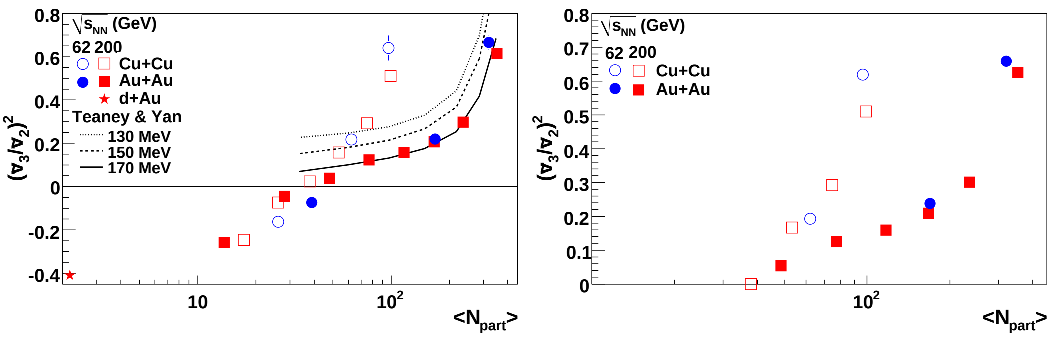

Figure F.5: Fits with the ridge described as a Gaussian (left) and as a v_3 term (right).

	A	v_1^2	v_2^2	$\frac{v_3^2}{v_2^2}$	$\frac{v_4^2}{v_2^2}$	Y_J	$\sigma_{\Delta\eta,J}$	$\sigma_{\Delta\phi,J}$	χ^2/NDF
$Cu + Cu$ $\sqrt{s_{NN}} = 62$ GeV									
0-10%	0.4538 ± 0.0003	-0.0115 ± 0.0006	0.0084 ± 0.0005	0.6396 ± 0.0579	0.2918 ± 0.0533	0.0576 ± 0.0037	0.3248 ± 0.0202	0.3444 ± 0.0197	0.87
10-30%	0.2698 ± 0.0002	-0.0218 ± 0.0005	0.0228 ± 0.0005	0.2165 ± 0.0202	0.0648 ± 0.0200	0.0655 ± 0.0031	0.2402 ± 0.0106	0.3071 ± 0.0122	1.09
30-60%	0.1121 ± 0.0002	-0.0750 ± 0.0011	0.0471 ± 0.0010	-0.1629 ± 0.0210	0.1279 ± 0.0201	0.0774 ± 0.0028	0.2268 ± 0.0070	0.3092 ± 0.0090	1.15
$Au + Au$ $\sqrt{s_{NN}} = 62$ GeV									
0-10%	1.6515 ± 0.0007	0.0009 ± 0.0003	0.0063 ± 0.0003	0.6662 ± 0.0354	0.1449 ± 0.0325	0.0537 ± 0.0083	0.4015 ± 0.0681	0.2730 ± 0.0454	1.16
10-40%	0.8811 ± 0.0003	-0.0030 ± 0.0003	0.0258 ± 0.0002	0.2203 ± 0.0083	0.0746 ± 0.0082	0.0596 ± 0.0036	0.3543 ± 0.0224	0.3066 ± 0.0164	0.89
40-80%	0.2259 ± 0.0003	-0.0349 ± 0.0009	0.0456 ± 0.0008	-0.0732 ± 0.0177	0.0296 ± 0.0171	0.0671 ± 0.0039	0.2721 ± 0.0141	0.3252 ± 0.0156	0.99
$Cu + Cu$ $\sqrt{s_{NN}} = 200$ GeV									
0-10%	0.8962 ± 0.0002	-0.0036 ± 0.0002	0.0128 ± 0.0002	0.5108 ± 0.0120	0.0958 ± 0.0111	0.2540 ± 0.0034	0.2477 ± 0.0031	0.2574 ± 0.0030	1.28
10-20%	0.6148 ± 0.0002	-0.0073 ± 0.0002	0.0212 ± 0.0002	0.2917 ± 0.0092	0.0572 ± 0.0090	0.2639 ± 0.0033	0.2386 ± 0.0027	0.2623 ± 0.0029	1.51
10-30%	0.4207 ± 0.0002	-0.0159 ± 0.0003	0.0296 ± 0.0003	0.1581 ± 0.0090	0.1036 ± 0.0088	0.2708 ± 0.0034	0.2346 ± 0.0024	0.2569 ± 0.0028	1.27
30-40%	0.2832 ± 0.0002	-0.0304 ± 0.0005	0.0379 ± 0.0004	0.0234 ± 0.0108	0.1043 ± 0.0105	0.2813 ± 0.0039	0.2238 ± 0.0025	0.2567 ± 0.0028	1.30
40-50%	0.1861 ± 0.0002	-0.0497 ± 0.0007	0.0506 ± 0.0006	-0.0738 ± 0.0120	0.1182 ± 0.0115	0.2925 ± 0.0042	0.2157 ± 0.0023	0.2426 ± 0.0027	1.19
50-60%	0.1190 ± 0.0002	-0.0872 ± 0.0011	0.0657 ± 0.0010	-0.2459 ± 0.0156	0.1182 ± 0.0146	0.2763 ± 0.0047	0.2174 ± 0.0026	0.2670 ± 0.0033	1.30

Table F.2: Fit parameters for data in Figure F.6, Figure F.7, Figure F.8, Figure F.9, Figure F.10, and Figure F.12.

	A	v_1^2	v_2^2	$\frac{v_3^2}{v_2^2}$	$\frac{v_4^2}{v_2^2}$	Y_J	$\sigma_{\Delta\eta,J}$	$\sigma_{\Delta\phi,J}$	χ^2/NDF
$Au + Au$ $\sqrt{s_{NN}} = 200$ GeV									
0-12%	3.3478 ± 0.0001	0.0009 ± 0.0000	0.0081 ± 0.0000	0.6146 ± 0.0028	0.2487 ± 0.0026	0.1917 ± 0.0011	0.3609 ± 0.0028	0.2986 ± 0.0017	1.59
10-20%	2.3607 ± 0.0002	0.0009 ± 0.0001	0.0225 ± 0.0001	0.2976 ± 0.0027	0.1006 ± 0.0026	0.2232 ± 0.0030	0.3116 ± 0.0046	0.2589 ± 0.0033	1.08
20-30%	1.6200 ± 0.0002	-0.0007 ± 0.0001	0.0344 ± 0.0001	0.2070 ± 0.0024	0.0687 ± 0.0023	0.2342 ± 0.0029	0.2973 ± 0.0038	0.2667 ± 0.0029	1.16
30-40%	1.0621 ± 0.0002	-0.0036 ± 0.0001	0.0430 ± 0.0001	0.1571 ± 0.0027	0.0557 ± 0.0027	0.2509 ± 0.0032	0.2618 ± 0.0031	0.2547 ± 0.0029	1.13
40-50%	0.6639 ± 0.0002	-0.0088 ± 0.0002	0.0461 ± 0.0002	0.1235 ± 0.0040	0.0588 ± 0.0039	0.2694 ± 0.0034	0.2544 ± 0.0029	0.2518 ± 0.0029	1.08
50-60%	0.3901 ± 0.0002	-0.0205 ± 0.0003	0.0478 ± 0.0003	0.0387 ± 0.0063	0.0833 ± 0.0062	0.2846 ± 0.0037	0.2289 ± 0.0024	0.2525 ± 0.0029	1.05
60-70%	0.2126 ± 0.0002	-0.0450 ± 0.0006	0.0546 ± 0.0006	-0.0451 ± 0.0102	0.0933 ± 0.0098	0.2908 ± 0.0042	0.2250 ± 0.0025	0.2500 ± 0.0030	0.97
70-80%	0.1066 ± 0.0002	-0.1018 ± 0.0013	0.0754 ± 0.0012	-0.2587 ± 0.0162	0.1255 ± 0.0148	0.2948 ± 0.0053	0.2263 ± 0.0029	0.2494 ± 0.0031	1.03
$d + Au$ $\sqrt{s_{NN}} = 200$ GeV									
MB	0.0923 ± 0.0002	-0.1300 ± 0.0016	0.0694 ± 0.0014	-0.4105 ± 0.0223	0.1332 ± 0.0194	0.2689 ± 0.0036	0.2469 ± 0.0030	0.2583 ± 0.0029	1.20

Table F.3: Fit parameters for data in Figure F.6, Figure F.7, Figure F.8, Figure F.9, Figure F.10, and Figure F.12.

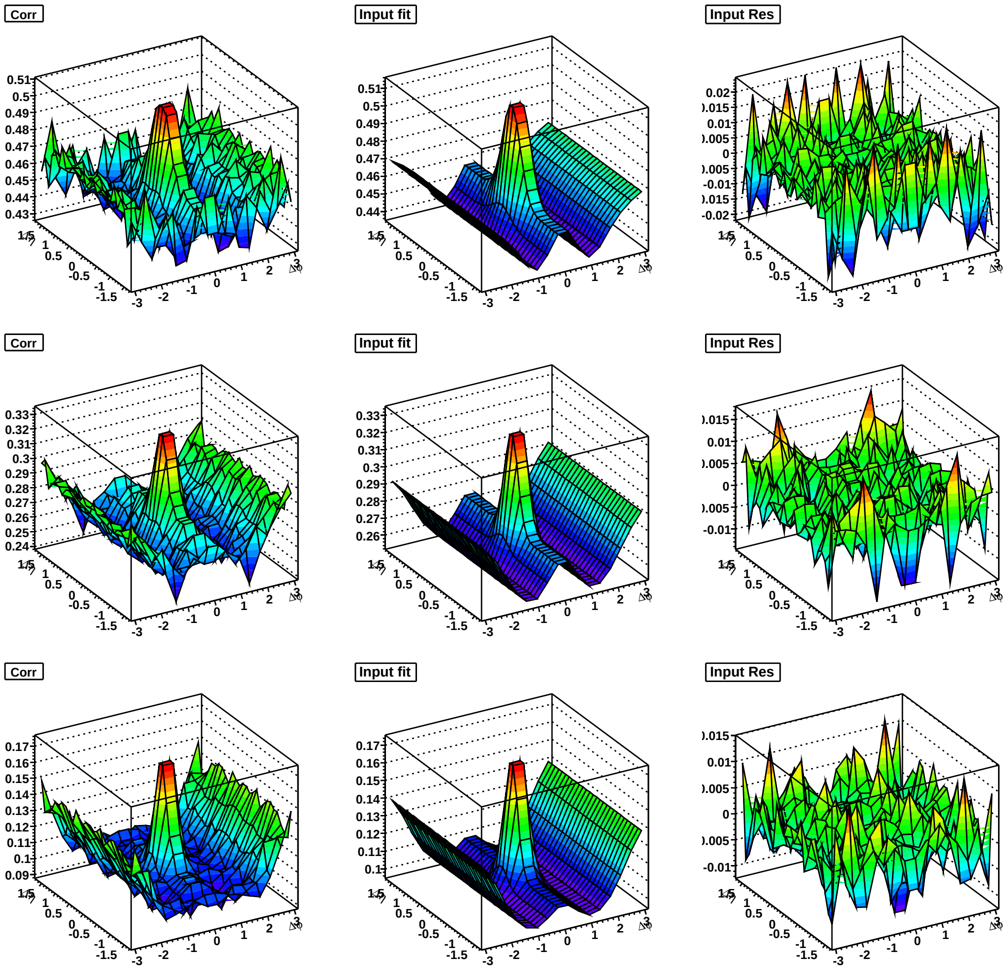

Figure F.6: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-10% (top), 10-30% (middle), and 30-60% (bottom) $Cu + Cu$ $\sqrt{s_{NN}}$ = 62 GeV

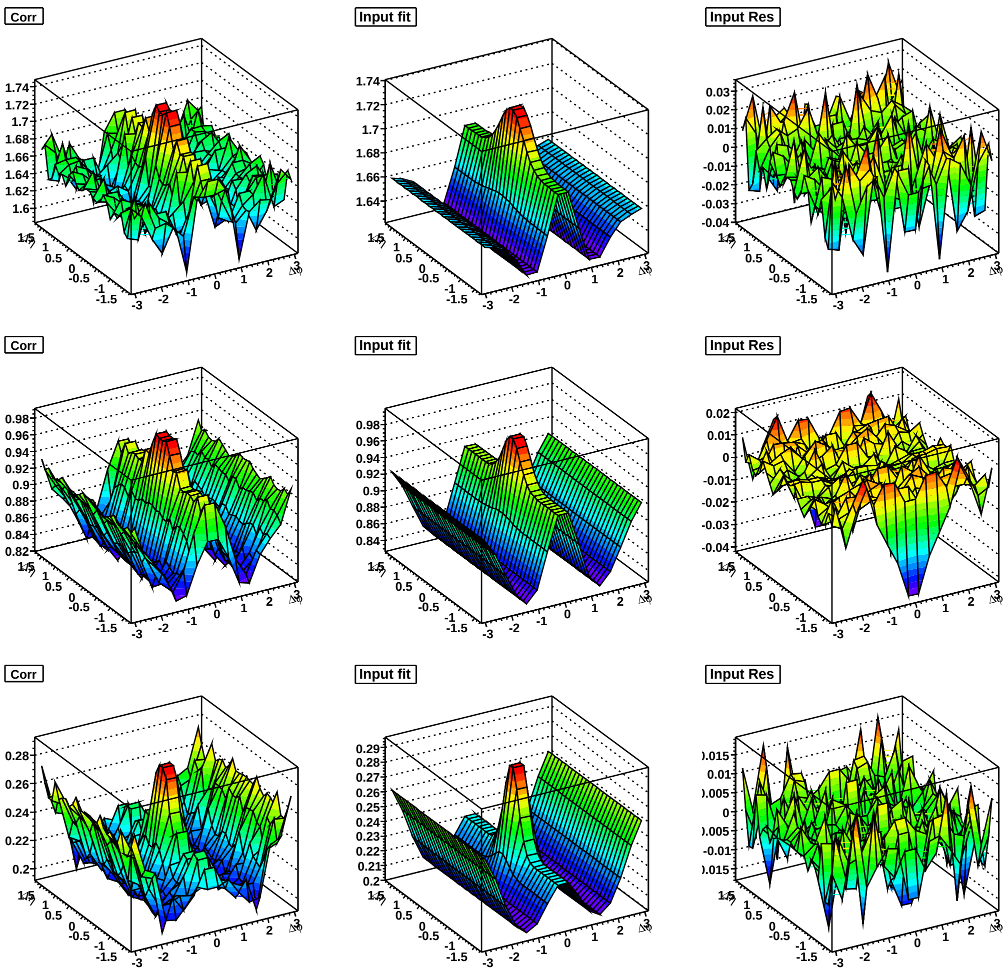

Figure F.7: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-10% (top), 10-40% (middle), and 40-80% (bottom) $Au + Au$ $\sqrt{s_{NN}} = 62$ GeV

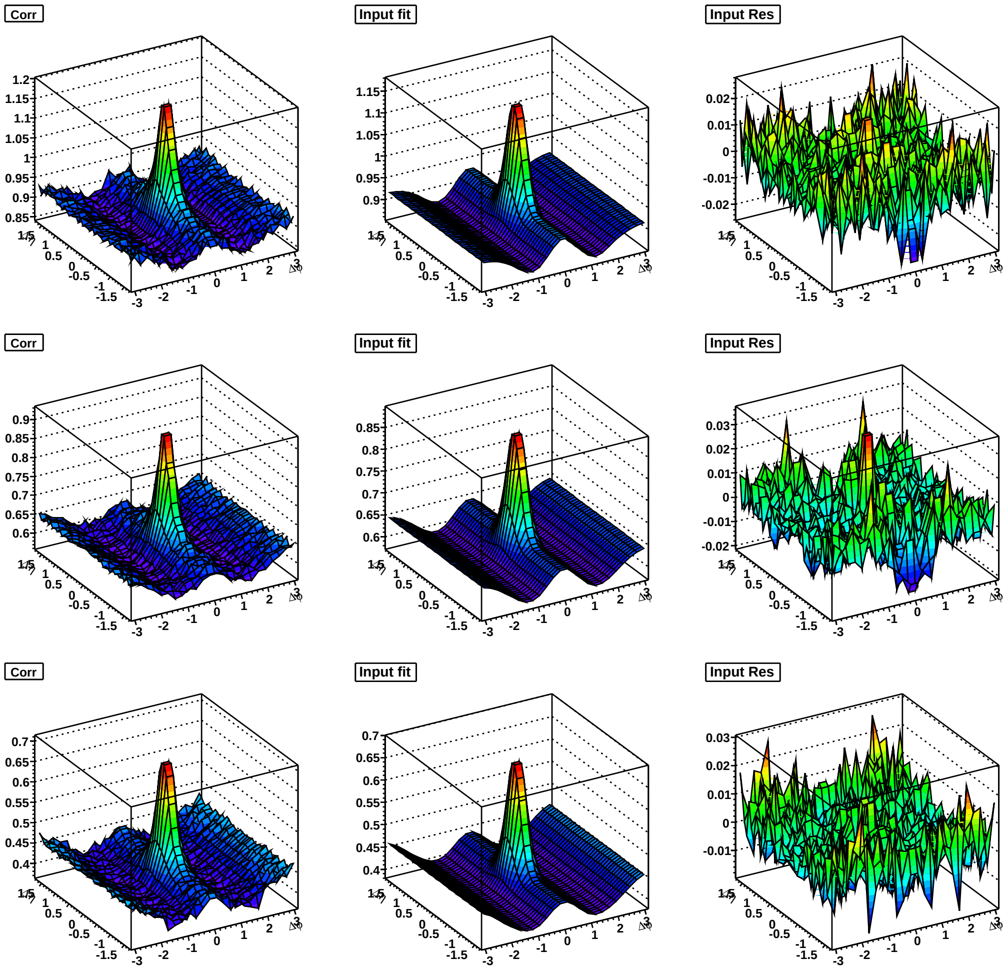

Figure F.8: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-10% (top), 10-20% (middle), and 20-30% (bottom) $Cu + Cu$ $\sqrt{s_{NN}} = 200$ GeV

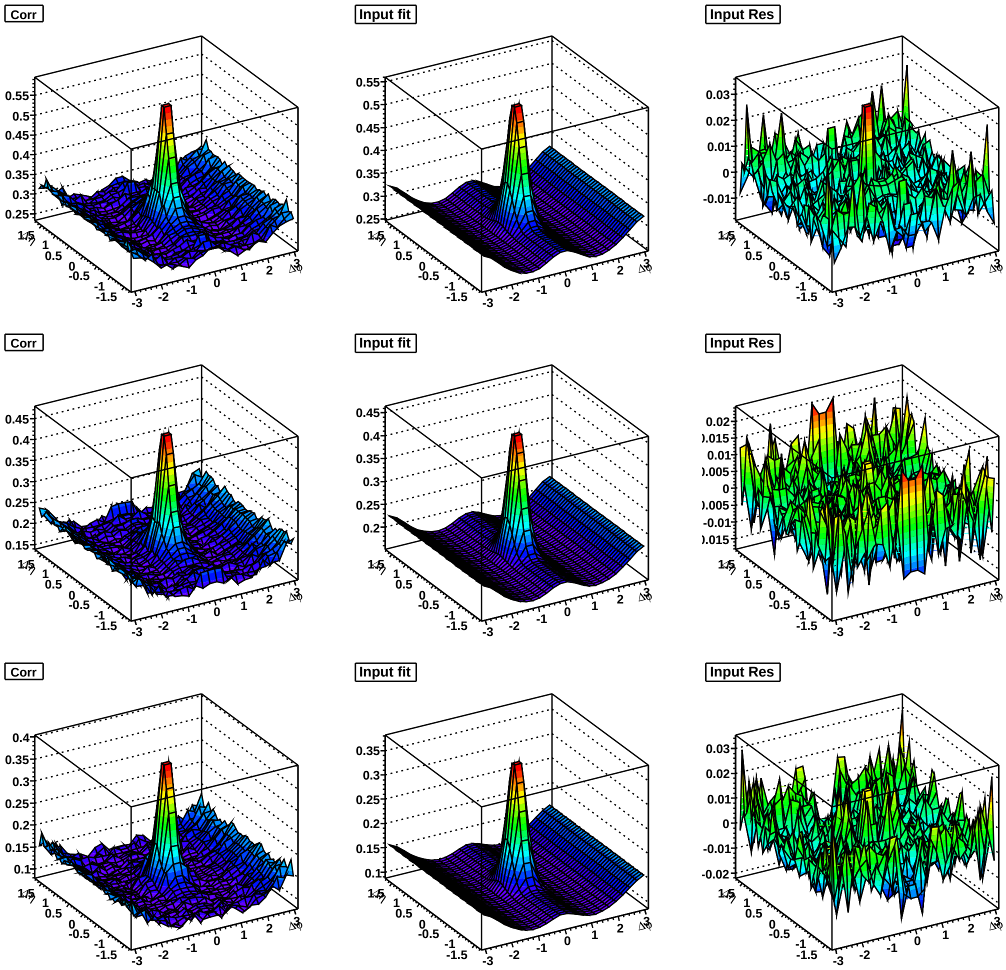

Figure F.9: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 30-40% (top), 40-50% (middle), and 50-60% (bottom) $Cu + Cu$ $\sqrt{s_{NN}} = 200$ GeV

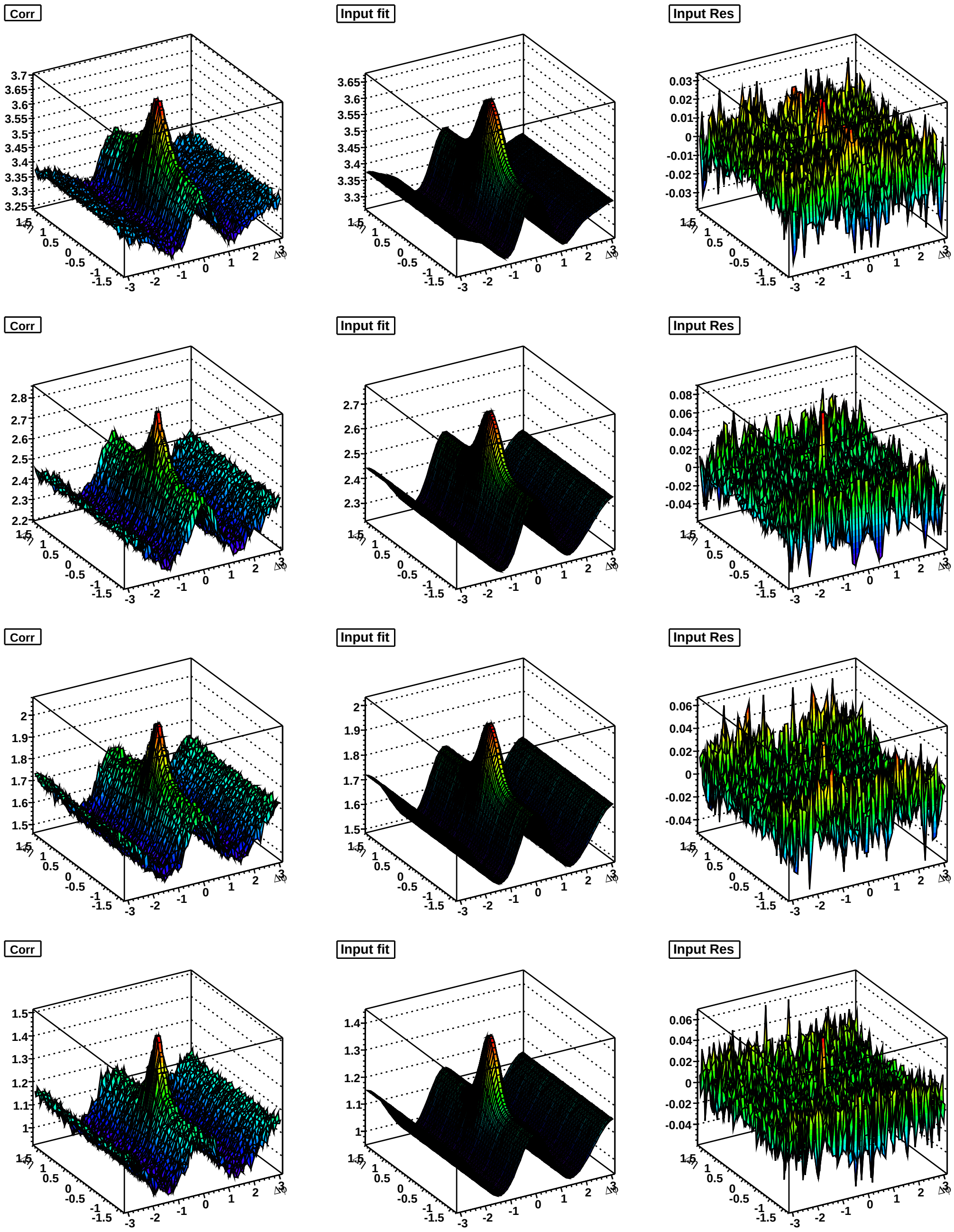

Figure F.10: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-12% (top), 10-20% (second down), 20-30% (third down), and 30-40% (bottom) $Au + Au$ $\sqrt{s_{NN}}$ = 200 GeV

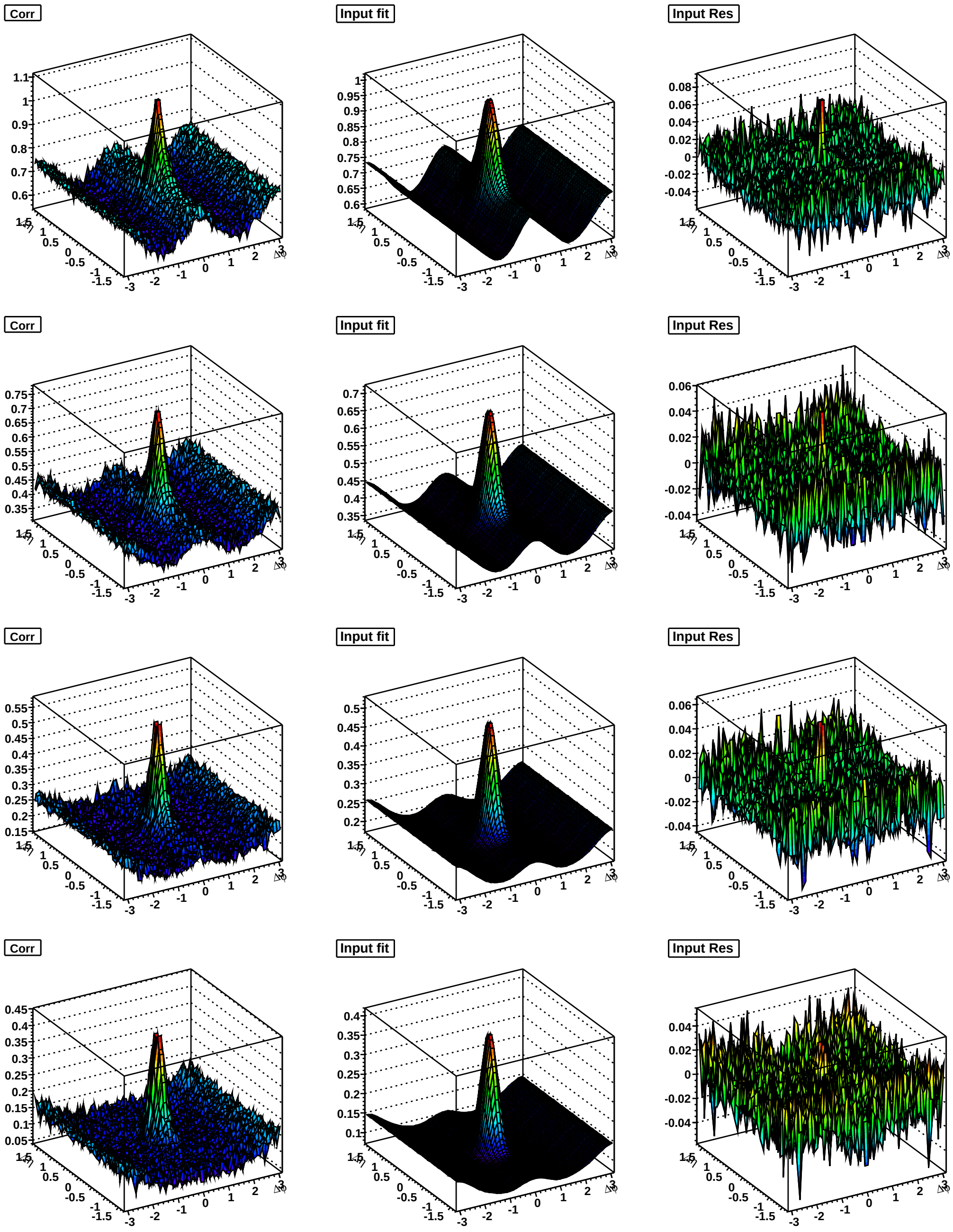

Figure F.11: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 40-50% (top), 50-60% (second down), 60-70% (third down), and 70-80% (bottom) $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV

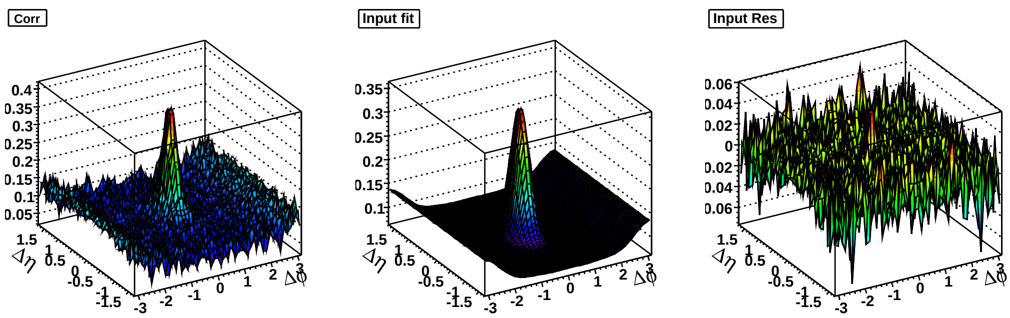

Figure F.12: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ and 3.0 $< p_T^{trigger} <$ 6.0 GeV/c in minimum bias $d + Au$ $\sqrt{s_{NN}} =$ 200 GeV

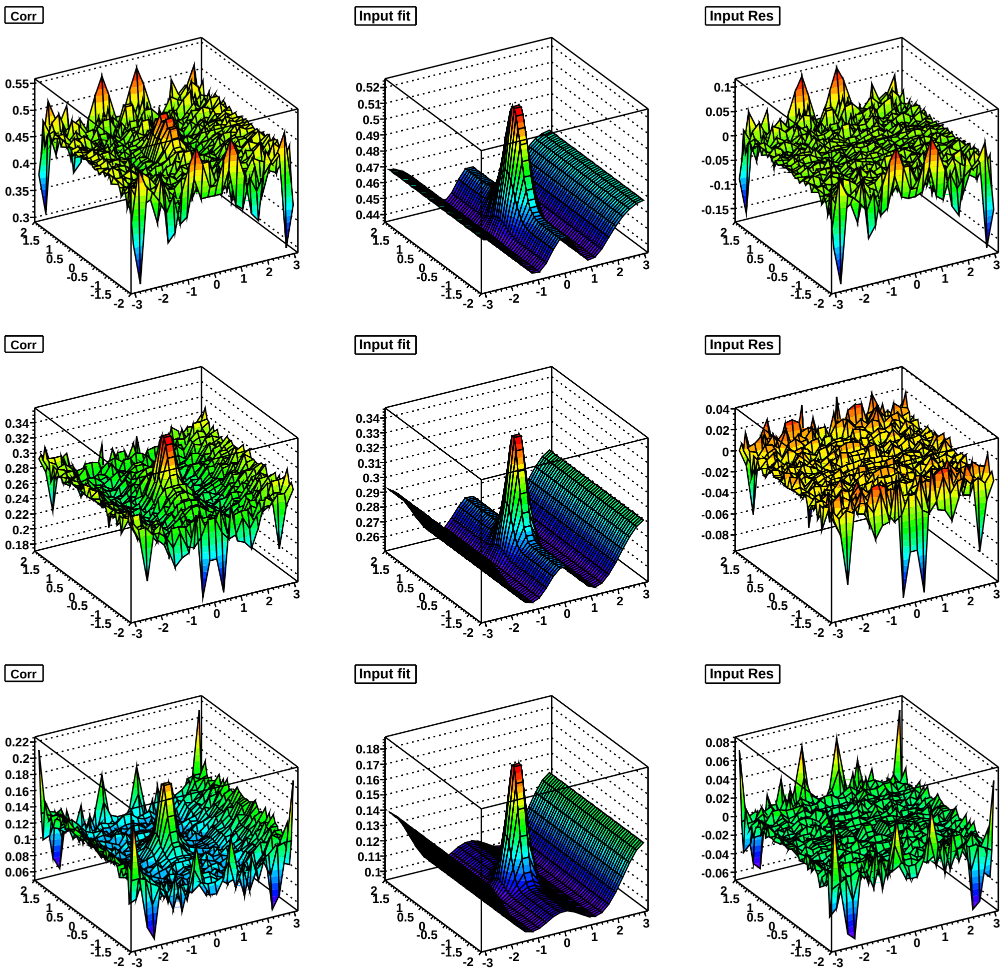

Figure F.13: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-10% (top), 10-30% (middle), and 30-60% (bottom) $Cu + Cu$ $\sqrt{s_{NN}} = 62$ GeV

	A	v_1^2	v_2^2	$\frac{dY_R}{d\Delta\eta}$	$\sigma_{\Delta\phi,R}$	Y_J	$\sigma_{\Delta\eta,J}$	$\sigma_{\Delta\phi,J}$	χ^2/NDF
$Cu + Cu$ $\sqrt{s_{NN}} = 62$ GeV									
0-10%	0.4422 ± 0.0036	-0.0296 ± 0.0054	-0.0036 ± 0.0022	0.0487 ± 0.0109	0.5839 ± 0.0547	0.0670 ± 0.0038	0.3204 ± 0.0185	0.2982 ± 0.0149	1.04
10-30%	0.2639 ± 0.0020	-0.0214 ± 0.0031	0.0066 ± 0.0012	0.0247 ± 0.0062	0.5936 ± 0.0566	0.0738 ± 0.0033	0.2319 ± 0.0094	0.2774 ± 0.0105	1.07
30-60%	0.1121 ± 0.0002	-0.0163 ± 0.0002	0.0107 ± 0.0002	0.0012 ± 1.4223	0.0000 ± 1.0649	0.0829 ± 0.0032	0.2171 ± 0.0065	0.2795 ± 0.0099	1.10
$Au + Au$ $\sqrt{s_{NN}} = 62$ GeV									
0-10%	1.6210 ± 0.0052	-0.0485 ± 0.0079	-0.0104 ± 0.0033	0.1346 ± 0.0159	0.5686 ± 0.0304	0.0564 ± 0.0072	0.3286 ± 0.0469	0.2867 ± 0.0280	1.11
10-40%	0.8545 ± 0.0036	-0.0497 ± 0.0054	0.0198 ± 0.0021	0.1137 ± 0.0107	0.5934 ± 0.0232	0.0684 ± 0.0044	0.2967 ± 0.0181	0.2761 ± 0.0159	0.97
40-80%	0.2260 ± 0.0003	-0.0153 ± 0.0020	0.0206 ± 0.0004	0.0000 ± 0.0251	1.2698 ± 0.7701	0.0665 ± 0.0039	0.2620 ± 0.0127	0.3240 ± 0.0158	1.09
$Cu + Cu$ $\sqrt{s_{NN}} = 200$ GeV									
0-10%	0.8379 ± 0.0091	-0.0968 ± 0.0125	-0.0190 ± 0.0037	0.2035 ± 0.0250	0.7164 ± 0.0236	0.2536 ± 0.0034	0.2495 ± 0.0032	0.2585 ± 0.0030	1.25
10-20%	0.5768 ± 0.0083	-0.0678 ± 0.0114	-0.0013 ± 0.0034	0.1327 ± 0.0228	0.7158 ± 0.0326	0.2637 ± 0.0033	0.2398 ± 0.0027	0.2630 ± 0.0029	1.45
10-30%	0.4138 ± 0.0010	-0.0254 ± 0.0016	0.0167 ± 0.0008	0.0333 ± 0.0032	0.5160 ± 0.0264	0.2708 ± 0.0034	0.2354 ± 0.0024	0.2573 ± 0.0028	1.25
30-40%	0.2829 ± 0.0002	-0.0177 ± 0.0003	0.0209 ± 0.0002	0.0025 ± 1.4572	0.0003 ± 1.0766	0.2837 ± 0.0038	0.2297 ± 0.0024	0.2546 ± 0.0027	1.36
40-50%	0.1861 ± 0.0002	-0.0185 ± 0.0002	0.0186 ± 0.0002	0.0070 ± 0.9324	0.0003 ± 0.0032	0.2950 ± 0.0042	0.2164 ± 0.0022	0.2397 ± 0.0026	1.24
50-60%	0.1192 ± 0.0000	-0.0200 ± 0.0000	0.0156 ± 0.0000	0.0000 ± 1.2132	0.0160 ± 0.0002	0.2784 ± 0.0000	0.2118 ± 0.0000	0.2630 ± 0.0000	1.63

Table F.4: Fit parameters for data in Figure F.13, Figure F.14, Figure F.15, Figure F.16, Figure F.17, and Figure F.18.

	A	v_1^2	v_2^2	$\frac{v_3^2}{v_2^2}$	$\frac{v_4^2}{v_2^2}$	Y_J	$\sigma_{\Delta\eta,J}$	$\sigma_{\Delta\phi,J}$	χ^2/NDF
				$Au + Au\ \sqrt{s_{NN}} = 200$ GeV					
0-12%	3.2879 $\pm$ 0.0008	-0.0974 $\pm$ 0.0012	-0.0138 $\pm$ 0.0006	0.2814 $\pm$ 0.0025	0.5273 $\pm$ 0.0026	0.1853 $\pm$ 0.0012	0.3629 $\pm$ 0.0028	0.3183 $\pm$ 0.0019	1.98
10-20%	2.2940 $\pm$ 0.0022	-0.1093 $\pm$ 0.0034	0.0357 $\pm$ 0.0014	0.2962 $\pm$ 0.0067	0.5624 $\pm$ 0.0061	0.2173 $\pm$ 0.0030	0.3098 $\pm$ 0.0047	0.2705 $\pm$ 0.0036	1.10
20-30%	1.5698 $\pm$ 0.0022	-0.0877 $\pm$ 0.0033	0.0589 $\pm$ 0.0014	0.2206 $\pm$ 0.0066	0.5698 $\pm$ 0.0079	0.2308 $\pm$ 0.0029	0.2962 $\pm$ 0.0039	0.2735 $\pm$ 0.0031	1.15
30-40%	1.0342 $\pm$ 0.0017	-0.0557 $\pm$ 0.0026	0.0606 $\pm$ 0.0012	0.1279 $\pm$ 0.0052	0.5470 $\pm$ 0.0113	0.2476 $\pm$ 0.0032	0.2605 $\pm$ 0.0031	0.2598 $\pm$ 0.0031	1.10
40-50%	0.6488 $\pm$ 0.0015	-0.0379 $\pm$ 0.0023	0.0446 $\pm$ 0.0010	0.0683 $\pm$ 0.0047	0.5528 $\pm$ 0.0179	0.2686 $\pm$ 0.0034	0.2567 $\pm$ 0.0030	0.2537 $\pm$ 0.0029	1.10
50-60%	0.3875 $\pm$ 0.0005	-0.0207 $\pm$ 0.0008	0.0336 $\pm$ 0.0005	0.0138 $\pm$ 0.0018	0.4502 $\pm$ 0.0308	0.2861 $\pm$ 0.0037	0.2311 $\pm$ 0.0024	0.2514 $\pm$ 0.0028	1.09
60-70%	0.2126 $\pm$ 0.0000	-0.0191 $\pm$ 0.0000	0.0230 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	1.1770 $\pm$ 0.8399	0.2937 $\pm$ 0.0001	0.2263 $\pm$ 0.0001	0.2466 $\pm$ 0.0001	1.00
70-80%	0.1067 $\pm$ 0.0002	-0.0209 $\pm$ 0.0002	0.0161 $\pm$ 0.0002	0.0000 $\pm$ 0.1465	1.1972 $\pm$ 0.9002	0.2983 $\pm$ 0.0054	0.2198 $\pm$ 0.0027	0.2448 $\pm$ 0.0031	1.12

Table F.5: Fit parameters for data in Figure F.13, Figure F.14, Figure F.15, Figure F.16, Figure F.17, and Figure F.18.

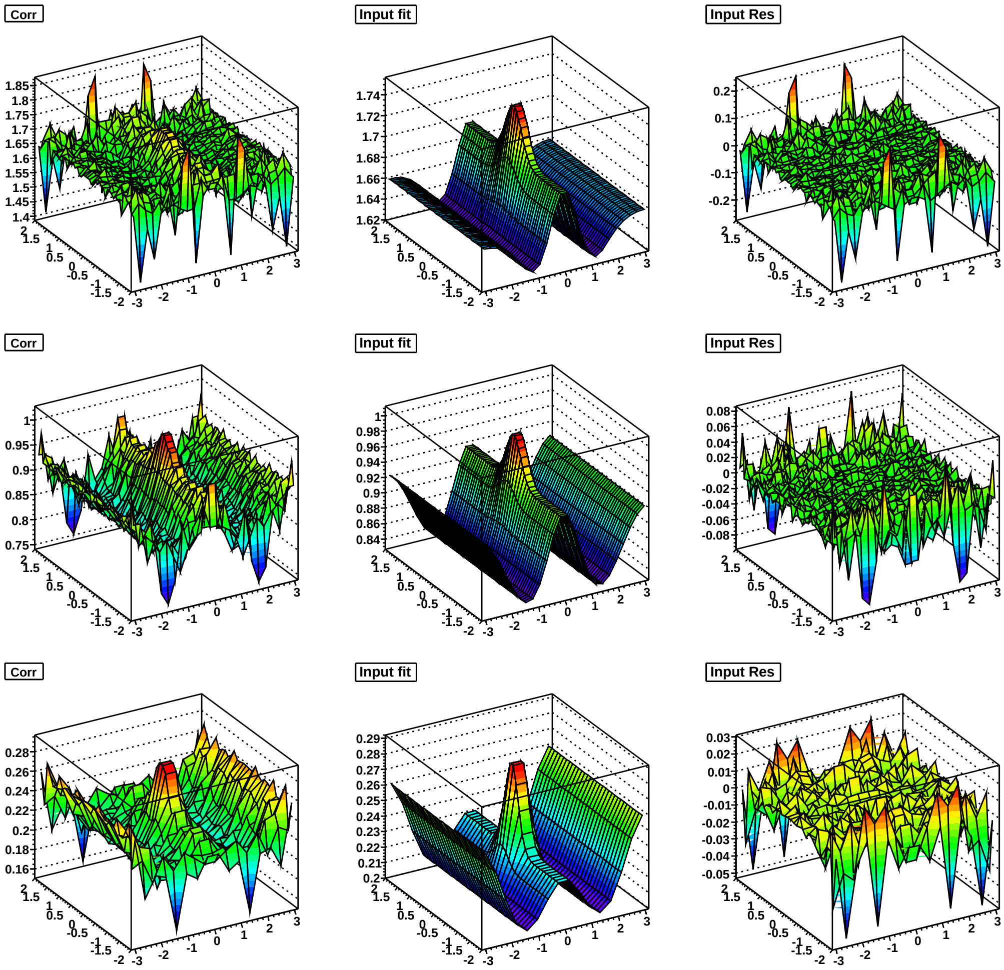

Figure F.14: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-10% (top), 10-40% (middle), and 40-80% (bottom) $Au + Au$ $\sqrt{s_{NN}} = 62$ GeV

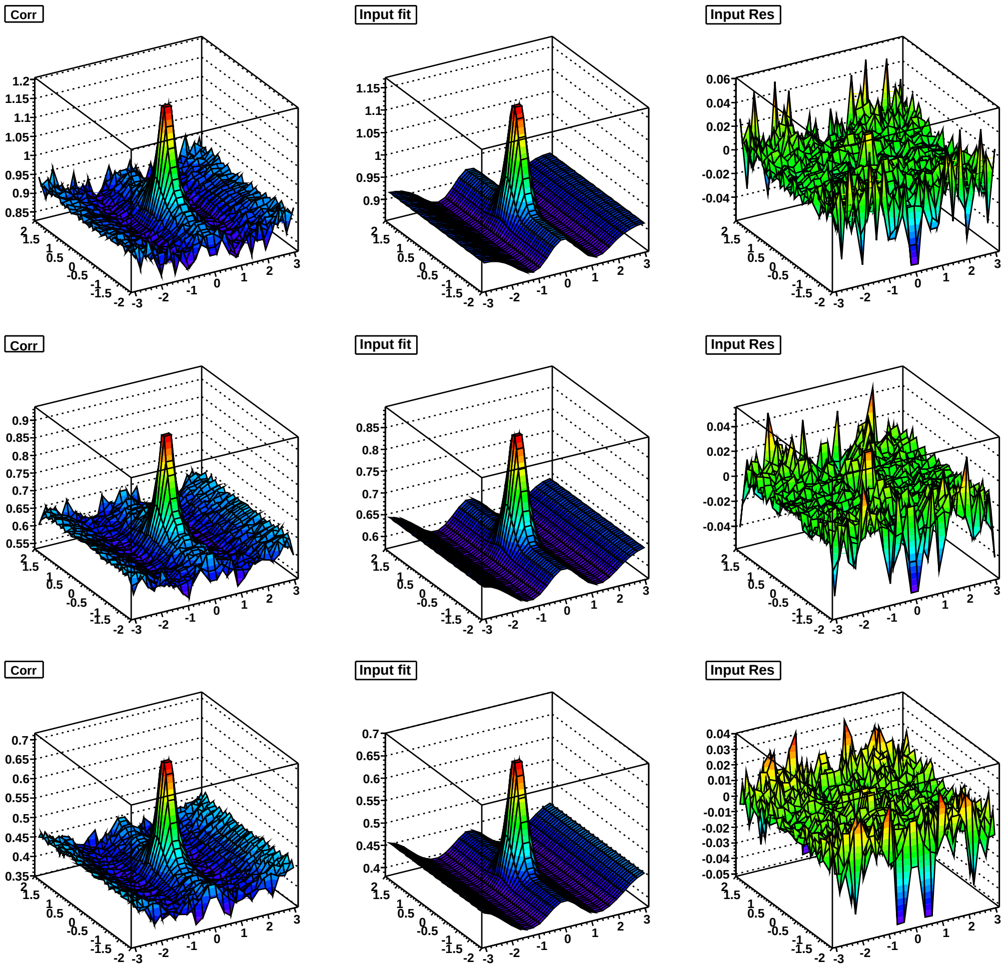

Figure F.15: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-10% (top), 10-20% (middle), and 20-30% (bottom) $Cu + Cu$ $\sqrt{s_{NN}} = 200$ GeV

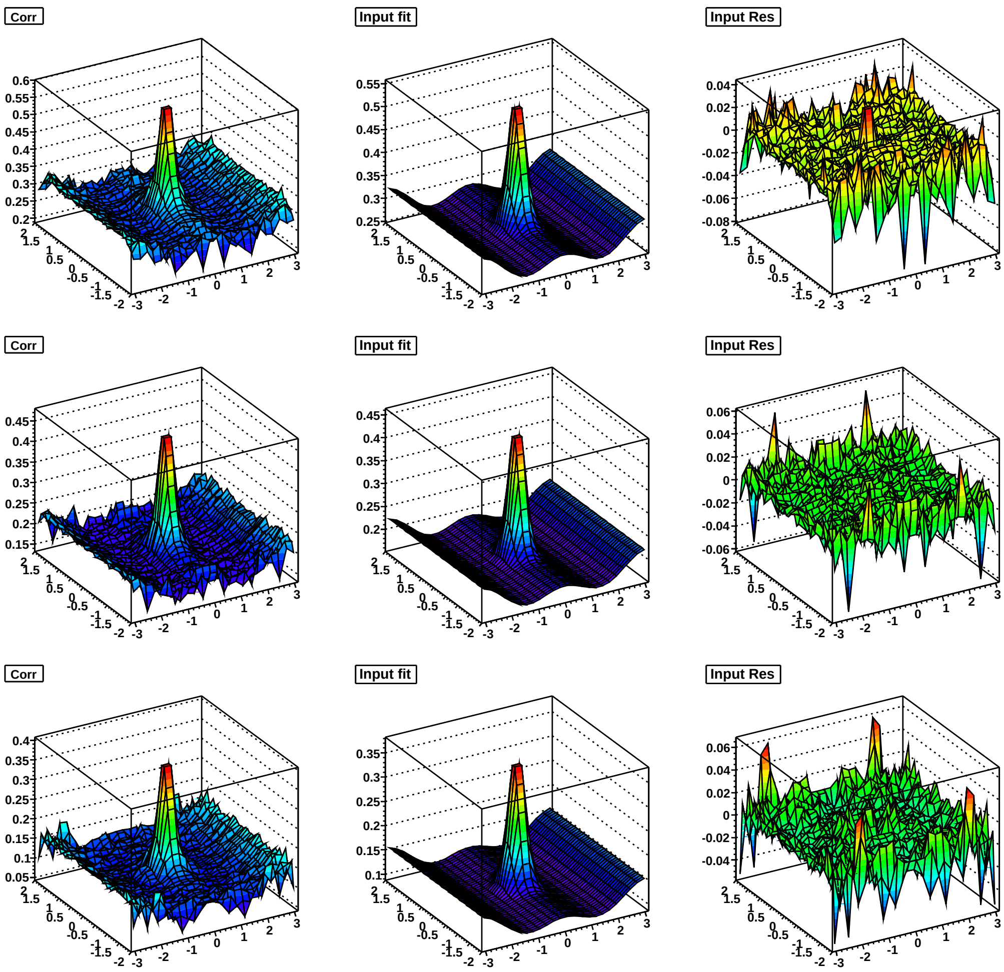

Figure F.16: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and $3.0 < p_T^{trigger} < 6.0$ GeV/c in 30-40% (top), 40-50% (middle), and 50-60% (bottom) $Cu + Cu$ $\sqrt{s_{NN}} = 200$ GeV

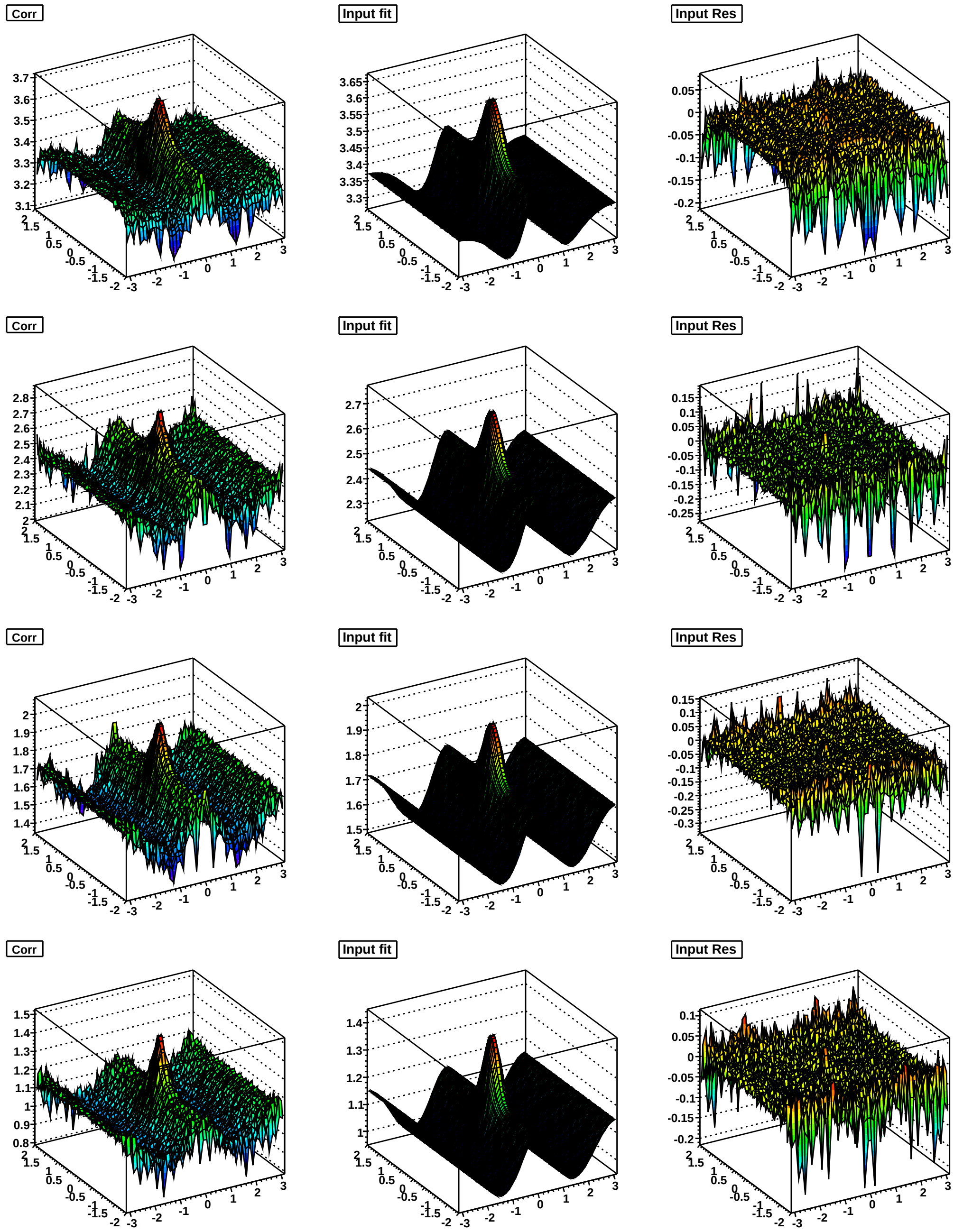

Figure F.17: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated}$ < $p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 0-12% (top), 10-20% (second down), 20-30% (third down), and 30-40% (bottom) $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV

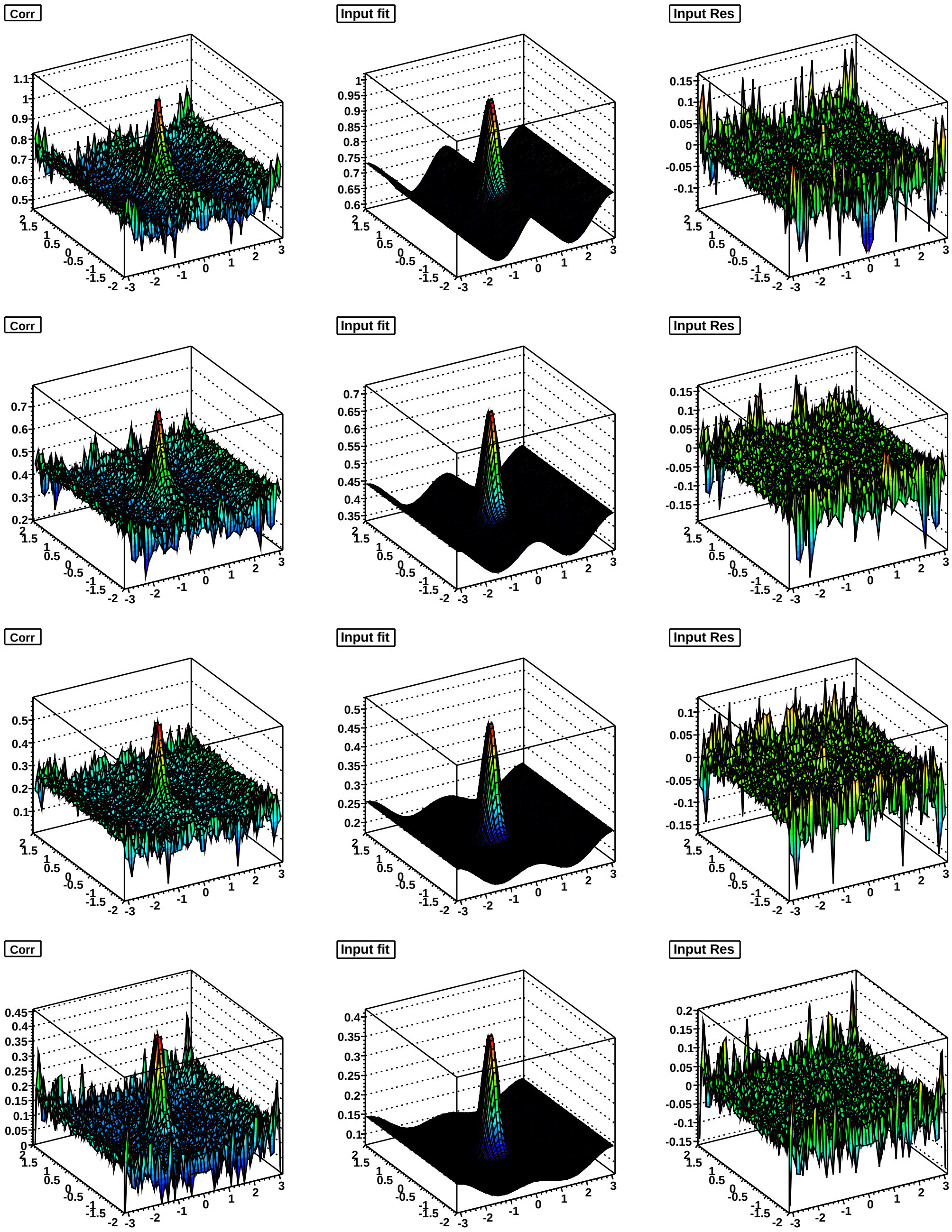

Figure F.18: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c < $p_T^{associated} < p_T^{trigger}$ and 3.0 < $p_T^{trigger}$ < 6.0 GeV/c in 40-50% (top), 50-60% (second down), 60-70% (third down), and 70-80% (bottom) $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV

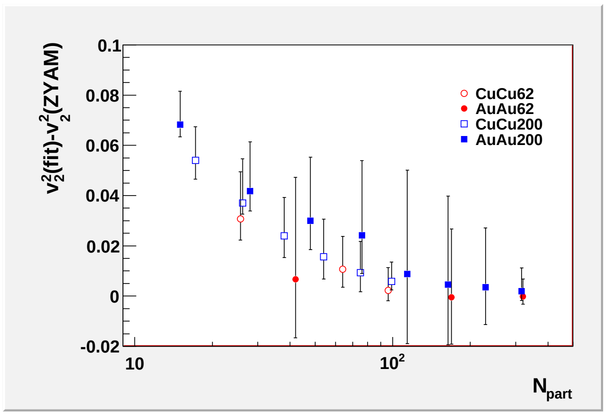

Figure F.19: The difference between the fit $\tilde{v_2}^2$ and the standard $v_2^{trigger} v_2^{assoc}$ used in the analysis.

System	$\sigma/\sigma_{\rm geo}$ [%]	$\langle v_2^{\rm trigger}\rangle$ [%]	$\langle v_2^{\rm assoc}\rangle$ [%]	$\langle v_2^{\rm trigger}\rangle\langle v_2^{\rm assoc}\rangle$	v_2^{fit}
$Cu+Cu$	0-10	$8.5^{+1.4}_{-2.5}$	$7.3^{+1.9}_{-0.7}$	$0.0062^{0.0091}_{0.0040}$	0.0084 ± 0.0005
62.4 GeV	10-30	$11.6^{+0.5}_{-3.2}$	$10.4^{+0.3}_{-1.8}$	$0.0121^{0.0130}_{0.0072}$	0.0228 ± 0.0005
	30-60	$14.5^{+1.0}_{-5.3}$	$11.8^{+0.3}_{-2.7}$	$0.0171^{0.0188}_{0.0084}$	0.0471 ± 0.0010
$Au+Au$	0-10	$8.5^{+0.3}_{-3.2}$	$7.6^{+0.3}_{-1.9}$	$0.0065^{0.0070}_{0.0030}$	0.0063 ± 0.0003
62.4 GeV	10-40	$17.5^{+0.3}_{-3.2}$	$15.0^{+0.3}_{-1.9}$	$0.0263^{0.0272}_{0.0187}$	0.0258 ± 0.0002
	40-80	$21.2^{+0.3}_{-5.9}$	$18.6^{+0.3}_{-3.4}$	$0.0394^{0.0406}_{0.0233}$	0.0456 ± 0.0008
$Cu+Cu$	0-10	$8.0^{+0.7}_{-3.0}$	$8.8^{+0.1}_{-2.2}$	$0.0070^{0.0077}_{0.0033}$	0.0128 ± 0.0002
200 GeV	10-20	$10.5^{+0.6}_{-2.5}$	$11.3^{+0.0}_{-1.8}$	$0.0119^{0.0125}_{0.0076}$	0.0212 ± 0.0002
	20-30	$11.2^{+0.7}_{-2.9}$	$12.5^{+0.1}_{-1.9}$	$0.0140^{0.0150}_{0.0088}$	0.0296 ± 0.0003
	30-40	$10.7^{+0.9}_{-2.6}$	$13.0^{+0.1}_{-2.2}$	$0.0139^{0.0152}_{0.0087}$	0.0380 ± 0.0004
	40-50	$11.2^{+2.2}_{-6.3}$	$12.6^{+0.5}_{-3.7}$	$0.0141^{0.0176}_{0.0044}$	0.0506 ± 0.0006
	50-60	$10.3^{+0.6}_{-2.7}$	$12.3^{+0.0}_{-2.6}$	$0.0127^{0.0134}_{0.0074}$	0.0657 ± 0.0010
$Au+Au$	0-12	$8.5^{+2.1}_{-2.1}$	$7.3^{+1.5}_{-1.5}$	$0.0062^{0.0093}_{0.0037}$	0.0081 ± 0.0001
200 GeV	10-20	$14.6^{+2.0}_{-2.0}$	$13.0^{+1.2}_{-1.2}$	$0.0190^{0.0236}_{0.0149}$	0.0225 ± 0.0001
	20-30	$17.8^{+2.1}_{-2.1}$	$16.5^{+1.2}_{-1.2}$	$0.0298^{0.0352}_{0.0240}$	0.0344 ± 0.0001
	30-40	$18.9^{+2.3}_{-2.3}$	$18.1^{+1.4}_{-1.4}$	$0.0342^{0.0413}_{0.0277}$	0.0430 ± 0.0001
	40-50	$11.5^{+2.3}_{-2.3}$	$19.0^{+2.5}_{-2.5}$	$0.0219^{0.0297}_{0.0152}$	0.0461 ± 0.0002
	50-60	$10.1^{+2.3}_{-2.3}$	$17.6^{+2.8}_{-2.8}$	$0.0178^{0.0253}_{0.0115}$	0.0478 ± 0.0003
	60-70	$8.5^{+2.2}_{-2.2}$	$15.4^{+2.9}_{-2.9}$	$0.0131^{0.0196}_{0.0079}$	0.0549 ± 0.0006
	70-80	$6.6^{+1.9}_{-1.9}$	$12.8^{+2.8}_{-2.8}$	$0.0084^{0.0133}_{0.0047}$	0.0754 ± 0.0012

Table F.6: v_2 values used in ZYAM method and from fit

Appendix G

$p_T^{trigger}$ dependence for different collision energies

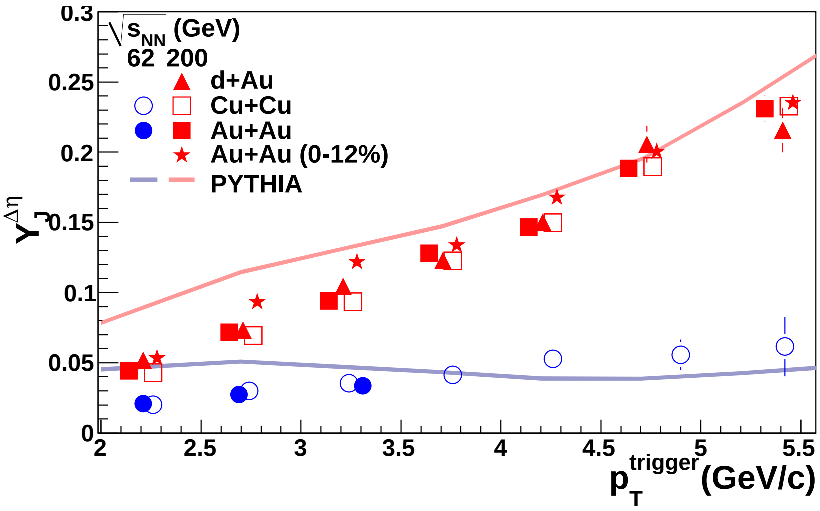

Figure G.1: $p_T^{trigger}$ dependence of Y_{Jet} for unidentified hadrons for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for 0-60% central $Cu + Cu$ and 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and minimum bias $d + Au$, 0-60% central $Cu + Cu$ and 40-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

All points are done using bin counting and the $\Delta\eta$ method except for the three lowest pt bins for central $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV.

For the lowest p_T central $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV data points, a fit in $\Delta\eta$ is

$< p_T^{trig} >$ (GeV/c)	Y_{Jet} $\Delta\eta$ bin counting	Y_{Jet} $\Delta\eta$ fit	fit-bc (σ)	Y_{Jet} $\Delta\phi$ bin counting	Y_{Jet} $\Delta\eta$ fit	fit-bc (σ)
2.2	0.053 ± 0.001	0.067 ± 0.002	5.7	0.042 ± 0.001	0.058 ± 0.001	11.3
2.7	0.097 ± 0.002	0.119 ± 0.005	4.3	0.075 ± 0.001	0.085 ± 0.002	4.9
3.2	0.120 ± 0.002	0.134 ± 0.004	3.0	0.107 ± 0.002	0.113 ± 0.003	1.8
3.7	0.133 ± 0.003	0.137 ± 0.005	0.7	0.133 ± 0.004	0.140 ± 0.005	1.2
4.2	0.164 ± 0.005	0.162 ± 0.007	-0.3	0.169 ± 0.006	0.173 ± 0.008	0.4
4.7	0.201 ± 0.007	0.198 ± 0.010	-0.2	0.204 ± 0.009	0.200 ± 0.011	-0.3
5.4	0.237 ± 0.009	0.222 ± 0.010	-1.1	0.263 ± 0.012	0.254 ± 0.015	-0.5

Table G.1: Y_{Jet} for different methods for central $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV

used, excluding small $\Delta\eta$ where a residual track merging effect is present. The fit produces a higher yield for these data points than bin counting, which is expected since the fit should correct for the residual track merging effect. The exact numbers for the bin counting and fit methods for central Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV are given in Table G.1. (The number of σ the data points are apart is approximate because the errors are correlated and I assumed that they were independent for this quick and dirty calculation.) The values for the yield from $\Delta\phi$ are somewhat suspect because the dip is broader in $\Delta\eta$ than in $\Delta\phi$, therefore, a fit in $\Delta\phi$ will not correct for the residual dip as well. The bin counting method in $\Delta\phi$ completely neglects any residual track merging correction. At the highest bins in $p_T^{trigger}$, all methods agree, consistent with previous published results. (The fit in $\Delta\eta$ is systematically lower; by looking at the fits it is evident that a Gaussian fit does not quite describe the peak. However, this is still within error at these momenta.) At lower $p_T^{trigger}$ the methods differ. The yield in $\Delta\phi$ is determined assuming that there is no jet-like correlation for $\mid \Delta\eta \mid < 0.78$ and at lower $p_T^{trigger}$ the jet-like correlation is wider so the background may be oversubtracted. An alternate plot of Figure G.1 is shown in Figure G.2 showing the difference between the fit and the bin counting methods in $\Delta\eta$ for the central $Au + Au$ data at $\sqrt{s_{NN}} = 200$ GeV. Based on these comparisons of the different methods, it was decided that using the fit method for the lowest three points was the best way to correct for the residual track merging effect.

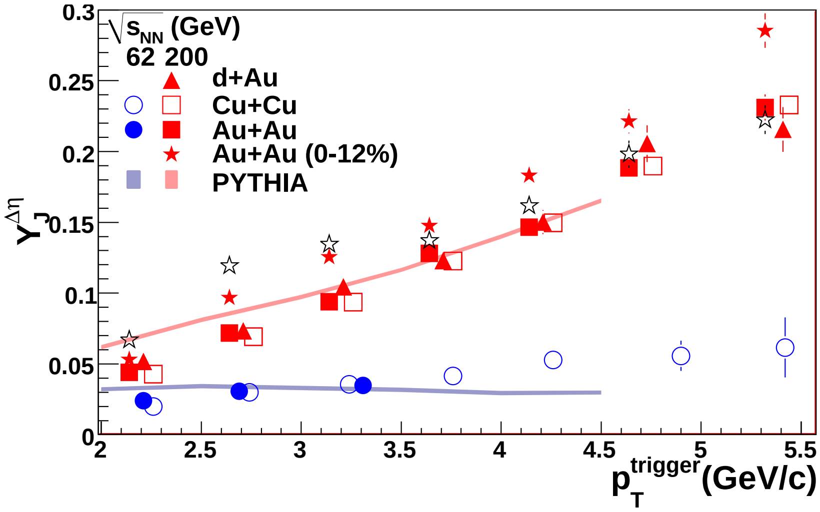

Figure G.2: $p_T^{trigger}$ dependence of Y_{Jet} for unidentified hadrons for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for 0-60% central $Cu + Cu$ and 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and minimum bias $d + Au$, 0-60% central $Cu + Cu$ and 40-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Red solid stars show Y_{Jet} for the $\Delta\eta$ method using bin counting and open black stars show Y_{Jet} for the $\Delta\eta$ method using a Gaussian fit in $\Delta\eta$.

G.A Fits for $p_T^{trigger}$ dependence in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

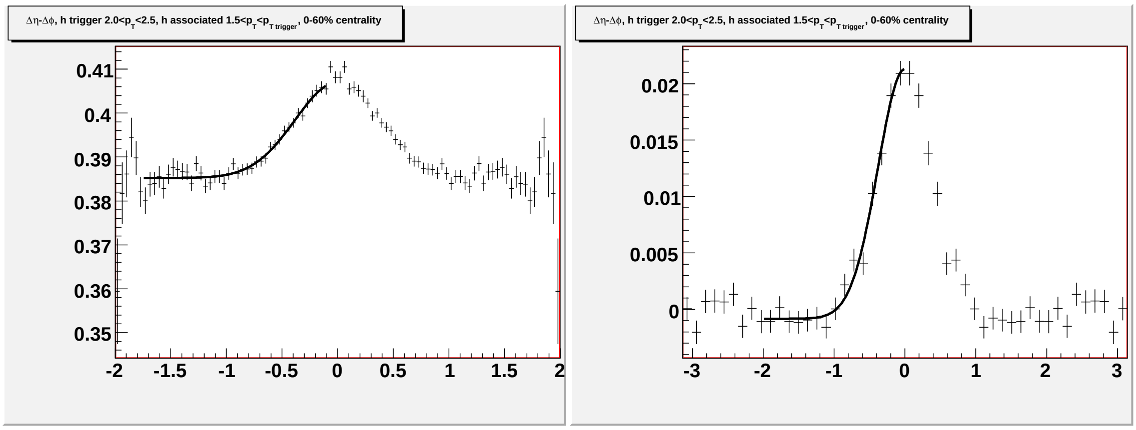

Figure G.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

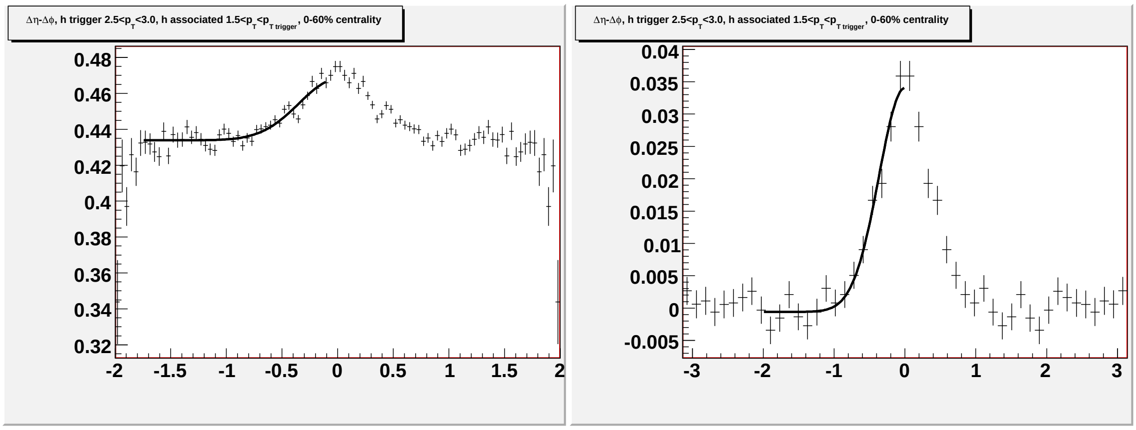

Figure G.4:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

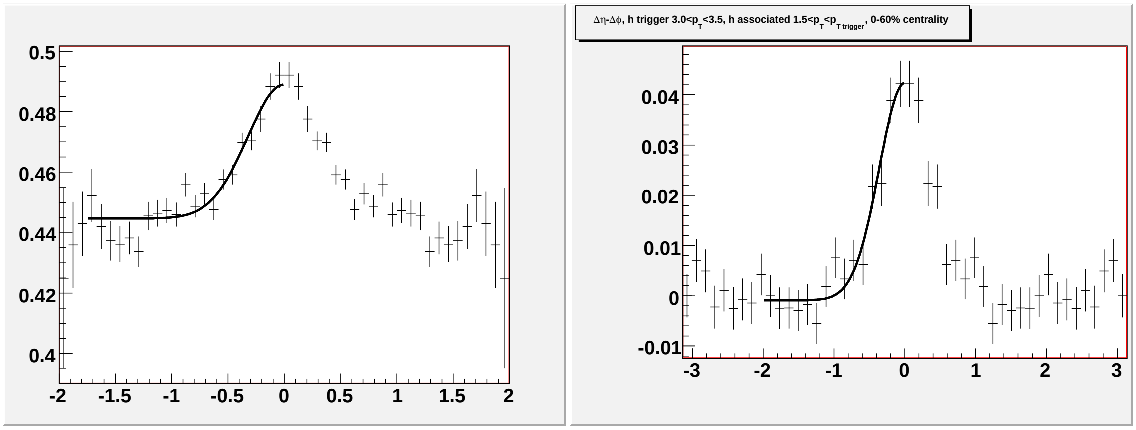

Figure G.5:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

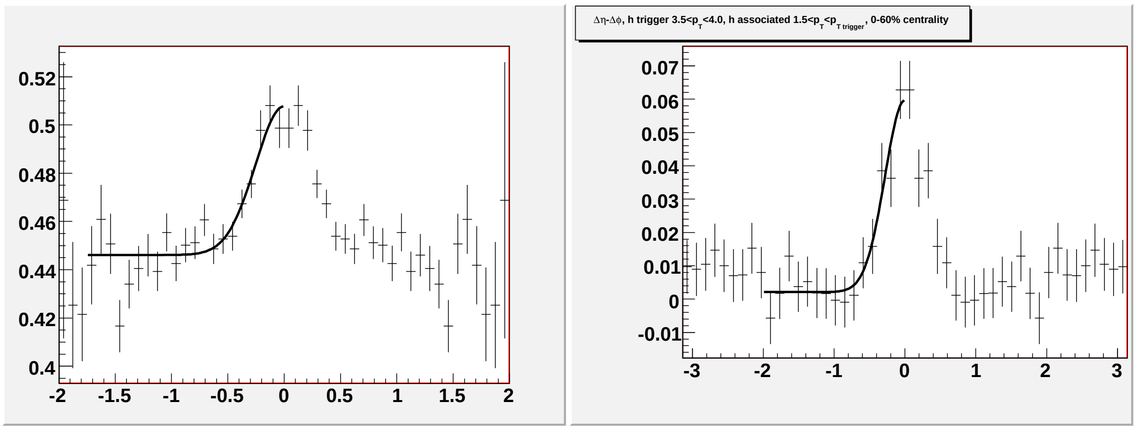

Figure G.6:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

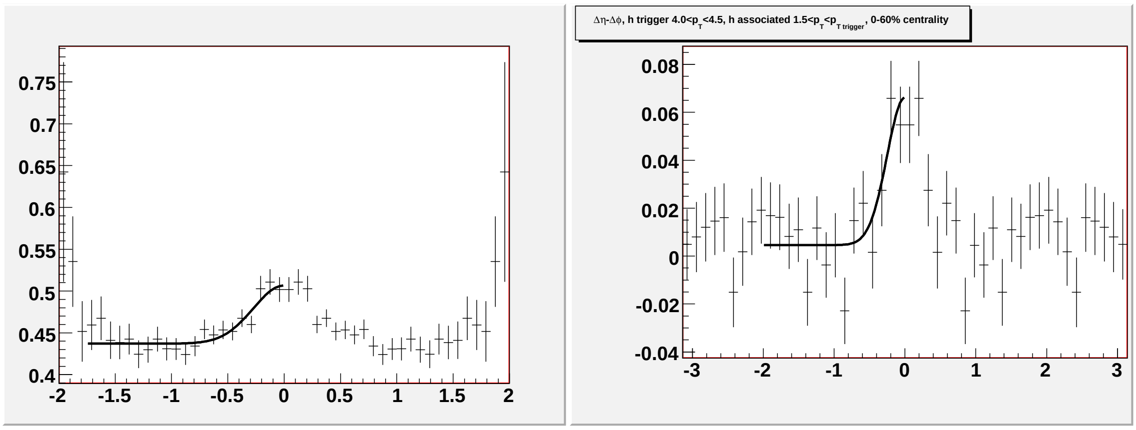

Figure G.7:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

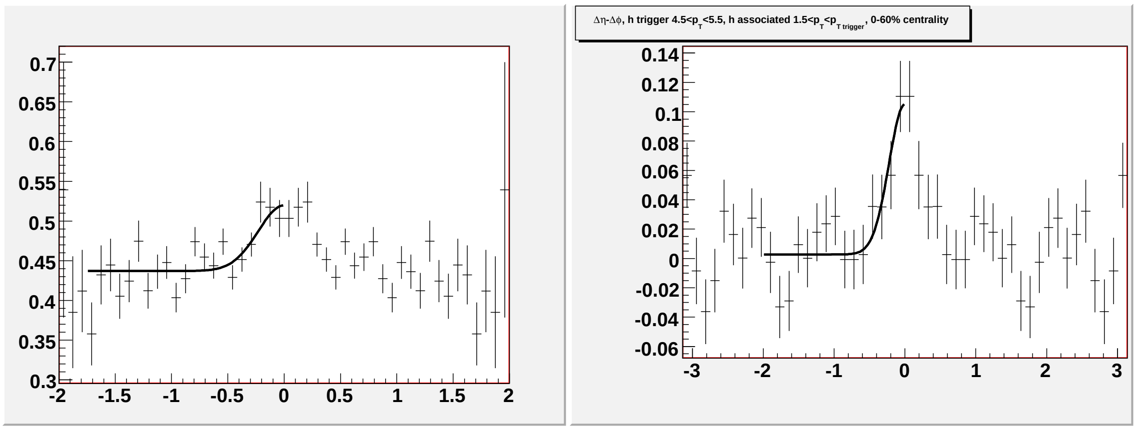

Figure G.8:]

Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over 0 $< |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.5 < p_T^{trigger} < 5.5$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c.

G.B Fits for $p_T^{trigger}$ dependence in Au+Au collisions at $\sqrt{s_{NN}}$ = 62 GeV

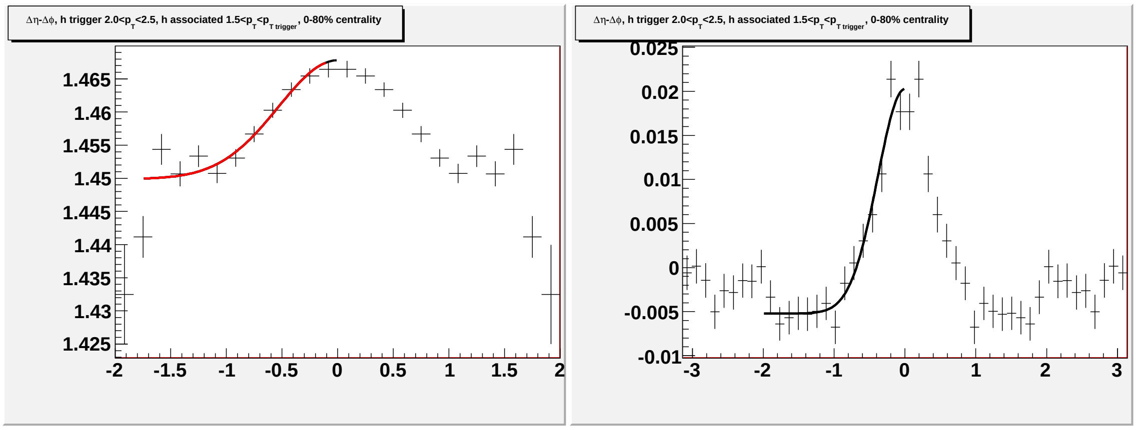

Figure G.9: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

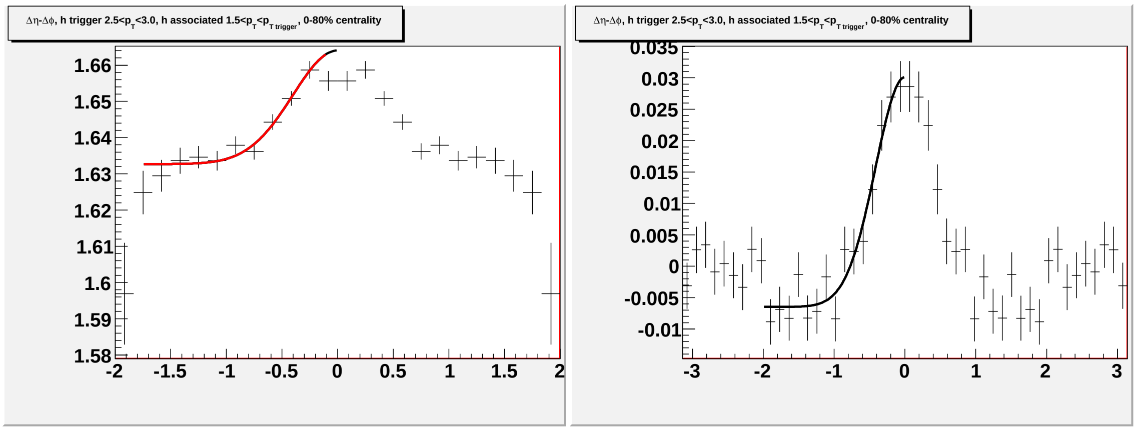

Figure G.10: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

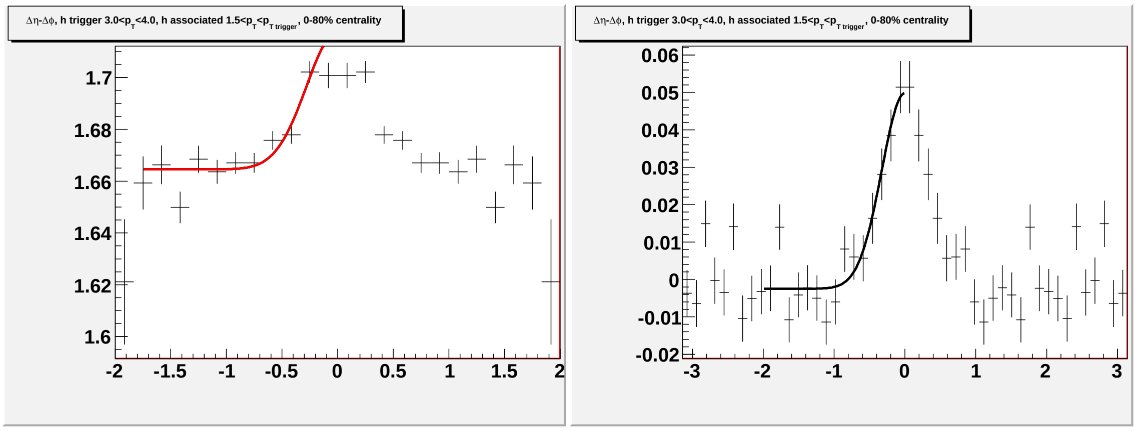

Figure G.11: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

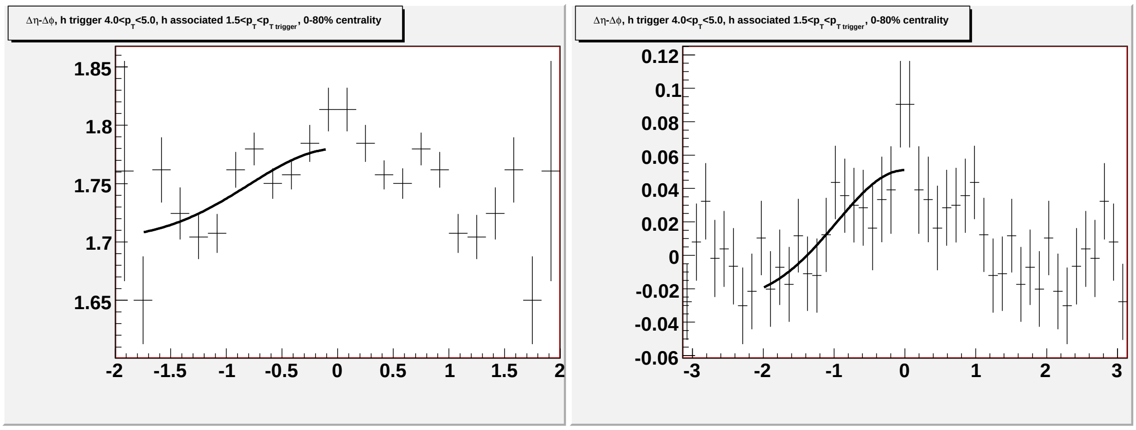

Figure G.12: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

G.C Fits for $p_T^{trigger}$ dependence in Cu+Cu collisions at $\sqrt{s_{NN}}$ = 200 GeV

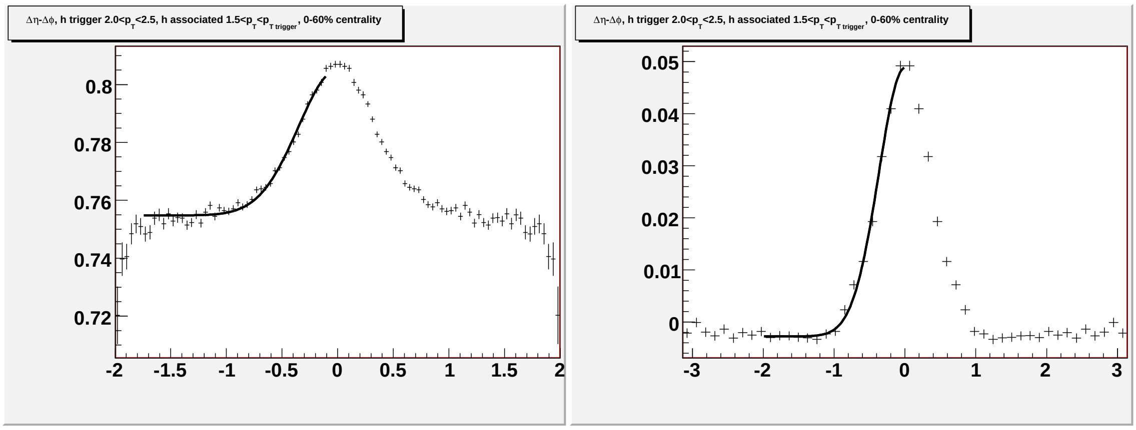

Figure G.13: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

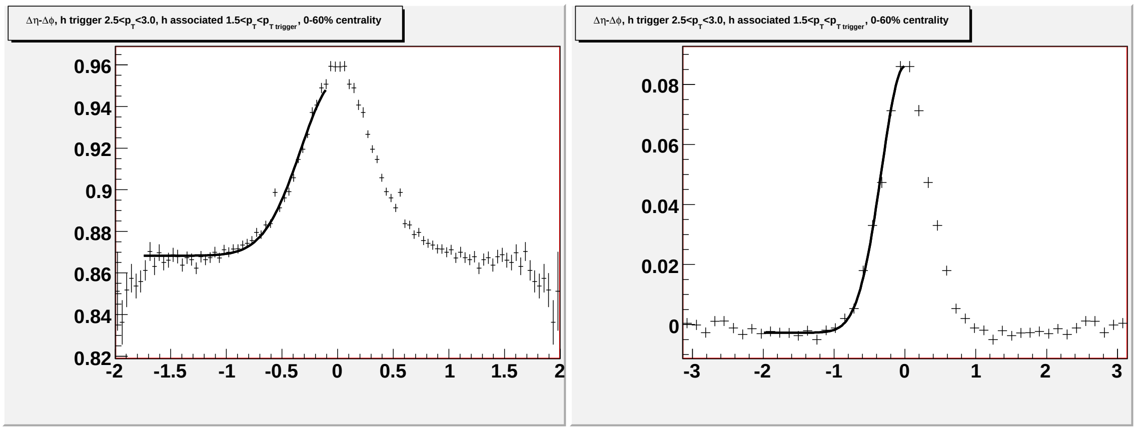

Figure G.14: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

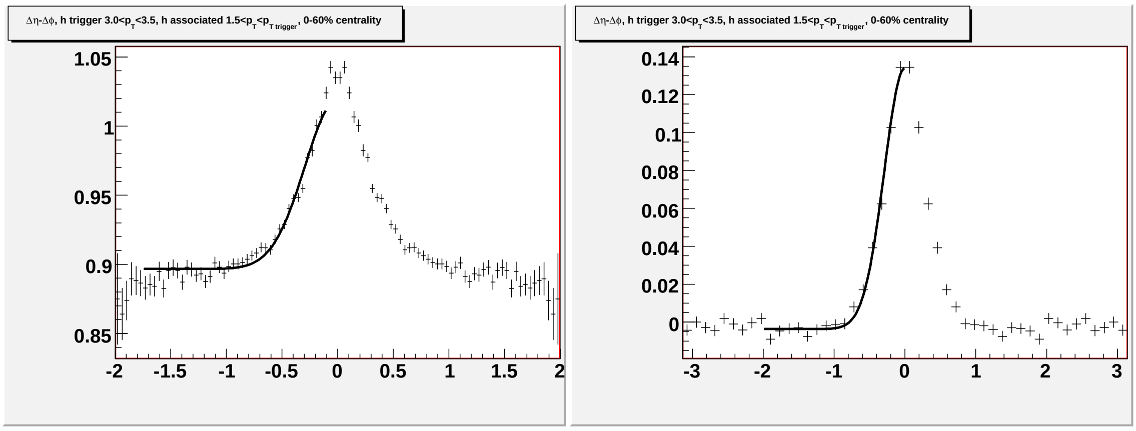

Figure G.15: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c.

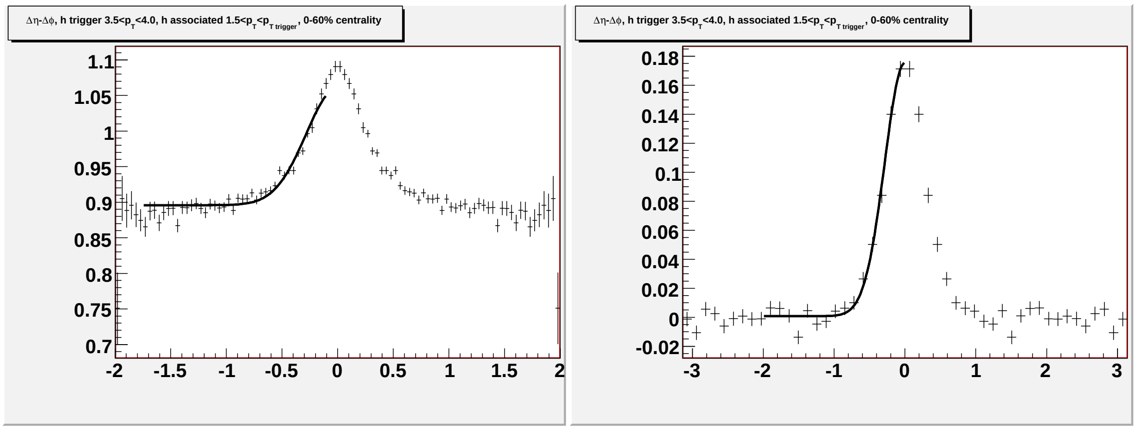

Figure G.16: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c.

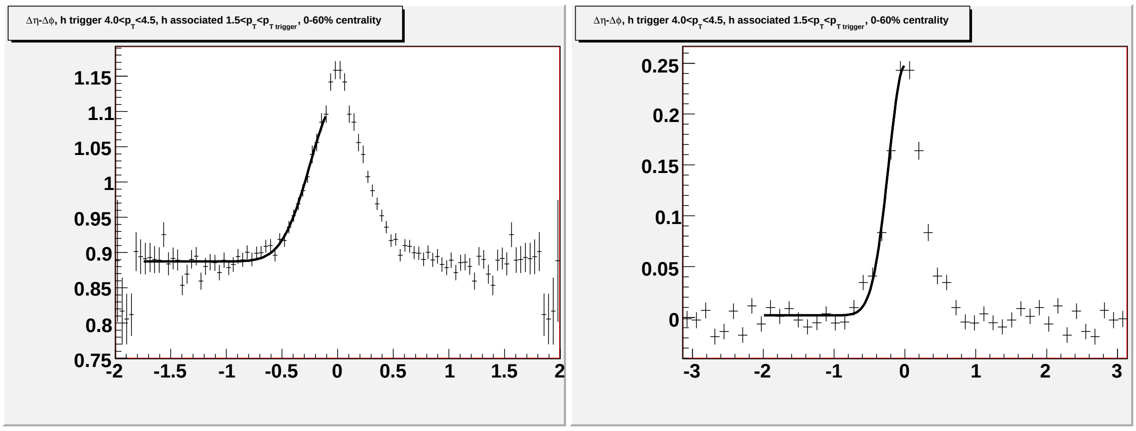

Figure G.17: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

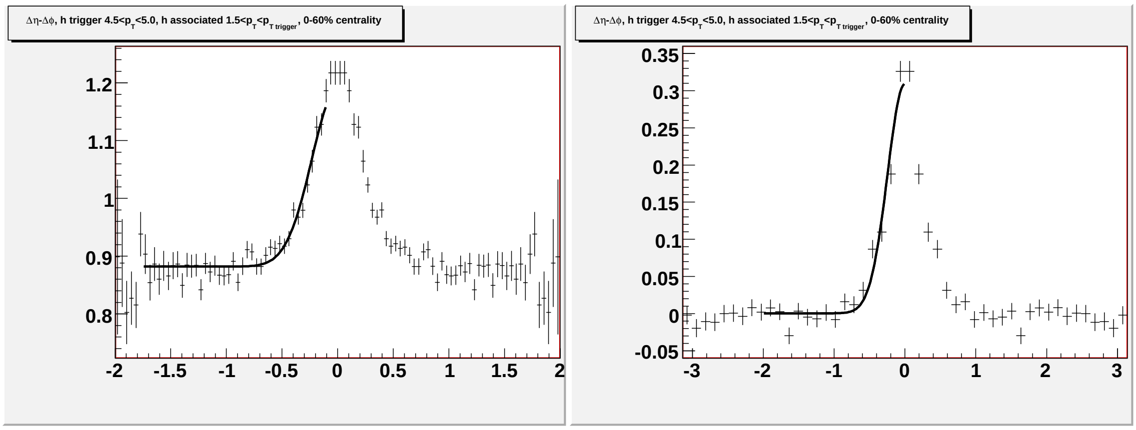

Figure G.18: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.5 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

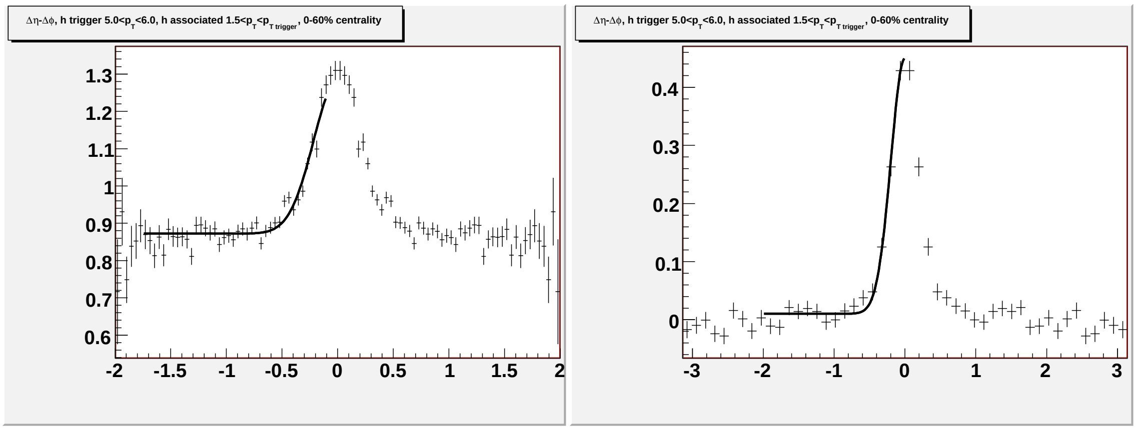

Figure G.19: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $5.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c.

G.D Fits for $p_T^{trigger}$ dependence in 40-80% $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV

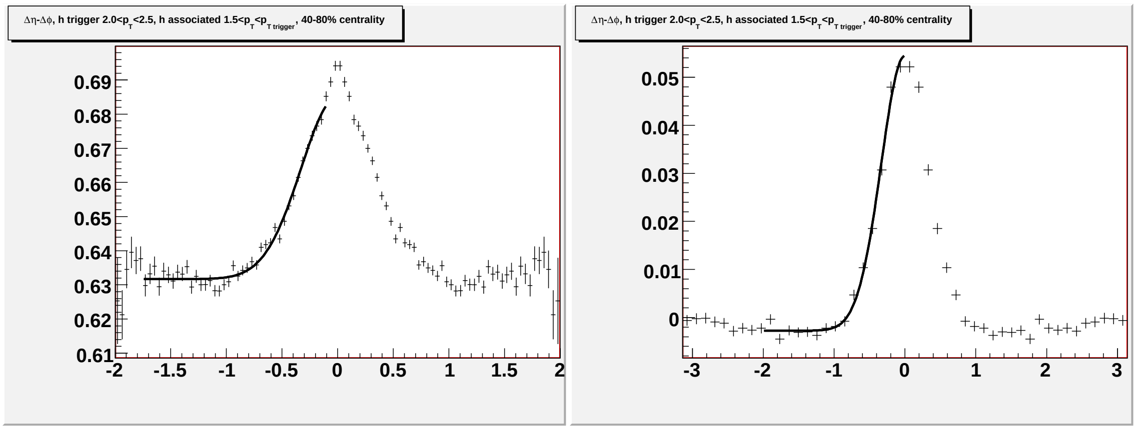

Figure G.20: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

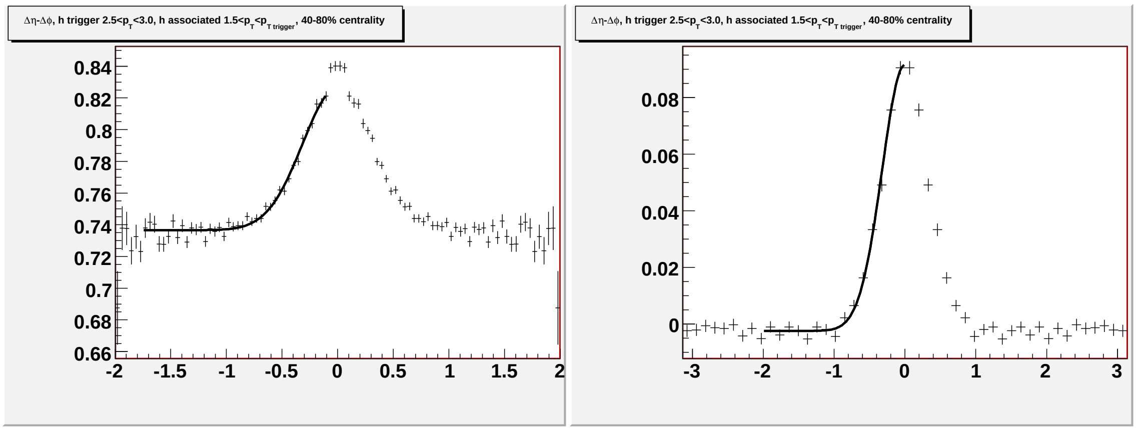

Figure G.21: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

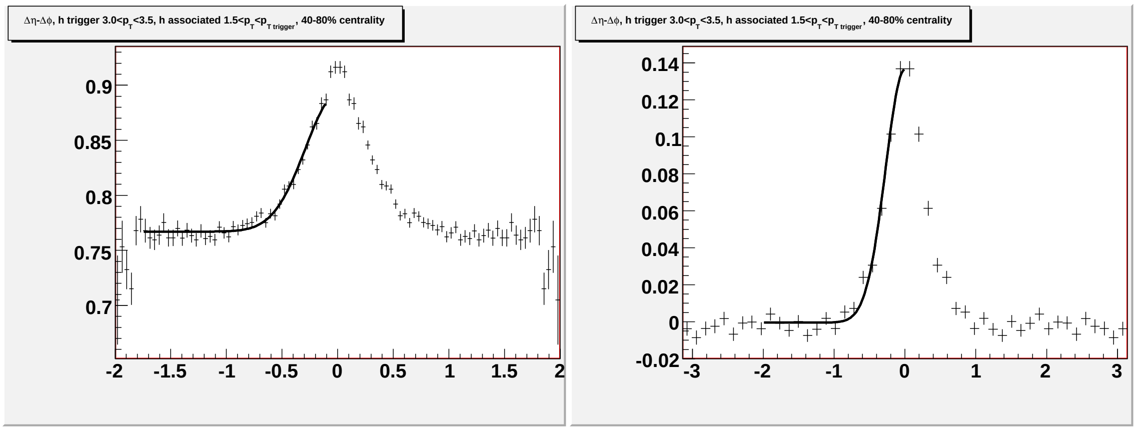

Figure G.22: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

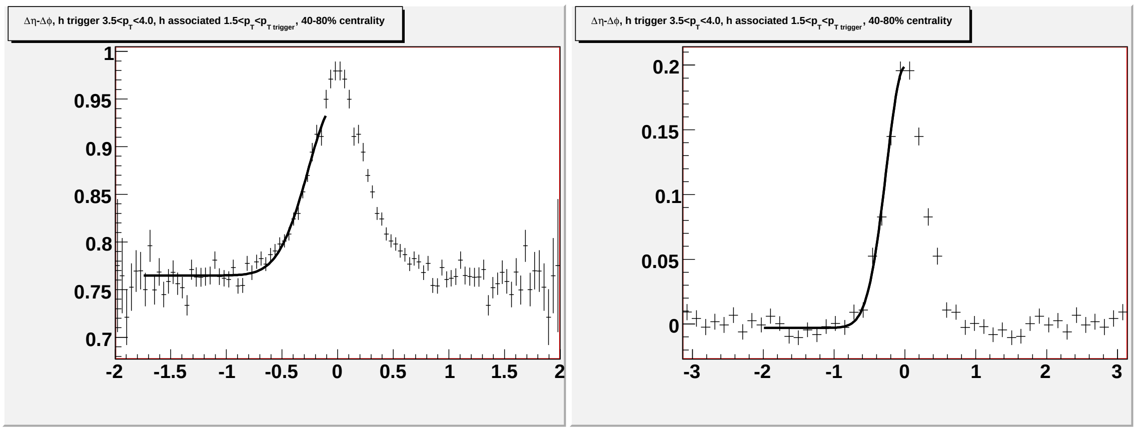

Figure G.23: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

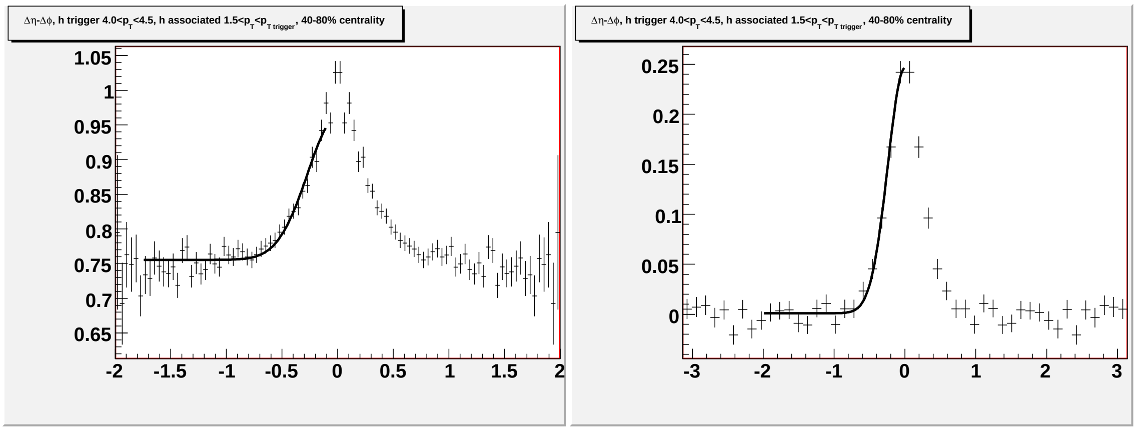

Figure G.24: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

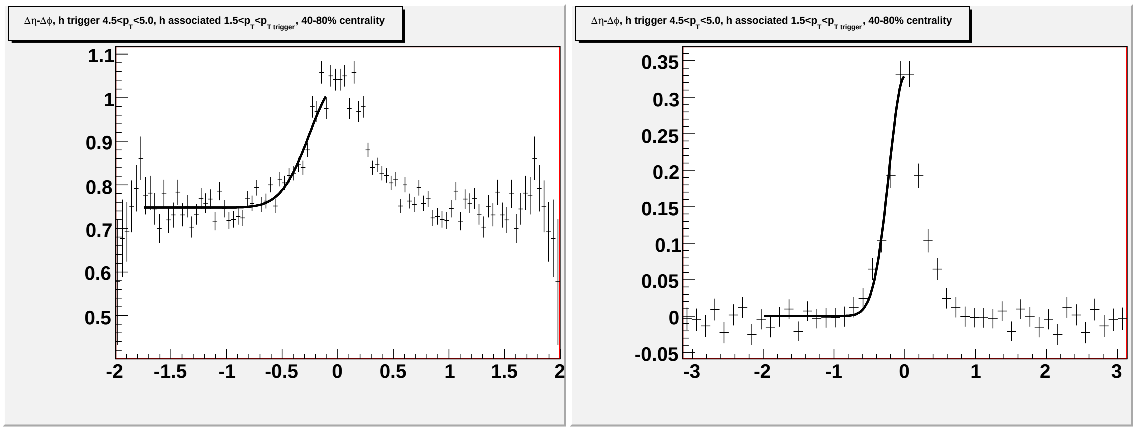

Figure G.25: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.5 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

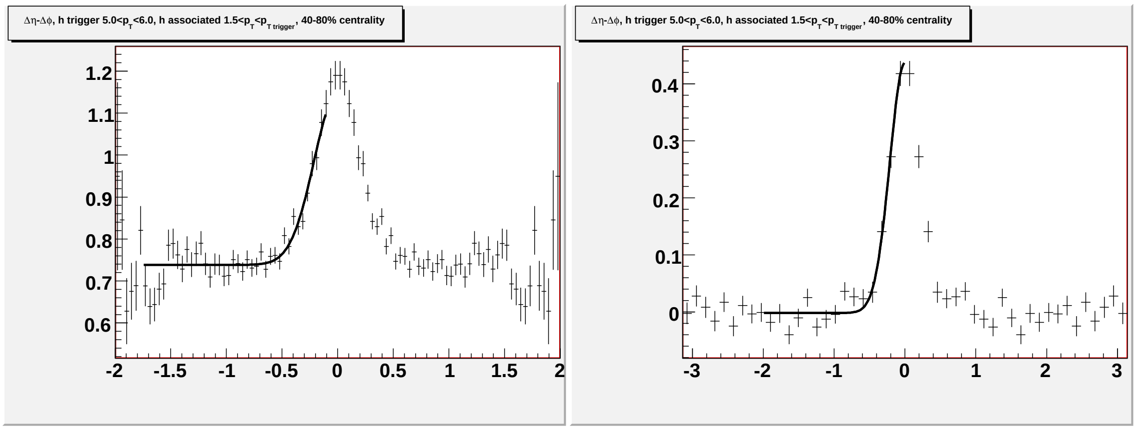

Figure G.26: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $5.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

G.E Fits for $p_T^{trigger}$ dependence in 0-12% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

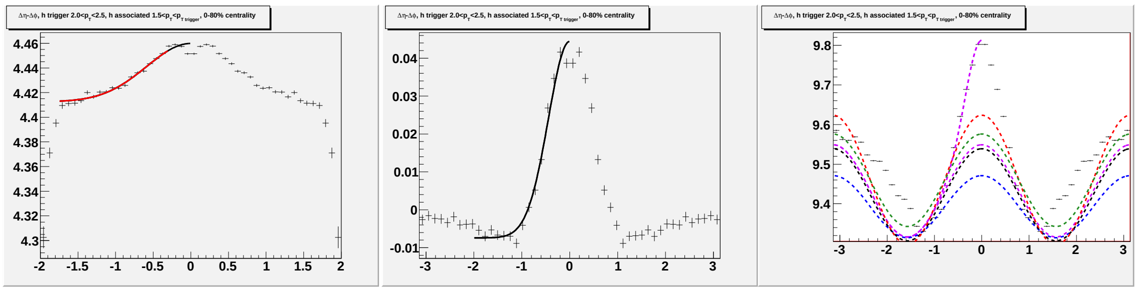

Figure G.27: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

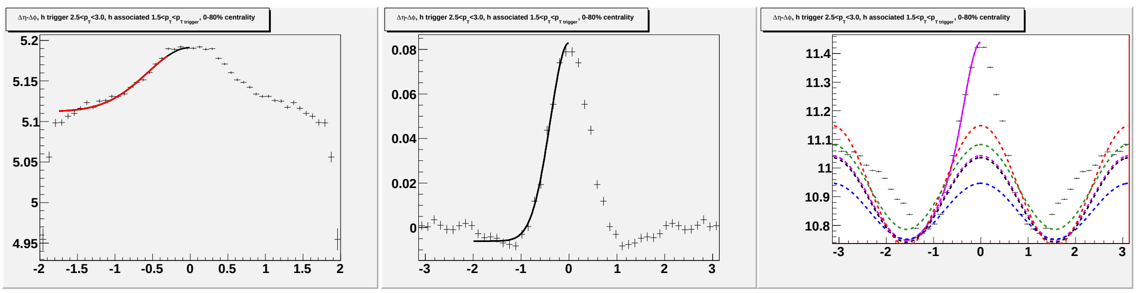

Figure G.28: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

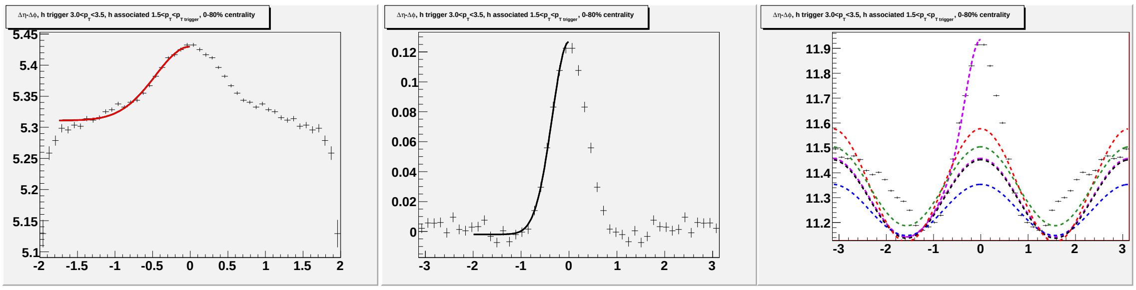

Figure G.29: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

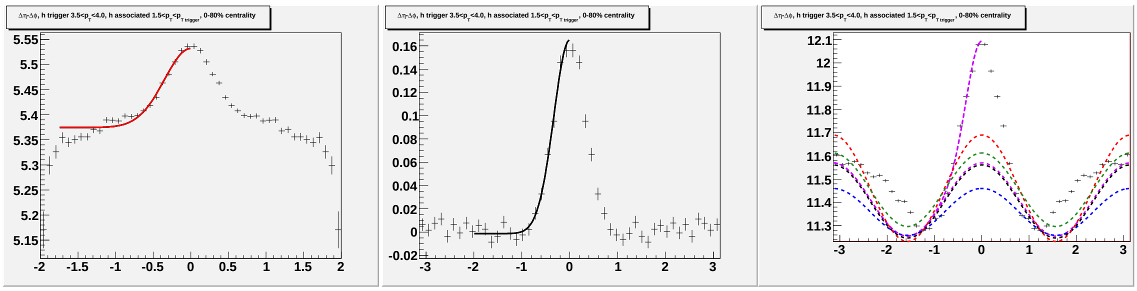

Figure G.30: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

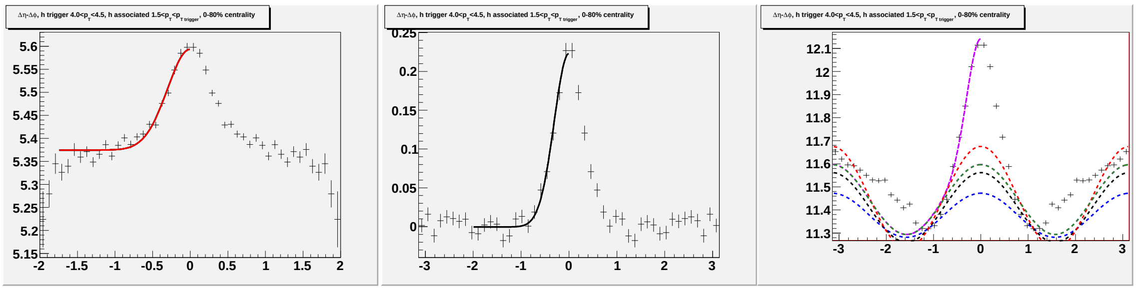

Figure G.31: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

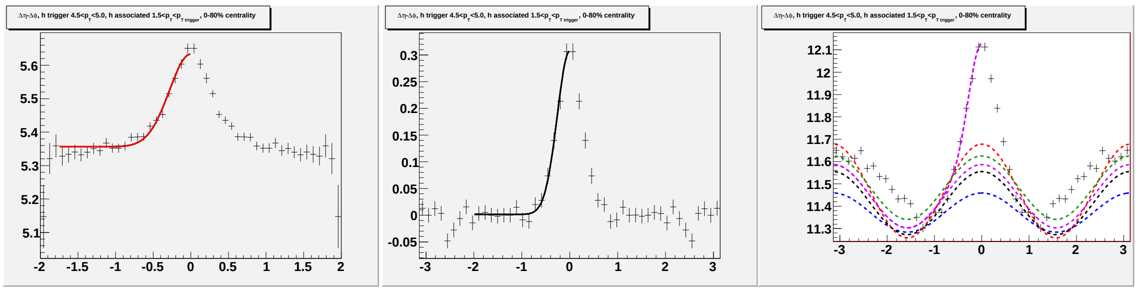

Figure G.32: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $4.5 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

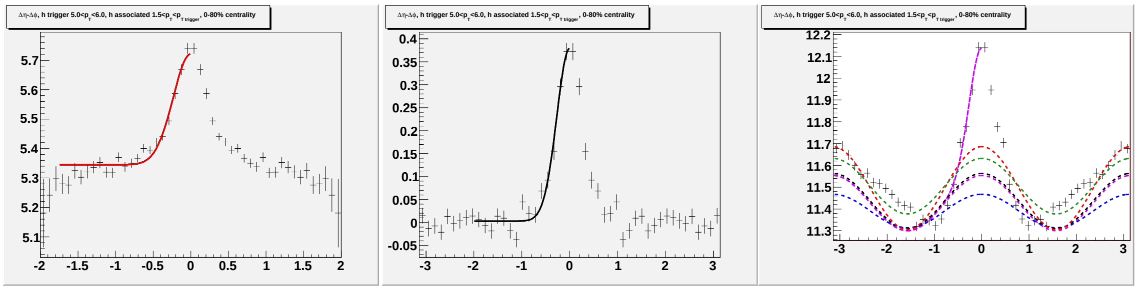

Figure G.33: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta \;| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta \;| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta \;| < 1.75$ (f) 1-D correlation for $5.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

G.F Fits for $p_T^{trigger}$ dependence in $d + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV

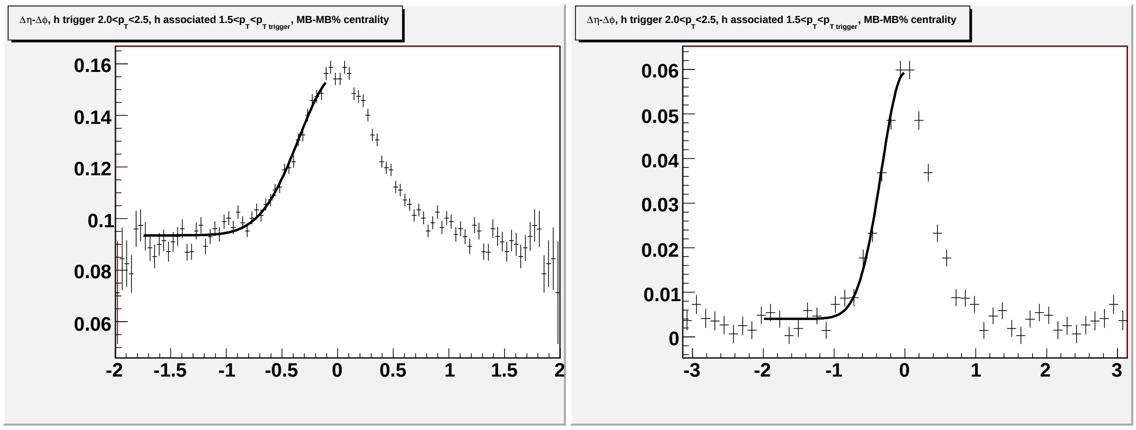

Figure G.34: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.0 < p_T^{trigger} < 2.5$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c.

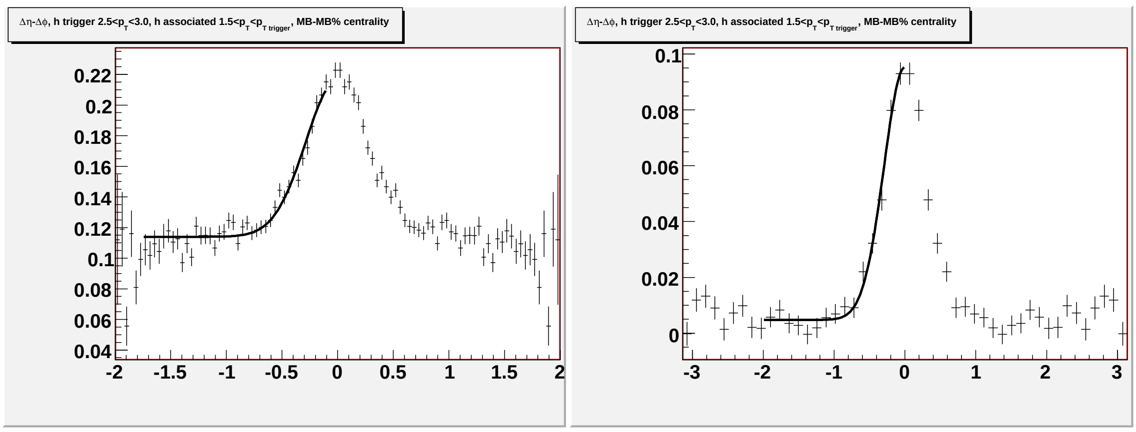

Figure G.35: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta$ $| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $2.5 < p_T^{trigger} < 3.0$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c.

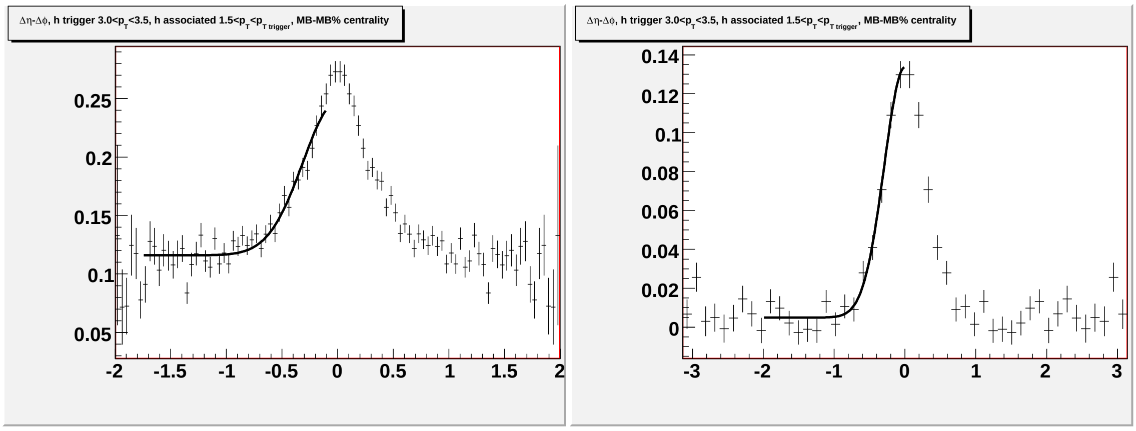

Figure G.36: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta \,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta \,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta \,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 3.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

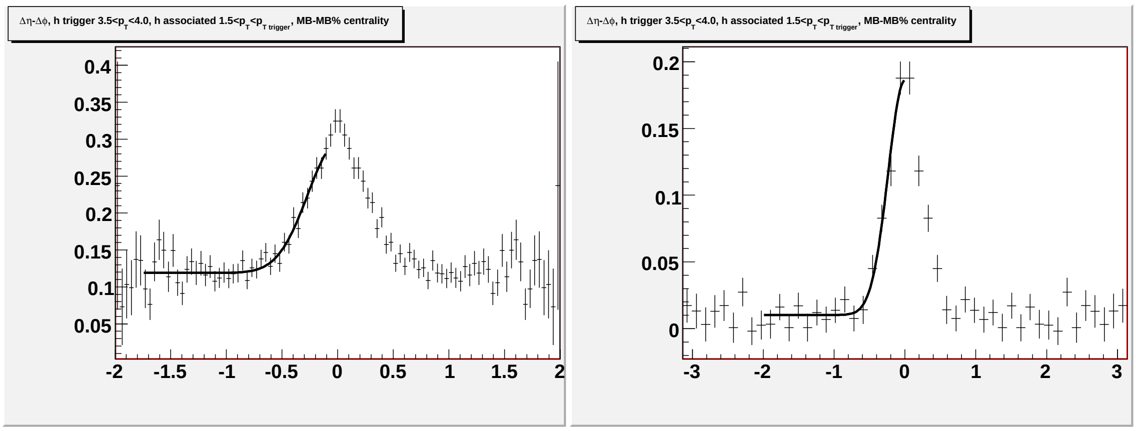

Figure G.37: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta \,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta \,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta \,| < 1.75$ (f) 1-D correlation for $3.5 < p_T^{trigger} < 4.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

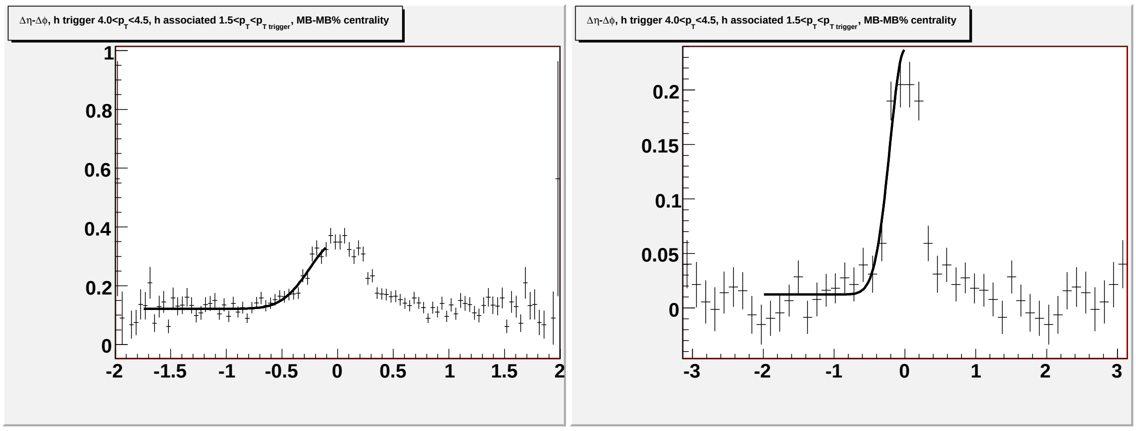

Figure G.38: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.0 < p_T^{trigger} < 4.5$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

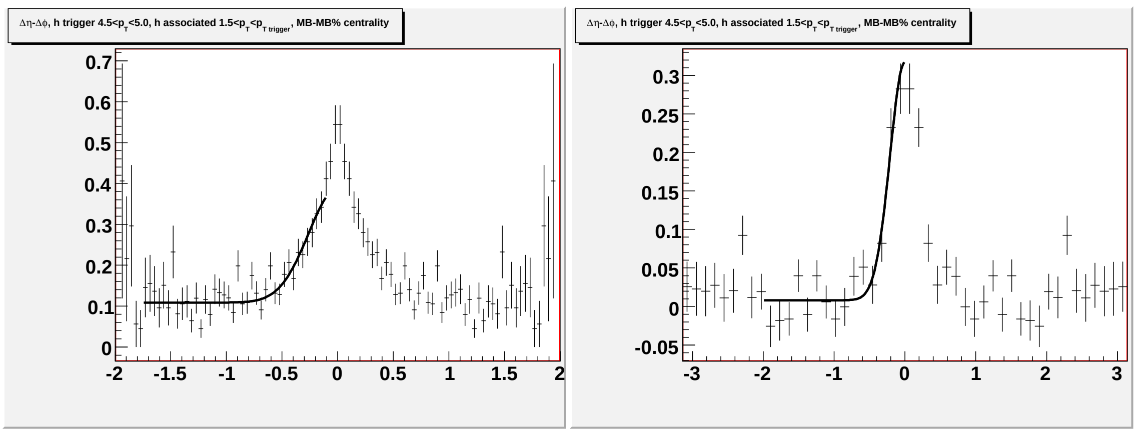

Figure G.39: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $4.5 < p_T^{trigger} < 5.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

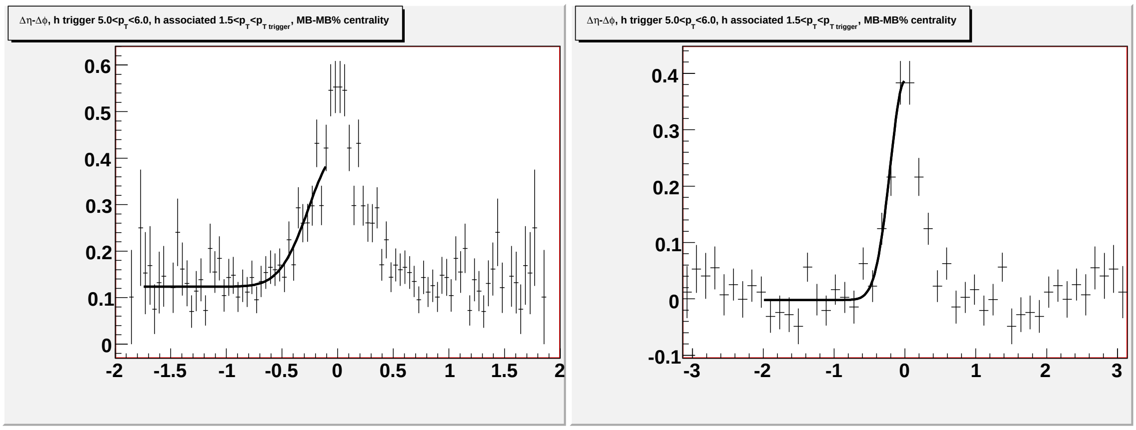

Figure G.40: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $5.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c.

Appendix H

N_{part} dependence for different collision energies

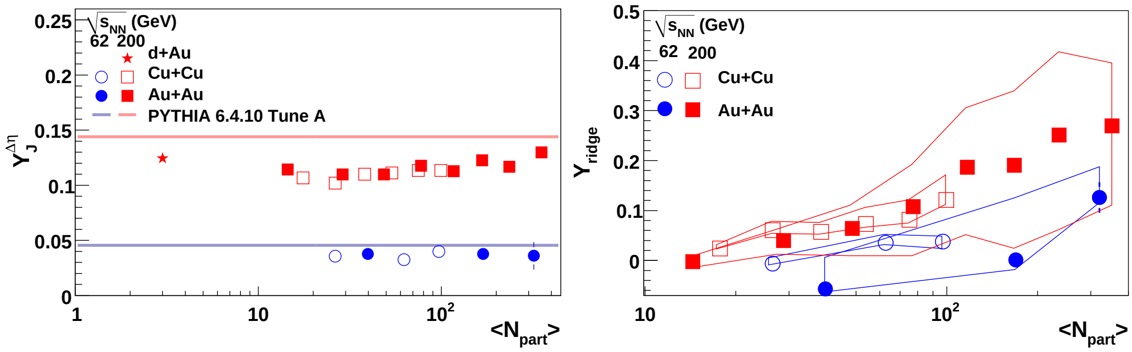

Figure H.1: N_{part} dependence of Y_{Jet} for unidentified hadrons for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for unidentified hadrons for $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $d+Au$, $Cu+Cu$ and $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

All points are done using bin counting and the $\Delta\eta$ method except for the central $\sqrt{s_{NN}} = 62$ GeV. For this data point, the fit is used because there is clearly a residual track merging effect. Because of the residual track merging effect, the width in $\Delta\eta$ is thrown out for the width plot.

H.A Fits for N_{part} dependence in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

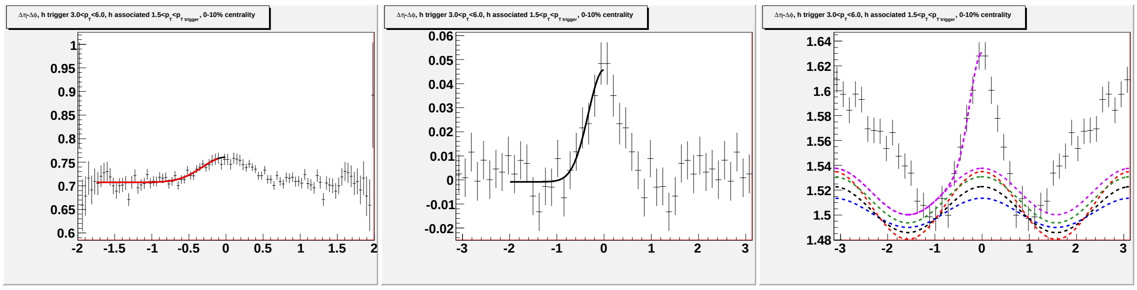

Figure H.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over 0 $< |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-10% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

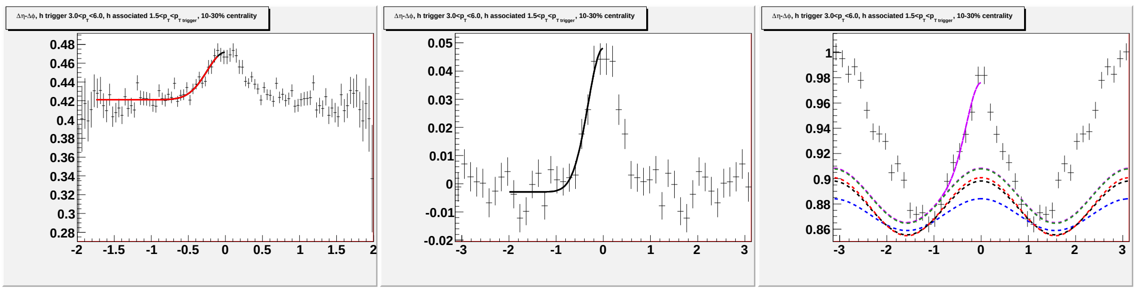

Figure H.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over 0 $< |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-30% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

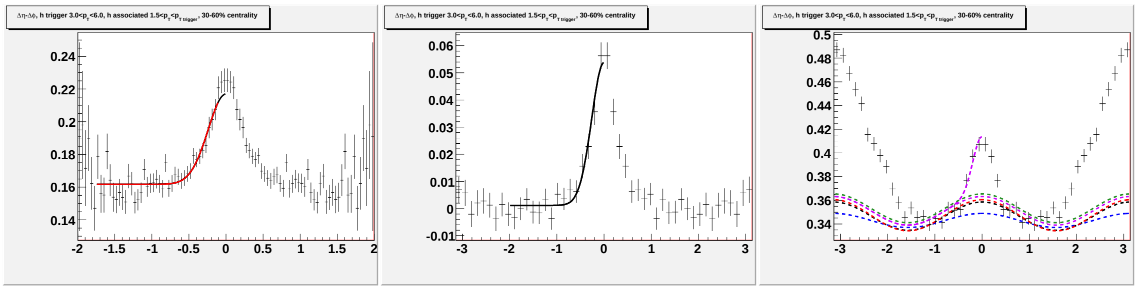

Figure H.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 30-60% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

H.B Fits for N_{part} dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

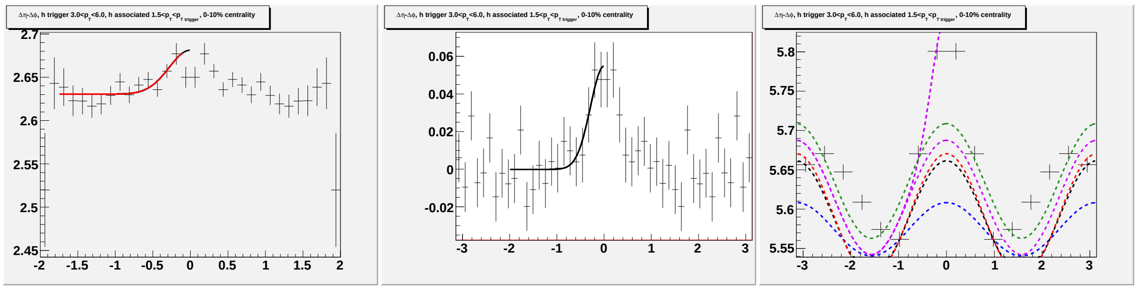

Figure H.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-10% $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

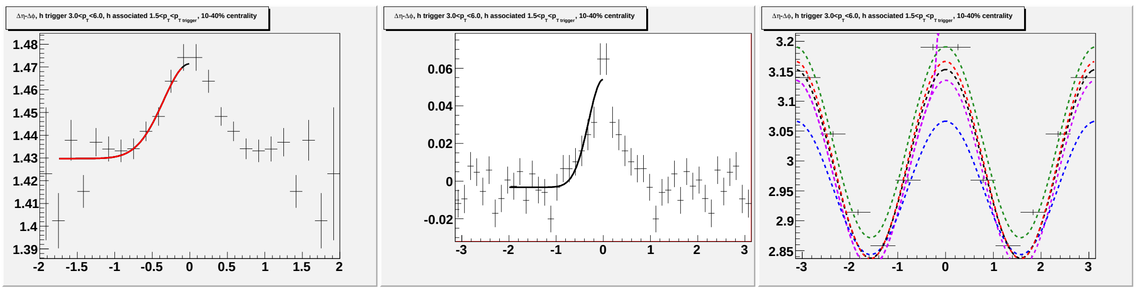

Figure H.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-40% $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

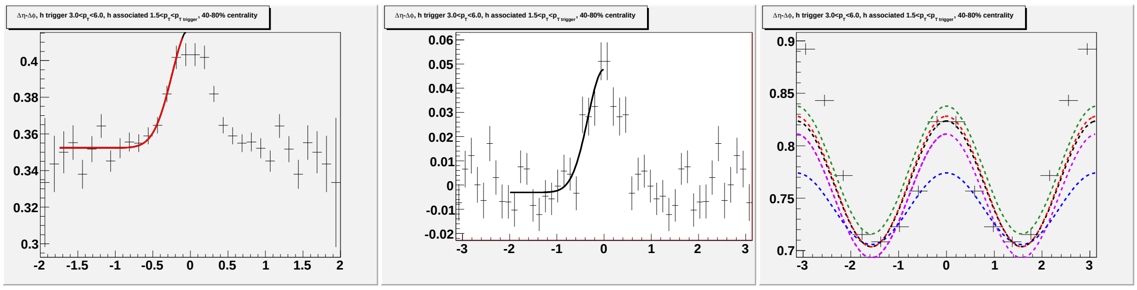

Figure H.7: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 40-80% $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV.

H.C Fits for N_{part} dependence in $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV

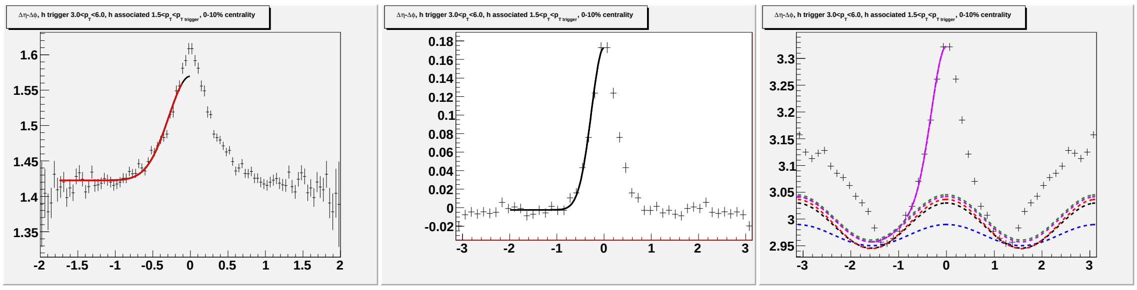

Figure H.8: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-10% $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

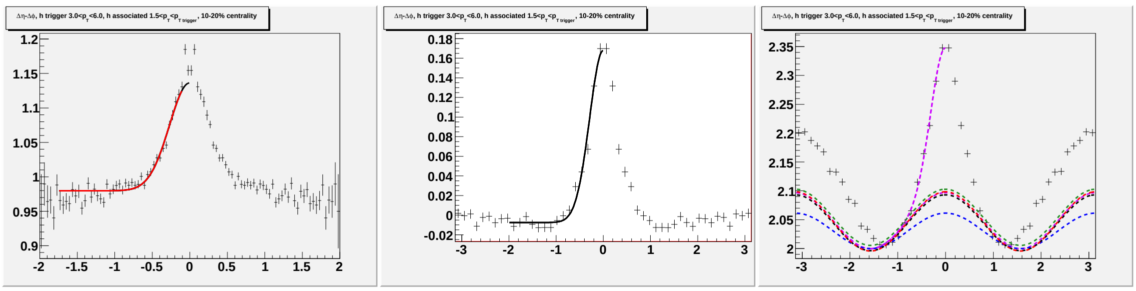

Figure H.9: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-20% $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

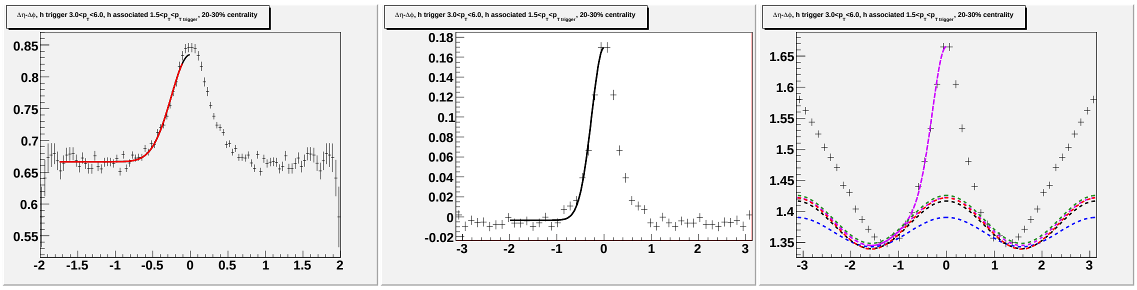

Figure H.10: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c for 20-30% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

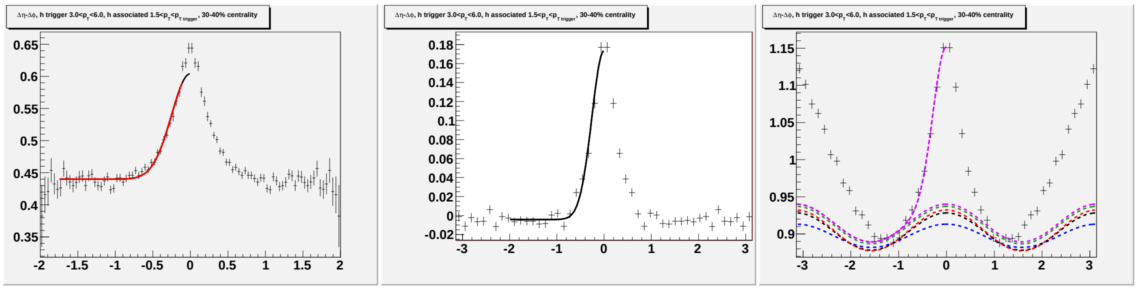

Figure H.11: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c for 30-40% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

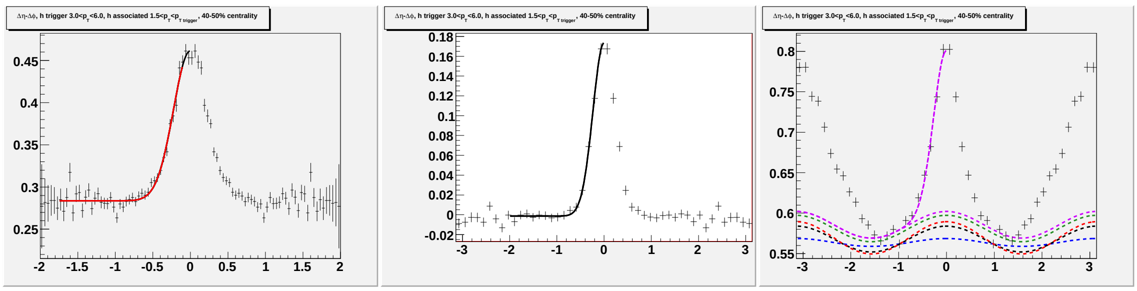

Figure H.12: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and 1.5 $< p_T^{associated} < p_T^{trigger}$ GeV/c for 40-50% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

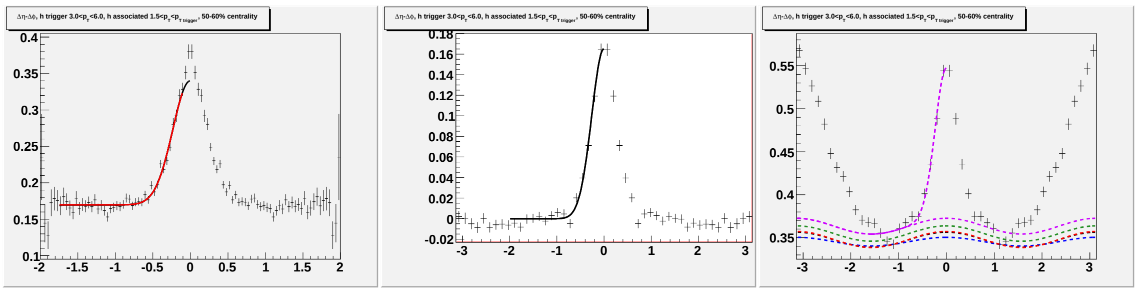

Figure H.13: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 50-60% $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

H.D Fits for N_{part} dependence in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV

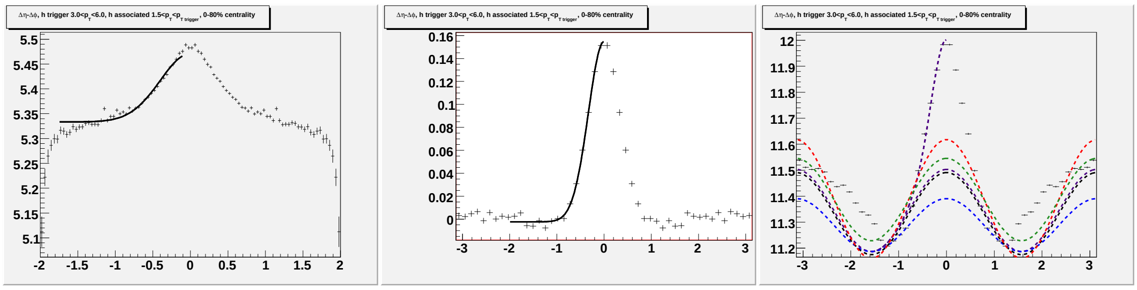

Figure H.14: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 0-12% $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

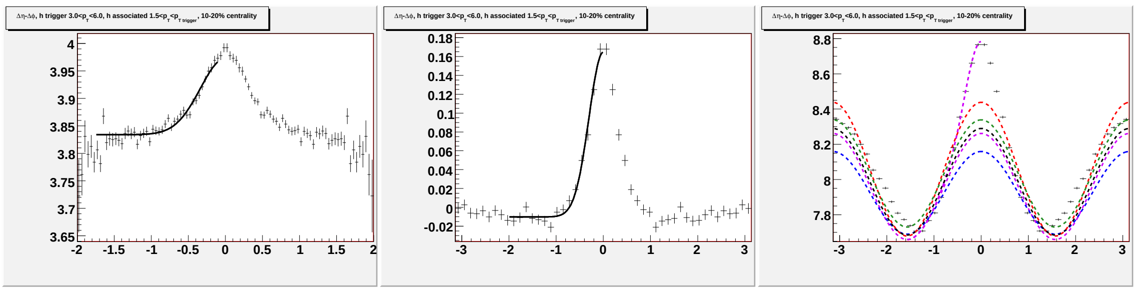

Figure H.15: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 10-20% $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV.

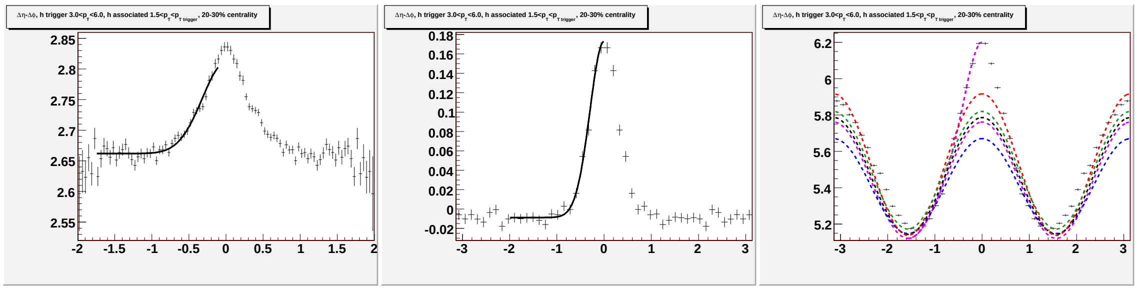

Figure H.16: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 20-30% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

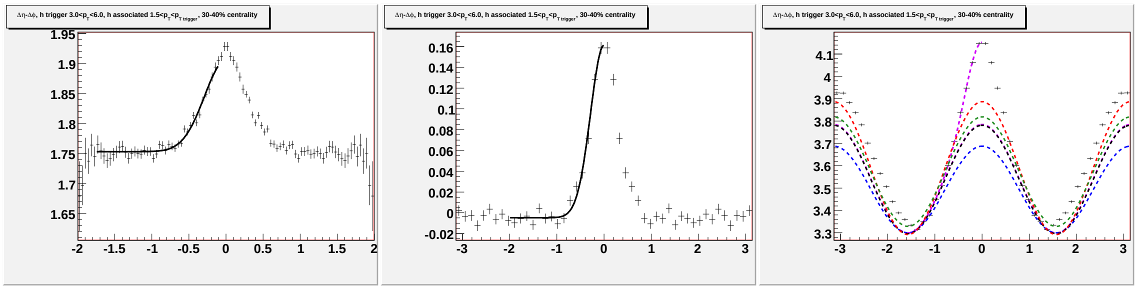

Figure H.17: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 30-40% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

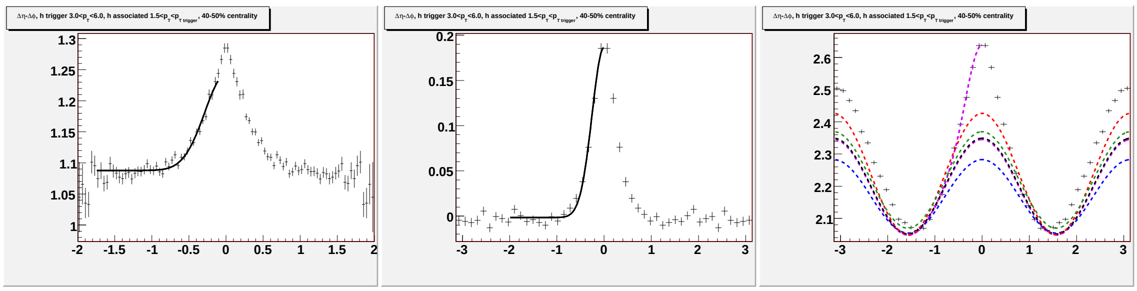

Figure H.18: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 40-50% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

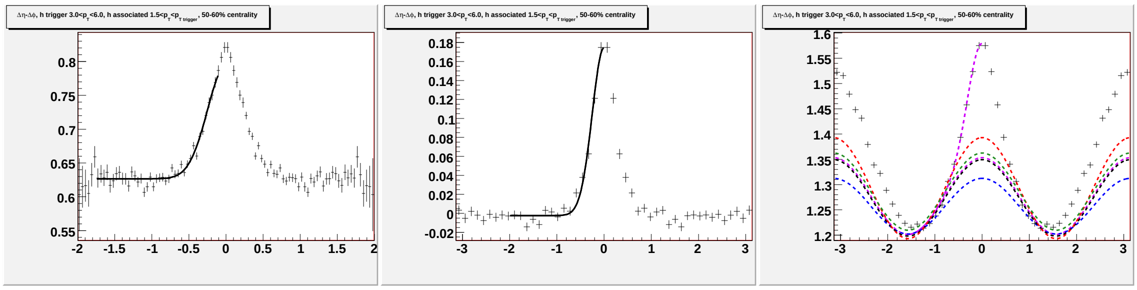

Figure H.19: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 50-60% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

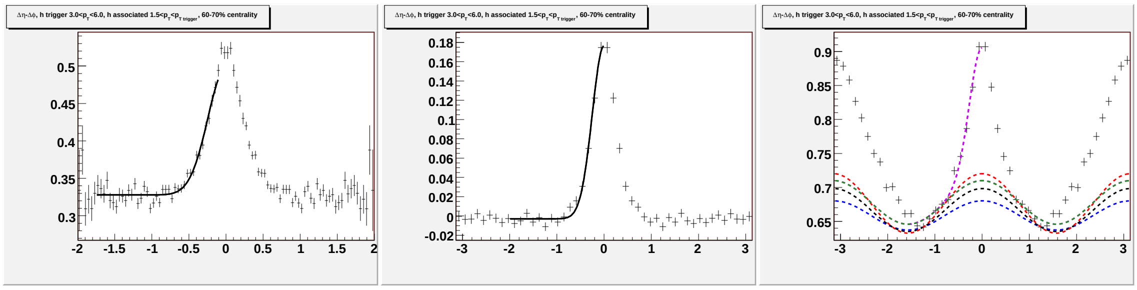

Figure H.20: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 60-70% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

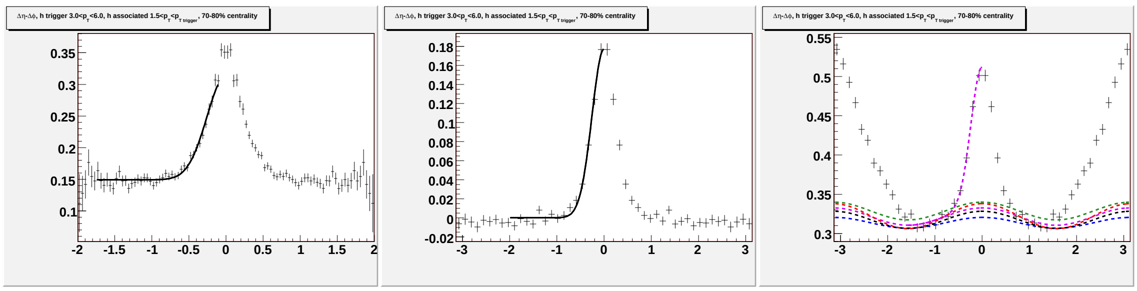

Figure H.21: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for 70-80% $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

H.E Fits for N_{part} dependence in $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

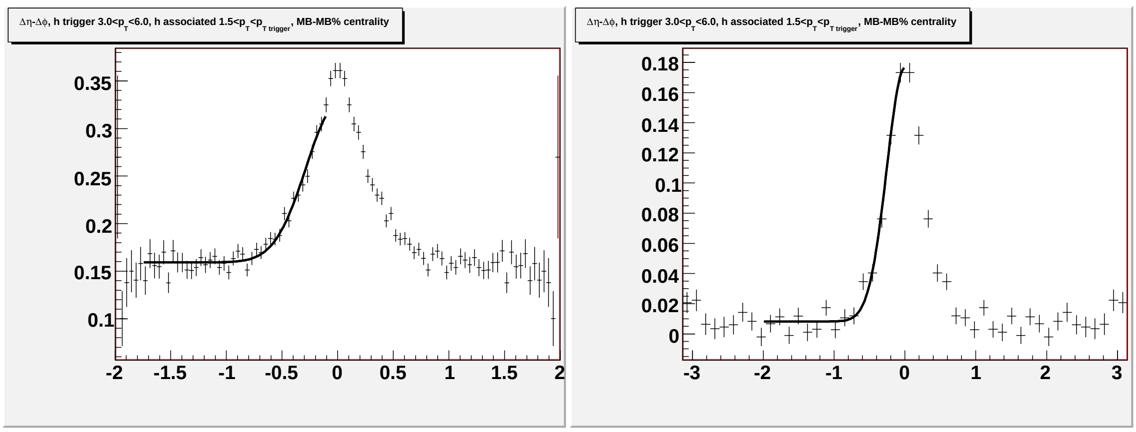

Figure H.22: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta \,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta \,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta \,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < p_T^{trigger}$ GeV/c for minimum bias $d + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

Appendix I

$p_T^{associated}$ dependence for different collision energies

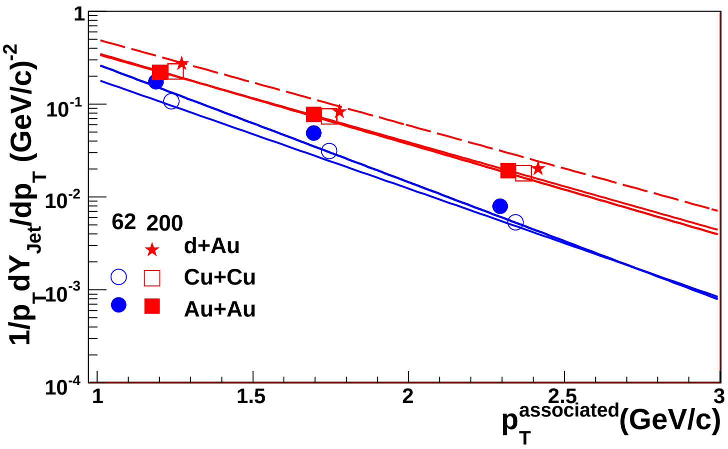

Figure I.1: $p_T^{associated}$ dependence of Y_{Jet} for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for $3.0 < p_T^{trigger} <$ 6.0 GeV/c for 0-60% central $Cu+Cu$ and 0-80% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and minimum bias $d+Au$, 0-60% central $Cu+Cu$ and 0-12% central $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

This figure still needs updated dAu data for both the yields and the widths. All points are done using bin counting and the $\Delta\eta$ method.

I.A Fits for $p_T^{associated}$ dependence in $Cu + Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV

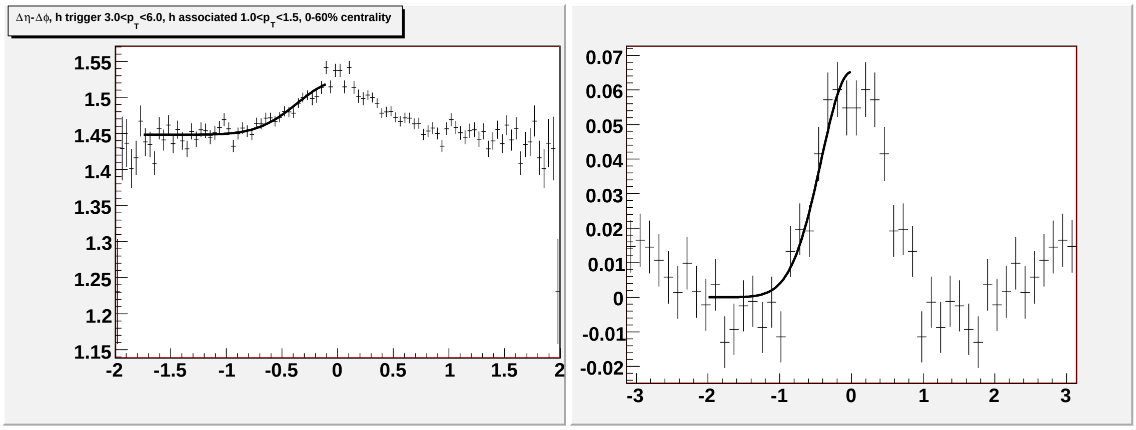

Figure I.2: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

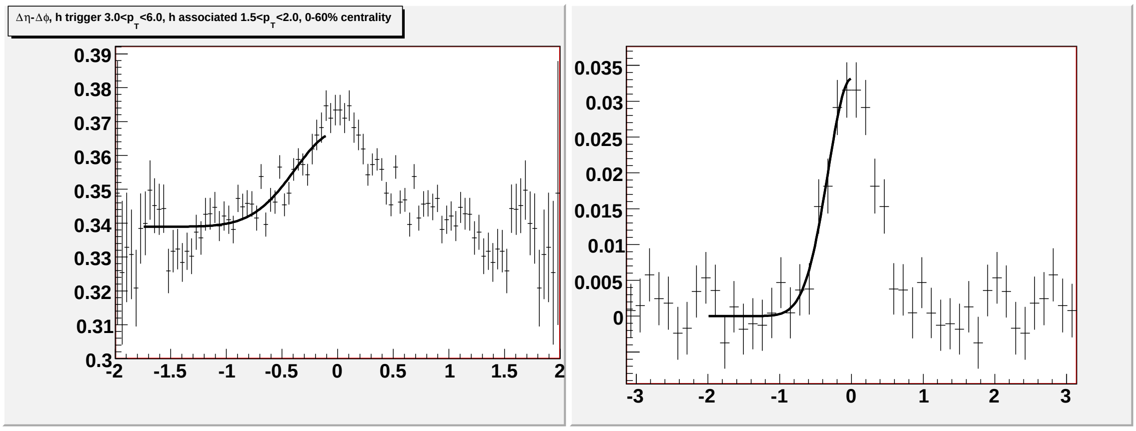

Figure I.3: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

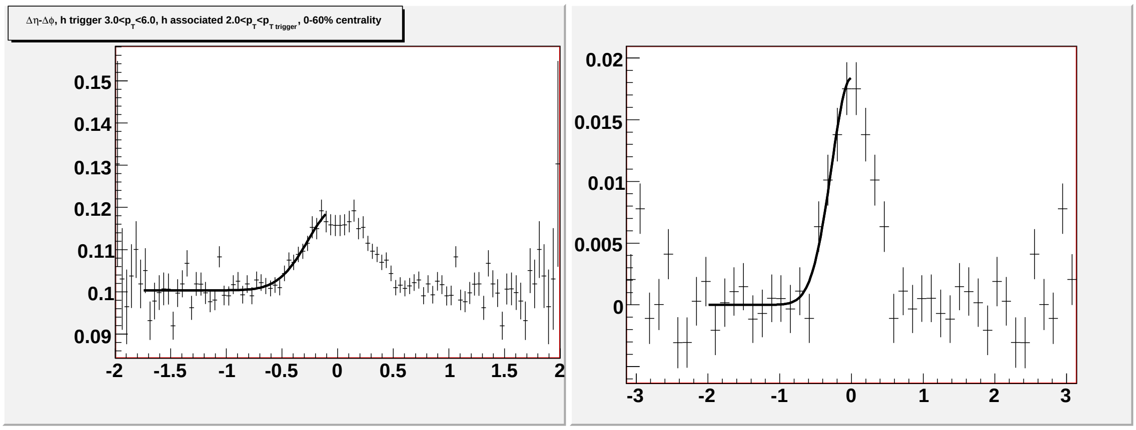

Figure I.4: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

I.B Fits for $p_T^{associated}$ dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV

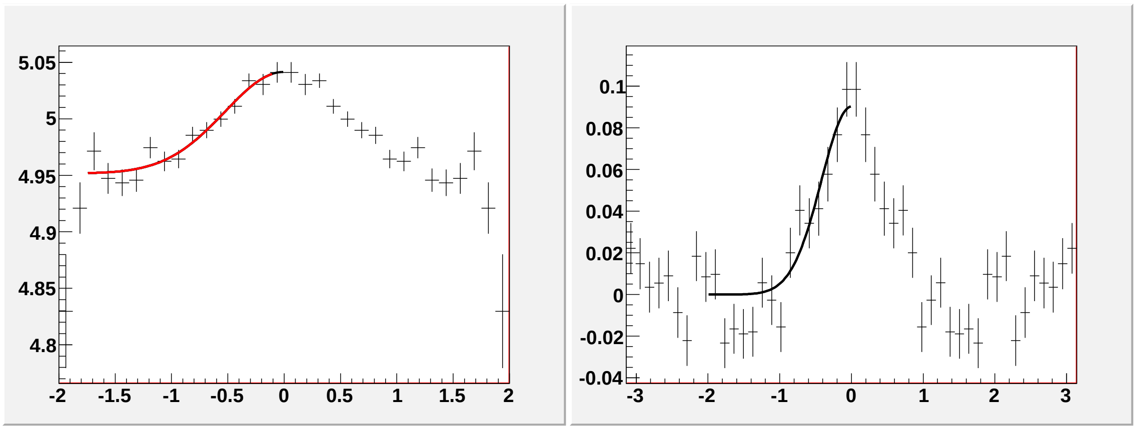

Figure I.5: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

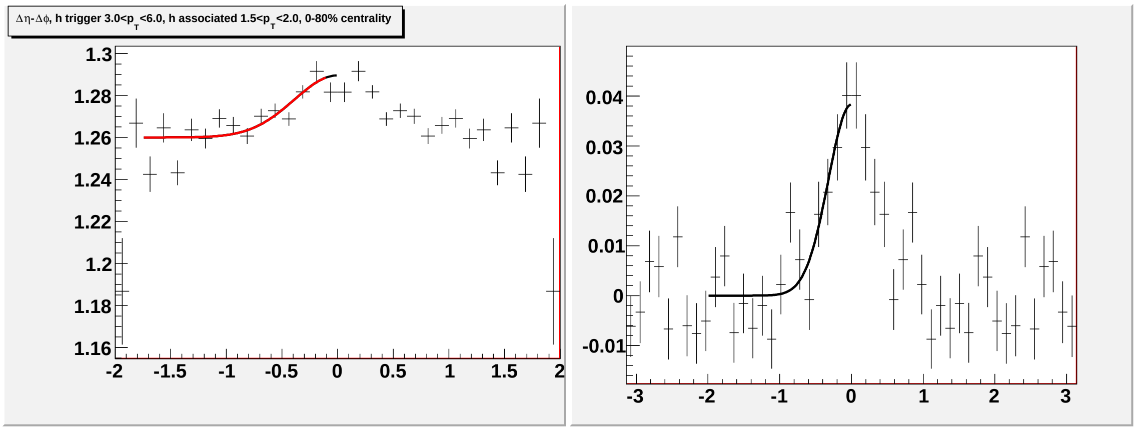

Figure I.6: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

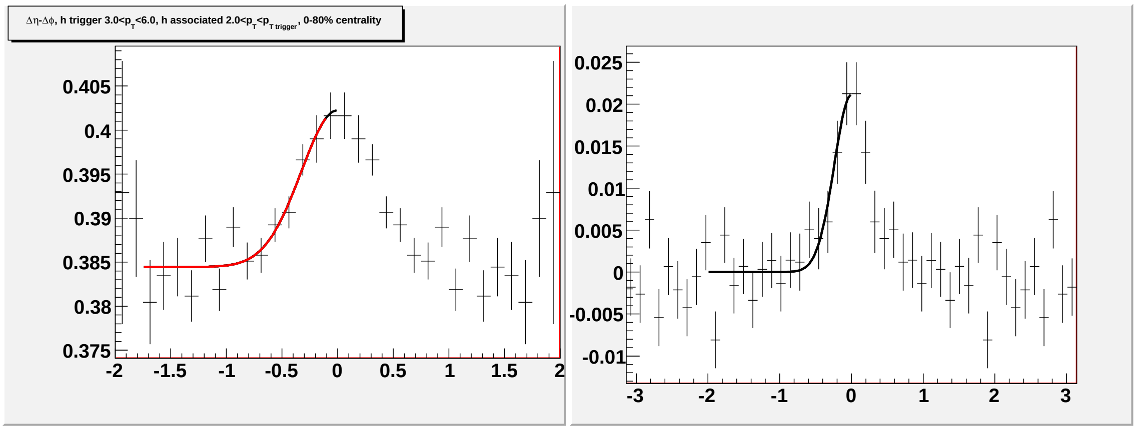

Figure I.7: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\ | < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\ | < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\ | < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

I.C Fits for $p_T^{associated}$ dependence in $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 200$ GeV

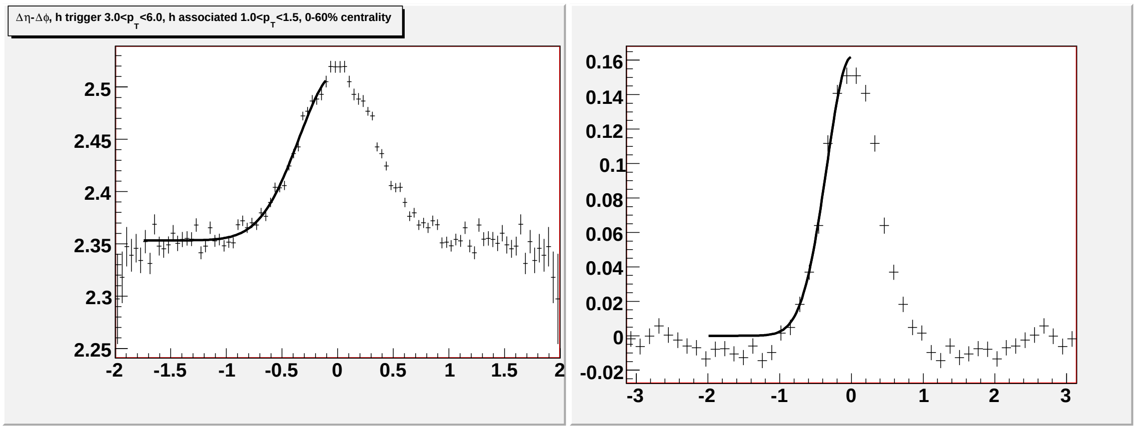

Figure I.8: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

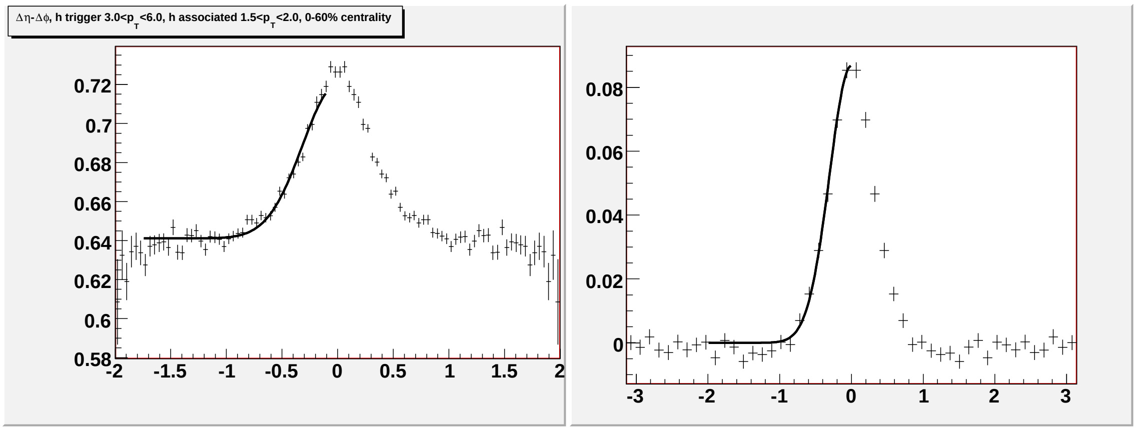

Figure I.9: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

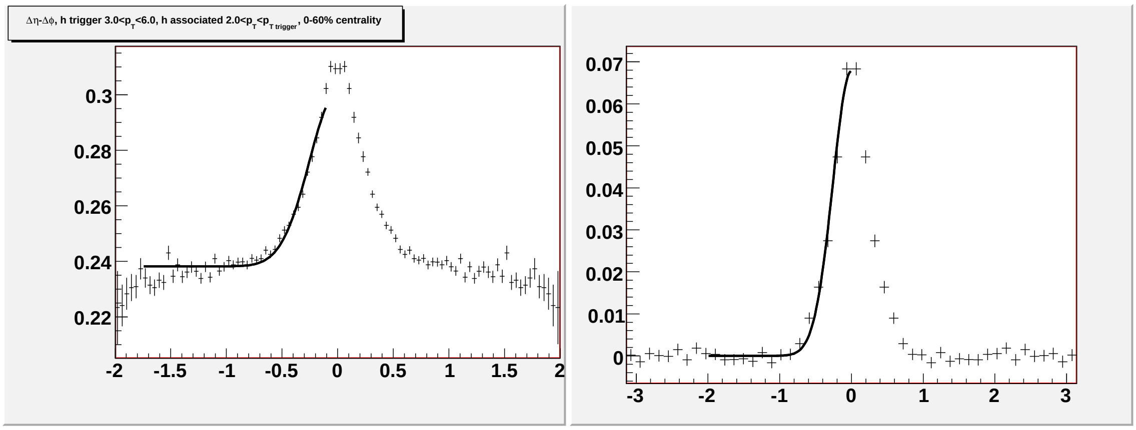

Figure I.10: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

I.D Fits for $p_T^{associated}$ dependence in minimum bias $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV

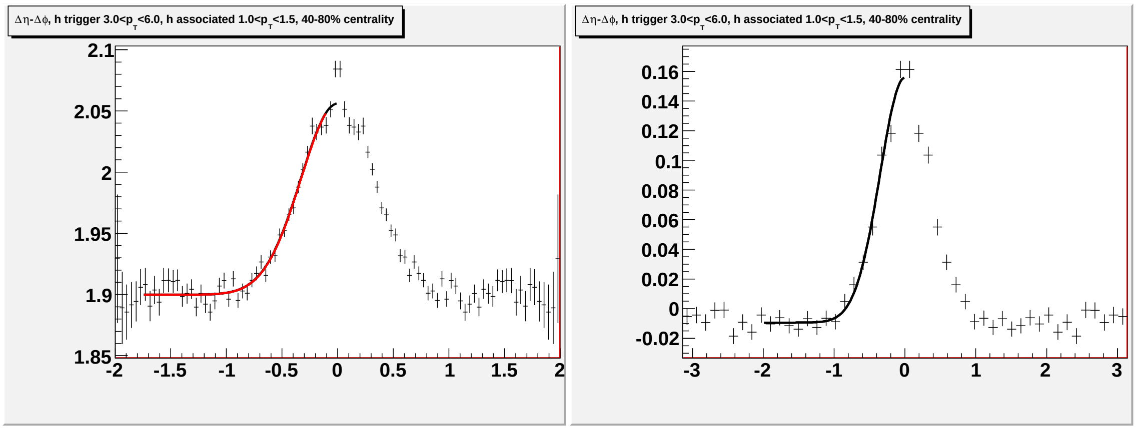

Figure I.11: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} <$ 1.5 GeV/c.

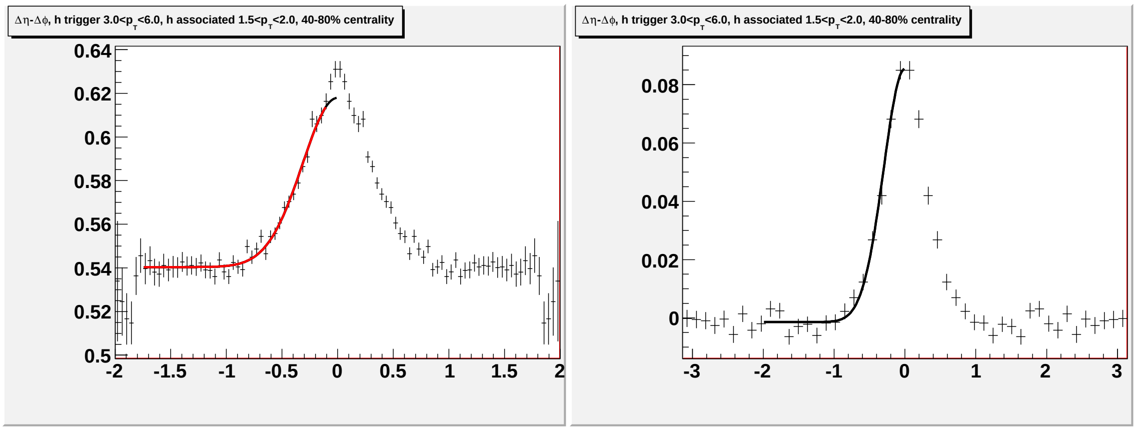

Figure I.12: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} <$ 2.0 GeV/c.

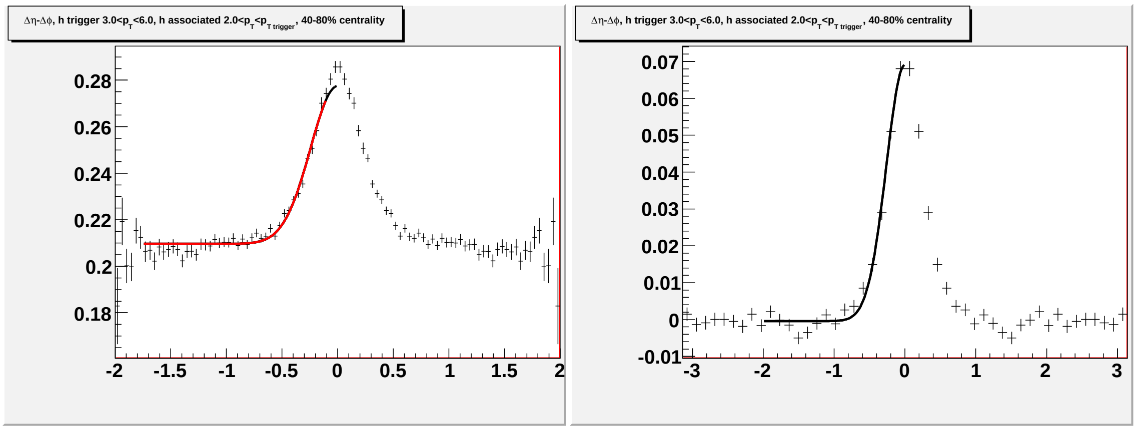

Figure I.13: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

I.E Fits for $p_T^{associated}$ dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

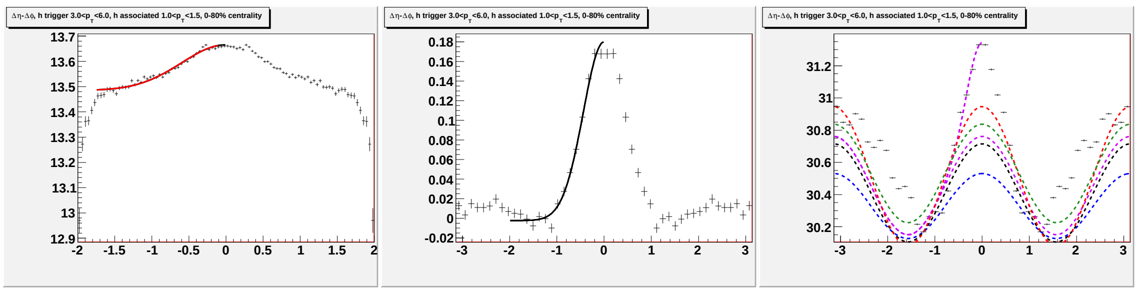

Figure I.14: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

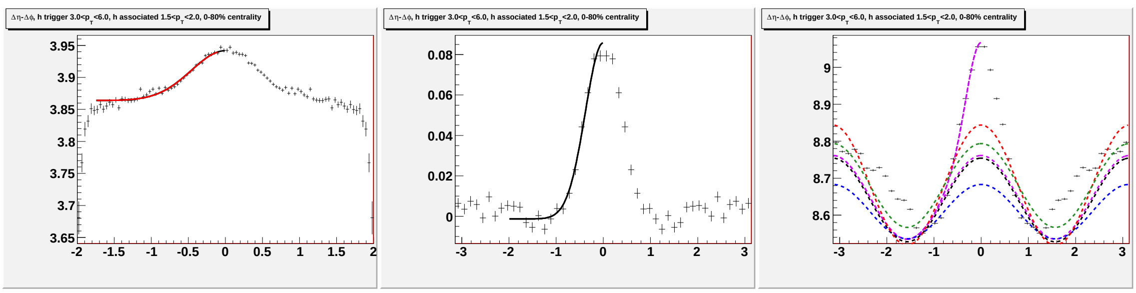

Figure I.15: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

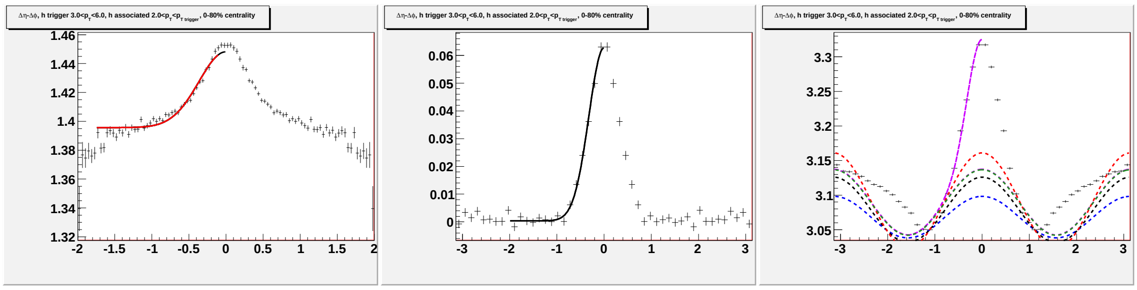

Figure I.16: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta\,| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta\,| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta\,| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

I.F Fits for $p_T^{associated}$ dependence in $d+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

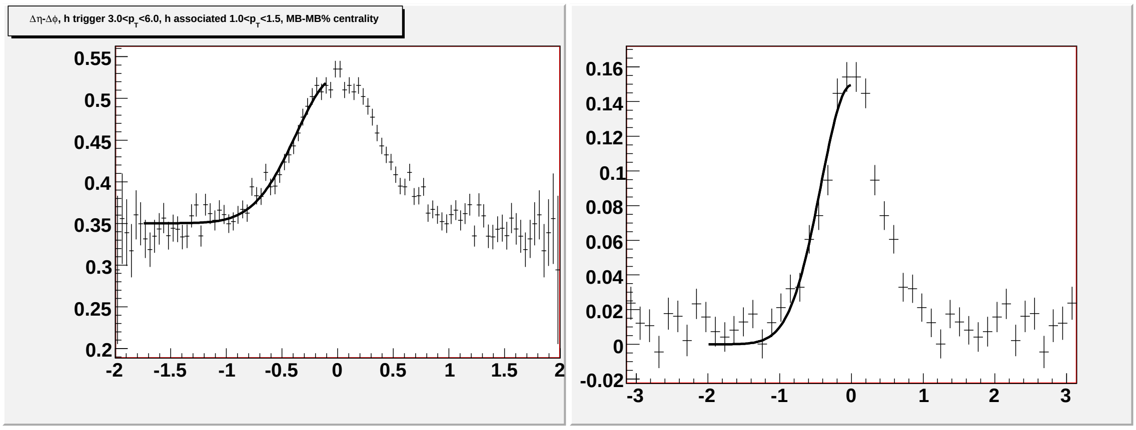

Figure I.17: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

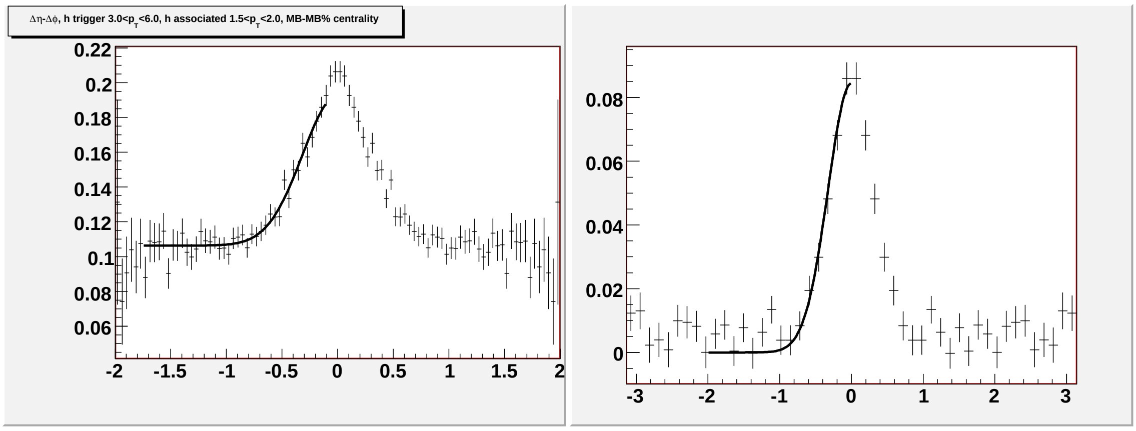

Figure I.18: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.5 < p_T^{associated} < 2.0$ GeV/c.

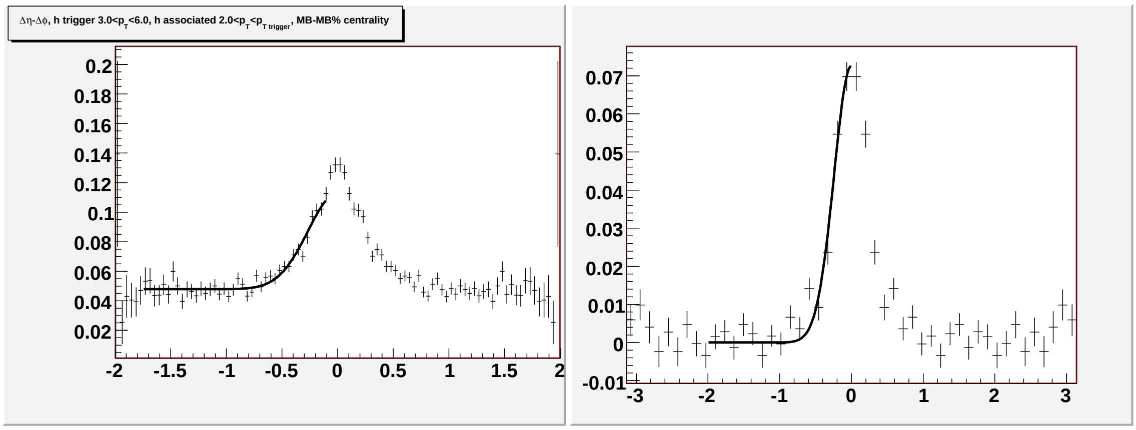

Figure I.19: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $2.0 < p_T^{associated} < 3.0$ GeV/c.

I.G Fits for $p_T^{associated}$ dependence in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV

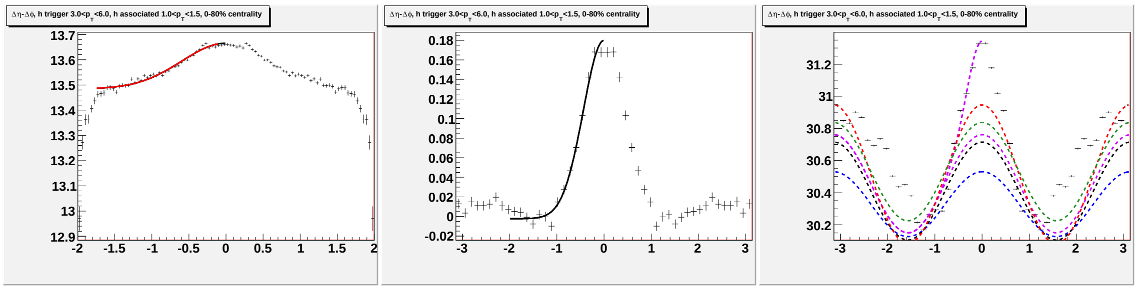

Figure I.20: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

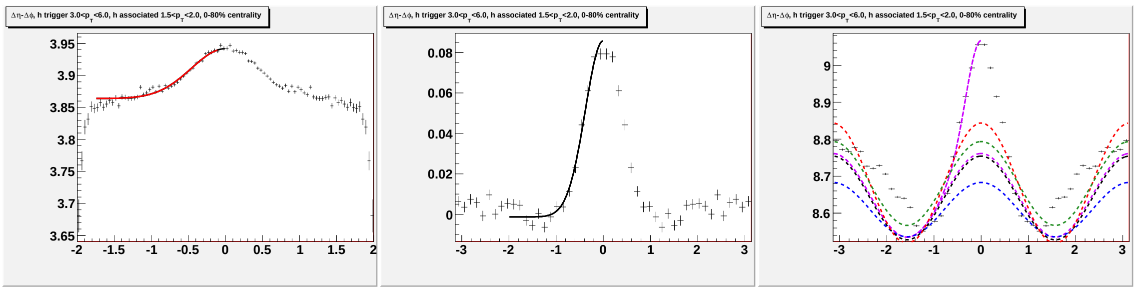

Figure I.21: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

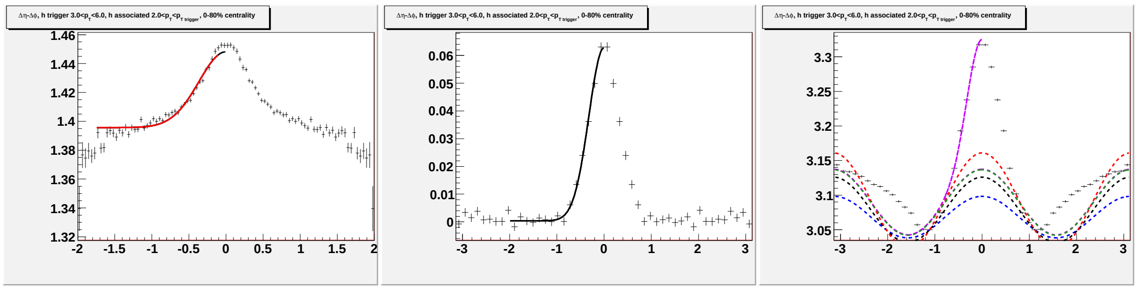

Figure I.22: Going left to right/top to bottom (a) jet-like correlation using $\Delta\eta$ method (b) jet-like correlation using $\Delta\phi$ method (c) correlation integrated over $|\Delta\eta| < 0.75$ (d) correlation integrated over $0.75 < |\Delta\eta| < 1.75$ (e) correlation integrated over $0 < |\Delta\eta| < 1.75$ (f) 1-D correlation for $3.0 < p_T^{trigger} < 6.0$ GeV/c and $1.0 < p_T^{associated} < 1.5$ GeV/c.

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